Generalized Existence Theory (GET): A Four-Phase Protocol Recursive Hierarchy — The Unique Syntax of All That Can Be Said

From Vacuum Fluctuations to Intelligent Civilization:A Computable, Programmable, Ad Hoc-Free, Cross-Scale, Necessary Unification

YUQI LION

Independent Researcher | Founder of Generalized Existence Theory (GET/SETs)

MomSayAI | GET Laboratory

Website: https://exist.chat

Email: yuqi@exist.chat

GitHub: Generalized-Existence-Theory

December 2025

Contents

Abstract

How does existence exist? How does it evolve?

This paper proposes Generalized Existence Theory (GET), starting from the meta-philosophical stance that "rule execution is phenomenon; without process, nothing exists," aiming to provide a computable answer to "how existence is possible": the sole dynamical ontology is the self-referential recursive execution of a protocol set L; form M is its stable external manifestation; uniformly expressed as $\Omega^{(M)} = \Phi(L)$.

GET's inputs are strictly limited to five empirical cornerstones: the energy–time constraint of quantum fluctuations, power-law statistics, the three laws of thermodynamics, mass–energy equivalence, and the dependence of structured existence on objective rule sets. From these cornerstones, we forcibly distill four principles of existence (difference as existence, difference must return to zero, difference must strive to persist, persistence exacts a cost), and from these derive four core mechanisms:

1. The Existence Equation——a comprehensive measure for all forms of existence:

$$\boldsymbol{\Psi_M(t) = \frac{E_{\text{solid}}(M,t)}{\tau_{\text{solid}}(L)} + \frac{E_{\text{active}}(M,t)}{\tau_{\text{active}}(L)} + \eta(L)\cdot e(t)}$$

where E_{solid} and E_{active} denote solidified-state and active-state energy respectively, and $e(t)$ is the instantaneous net flux. The total structural assets $E_{\text{locked}} = E_{\text{solid}} + E_{\text{active}}$ correspond to the mass m in traditional physics, satisfying $E_{\text{locked}} \approx mc^2$.

2. The Survival Criterion——the necessary and sufficient condition for a form to persist:

$$\boldsymbol{\frac{E_{\text{active}}(M,t)}{\tau_{\text{active}}(L)} + \eta(L)\cdot e(t) \geq 0}$$

3. The Geometric Selection Principle——the sole gateway for recursive upgrade: only when the net borrowing energy curve $E(t)$ and the threshold line form a qualified interval that meets the minimum window condition can a system transition to a higher recursion depth d.

4. The Four-Phase Cycle——the execution algorithm for recursive upgrade: Trial—Capture—Solidify—Generate. Complexity growth is driven **only** by this cycle.

Thus, GET derives a single causal chain from cosmic genesis to hierarchical evolution, and gives strongly falsifiable predictions: the non-gravitational interaction cross section of dark matter is strictly zero; the equation of state of dark energy holds exactly as $w_\Lambda = -1$, with strictly constant density ρ_Λ; gravity is defined as a geometric effect at $d = 0$. The conservation equations clarify the serial path of "matter $\to$ radiation $\to$ redshift freezing $\to$ dark energy background."

GET introduces no extra entities or ad hoc hypotheses, but starts from widely accepted empirical cornerstones to provide a unified grammar for how existence is possible and how it evolves, thereby integrating ontology, physics, and cognition within a single framework.

This paper establishes a four-level falsification system and provides the ultimate adjudication conditions. Any person or institution, using AGI and including brute-force enumeration, may participate: if a falsification condition is met, GET fails; if none is met, the assertions remain valid. GET is not another fitting model, but a public trial of the unique grammar of speakable existence.

Keywords: Generalized Existence Theory; protocol-recursive systems; dark matter; dark energy; nature of gravity; geometric selection principle; four-phase cycle; cross-scale unified theory; falsifiability

1 Introduction: From Fragmented Vocabularies to a Unified Grammar of Existence

How does existence exist? How does it evolve?

This question is metaphysical. Yet we answer it every day.

Although modern science has achieved high-precision descriptions at different scales, it has long lacked a unified generative grammar that can operate across scales. Quantum theory, cosmology, life science, and social science are each powerful, but at the level of "how could this be possible" they still exhibit a fragmented structure. When systematic tension arises between theory and observation, a common strategy is to introduce additional entities, tunable parameters, and mechanism patches to preserve the fit. This paper calls this the "ad hoc fitting paradigm," distinguishing it from the serious Newton–Einstein–Popper tradition: the latter starts from first principles, takes deductive closure as its method, and adopts strong falsifiability as the standard of adjudication.

Generalized Existence Theory (GET) is developed within this serious tradition. GET introduces no new a priori entities; its theoretical inputs are limited to five empirical cornerstones: the energy–time constraint and power-law statistics of quantum fluctuations, the three laws of thermodynamics, mass–energy equivalence, and the dependence of structured existence on rule sets. Grounded in the meta-philosophical stance that "rule execution is phenomenon; without process, nothing exists," GET proposes: the sole dynamical ontology is the self-referential recursive execution of a protocol set L; form M is its stable external manifestation; uniformly expressed as

$$\Omega^{(M)} = \Phi(L).$$

Within this framework, existence and evolvability are quantified simultaneously. The former is used to determine whether a system can persist via the existence-degree equation; the latter is given by a unified upgrade mechanism: complexity growth is driven **only** by "geometric selection + the four-phase cycle (Trial—Capture—Solidify—Generate)." Thus, cosmic genesis, category differentiation, and stratified evolution are no longer parallel patch narratives, but are written as different stages of the same causal chain.

This paper further provides adjudicable predictions: dark matter's non-gravitational cross section is strictly zero; dark energy's intrinsic equation of state $w_\Lambda = -1$ holds exactly, and its density ρ_Λ is strictly constant ($\dot{\rho}_\Lambda = 0$). Unlike the ΛCDM model's positioning of dark energy as a mysterious constant, GET restores it as the natural product of an extremely early-universe event (Category C). The observed increase in the dark-energy fraction over time arises entirely from the expansion dilution of ordinary matter and radiation, not from any dynamical evolution internal to dark energy. The conservation equations (see Axiom XI) make this serial path explicit. Gravity is defined as a geometric effect at $d = 0$.

Before entering the strict formal system, we will first forcibly distill four basic ”existence principles” from the five cornerstones (Section 2). They are not conclusions of later derivations, but the philosophical foundation and logical skeleton of GET: **difference is existence (origin), difference returns to zero (constraint), difference must seek survival (path), survival must pay a price (fate)**. All subsequent definitions, axioms, and corollaries are expansions and formalizations of these four principles.

To avoid any retreating explanations, this paper establishes a four-level falsification system (Section 4) and uses Corollary 22 as the ultimate adjudication. The goal is not to offer a locally fitted model, but a publicly auditable grammar of existence. Any person or institution may participate—using AGI, including but not limited to brute-force enumeration: if any falsification condition is hit, GET fails; if not, the assertion remains valid under the same adjudication protocol.

READING GUIDE

For first-time readers, it is recommended to start with **Appendix E, ”GET Reading and Verification Guide,”** avoiding all mathematical symbols and formal definitions, and to follow from everyday intuitions you already know—hungry, tired, etc.—step by step. Then return to Chapter 2’s methodological starting point.

---

2 Methodology: Deductive Science Returning to First Principles

2.1 Root of the Dilemma: Paradigm Drift from Deductive Science to Ad Hoc Fitting

In the history of science, the most penetrating theories tend to follow the same methodological skeleton: start from a small number of solid facts, organize knowledge with a minimal axiomatic system, and derive adjudicable conclusions through strict deduction.

Contemporary foundational research has been highly successful in local fitting precision, yet shows clear drift at the level of unified explanation: when systematic tension arises between theory and observation, the common strategy is to add new entities, free parameters, or patch mechanisms to preserve the fit—rather than returning to the axiomatic level to rebuild the logical chain. This paper calls that path the "ad hoc fitting paradigm." Its engineering value should not be denied, but its costs are rising ontological complexity, reduced deductive transparency, and weakened decisiveness.

GET's task is not to negate existing achievements, but to pull the problem back from "how to keep fitting" to "why it must be so."

2.2 GET's Methodological Stance: Returning to the Traditional Serious Paradigm

GET does not invent a new scientific standard; it returns to the traditional serious paradigm of Newton–Einstein–Popper and carries it through to the end. This paradigm contains three non-negotiable principles:

1. **First-principles inputs:** A theory's starting point must be high-consensus empirical facts that are repeatedly verifiable.
2. **Deductive closure:** Every step from inputs to conclusions is traceable, auditable, and reproducible.
3. **Strong falsifiability:** The theory must provide public, binary, executable failure conditions.

Accordingly, GET rejects the survival mode of "retreating parameters to preserve fidelity," and adopts an adjudication rule of "hit = defeat."

2.3 Deductive Starting Point: Five Empirical Cornerstones (The Only Inputs)

All inputs to GET are restricted to the following five items:

1. **Heisenberg uncertainty principle (the energy–time constraint of quantum fluctuations):** $\Delta E \cdot \Delta t \geq \hbar/2$.
2. **Extreme-number power-law statistical distribution (the energy-scale distribution of quantum fluctuations):** $P(\Delta E) \propto \Delta E^{-\alpha},\ \alpha > 1$.
3. **Rigid boundaries of the three laws of thermodynamics:**
   - **First law (energy conservation):** Energy can convert among forms, but the total remains conserved.
   - **Second law (entropy increase):** The entropy of an isolated system never decreases; processes have an irreversible direction.
   - **Third law (unattainability of absolute zero):** No finite sequence of operations can bring a system to absolute zero.
4. **Einstein's mass–energy equivalence relation:** $E = mc^2$, revealing that energy and mass are two morphological manifestations of the same essence.
5. **Rule dependence of structured existence:** All stable non-trivial structures (from fundamental particles to life and society) in their existence, maintenance, and interactions necessarily depend on an internal, hierarchical, describable rule set that does not depend on consciousness for its existence or transfer (e.g., physical laws, genetic codes, social contracts). The execution and maintenance of these rules require continuous energy input. **Rules themselves are objective existents.**

Beyond these five cornerstones, GET introduces no independent a priori entities, free parameters, or ad hoc patches. This means every subsequent definition, axiom, and corollary must be completed within the closure of these inputs; any additional independent assumption constitutes a negation of GET.

Closure criterion:

$$\exists \boldsymbol{X}\ \big(\text{causal}(\boldsymbol{X}) \wedge \boldsymbol{X} \notin \text{Closure}(\boldsymbol{L})\big) \Rightarrow \textbf{GET falsified.}$$

2.4 History's Echo: The Unfinished Half-Step

Twentieth-century physics reached astonishing heights: Heisenberg gave the quantum constraint; Einstein wrote down mass–energy unification. The former carved rigid boundaries for process; the latter revealed the equivalence of forms.

But in GET's view, history stopped at a crucial point—"half a step short": we treated these achievements as descriptive endpoints, yet did not continue to ask for their common generative grammar.

That "half-step" is not the addition of another entity, nor the introduction of another parameter set; it requires a reversal of perspective: from "why phenomena can be described by these equations," to "how these equations must necessarily emerge from one and the same underlying execution."

What GET does is precisely to take that half-step: it reorganizes repeatedly verified cornerstones—from parallel laws into a unified grammar—reducing "laws" and "phenomena" to two descriptions of the same running process. Thus the theoretical goal shifts from "improving fits" to "restoring necessity."

2.5 Meta-Philosophical Anchoring: Rule Execution Is Phenomenon

GET anchors itself with four foundational questions:

1. Which comes first, phenomena or laws?
2. Is Δt a point or a process?
3. Do laws exist objectively prior to observers?
4. Can true vacuum be strictly regarded as a system?

GET's unified answer is:

Rule execution is phenomenon; without process, nothing exists.

Accordingly, true vacuum is defined as the statistical steady-state manifestation of a baseline dynamical system L_0; "existence" is defined as the stable external manifestation of rules executing self-referentially and recursively, rather than a pile of static entities. **A "system" is a closed collection of processes with rules that keep running—true vacuum satisfies this definition, and therefore can be strictly regarded as a system.**

This stance directly yields the claim in Principle 1 (Section 2.6): existence is difference being executed.

The unified answers to the four questions are formalized in Definitions A1/A2.

2.6 From Cornerstones to Grammar: Methodological Commitments

Under the anchoring above, GET imposes four constraints on itself:

1. **Ontological minimalism:** the only dynamical ontology is the protocol-set system L.

2. **Zero ad hoc logic:** no independent patch variables are introduced during derivation.
3. **Cross-scale single grammar:** the same core structure runs through quantum, cosmos, life, and society.
4. **Public adjudication:** a four-level falsification system and Corollary 22 execute final adjudication.

Therefore, GET is not an "interpretive framework hard to refute," but a deductive system that actively accepts the constraint of failure conditions.

This methodology will be concretized for the first time in Section 2.7 as four "existence principles" — forcibly distilled from the five cornerstones and serving as the top-level outline for the entire axiomatic system.

2.7 Ultimate Test Stance

GET's scientific status is determined by its adjudication mechanism, not by rhetorical intensity. Its test logic is:

- If any final adjudication condition (any clause of Corollary 22) is hit, GET fails.
- If none is hit, the assertion continues to hold under the same adjudication protocol.

This shifts theoretical competition from "who fits better" to "who is more parsimonious, who is more transparent, and who dares to publish failure conditions."

Binary adjudication is executed via pre-registered criteria + pre-set adjudication standards to avoid post hoc interpretation. For experimental predictions (e.g., the dark-matter cross section), the adjudication standard is the community-consensus significance in particle physics (e.g., 5σ); for logical criteria (e.g., finding an existence not describable by $\Omega = \Phi(L)$), the standard is a peer-audited, unambiguous confirmation of an instance.

2.8 Closing

This chapter establishes GET's only legitimate starting point and its method of advance: take the five cornerstones as inputs, take deductive closure as the path, and take strong falsifiability as the boundary.

The next chapter will first present the four existence principles forcibly distilled from these five cornerstones (Section 3.2), then proceed into the definition system, completing the formal modeling from ontological grammar to computable structure:

$$\boldsymbol{\Omega^{(M)} = \Phi(L), \qquad \Psi_M(t) = \frac{E_{\text{solid}}(M,t)}{\tau_{\text{solid}}(L)} + \frac{E_{\text{active}}(M,t)}{\tau_{\text{active}}(L)} + \eta(L) \cdot e(t).}$$

3 Theory Construction: From the Five Cornerstones to a Hierarchical Recursive Grammar

3.1 Empirical Cornerstones and Meta-Philosophical Declaration

3.1.1 The Five Empirical Cornerstones

We take the following five universally observed empirical facts—repeatedly verified by modern physics—as the **only inputs** for constructing GET:

1. **Heisenberg uncertainty principle (energy–time constraint of quantum fluctuations)** Experiments confirm that true vacuum contains continuous random quantum fluctuations. For a single fluctuation, the energy scale ΔE and its duration scale Δt obey a rigid constraint:
$$\Delta E \cdot \Delta t \geq \frac{\hbar}{2}$$
This relation is an **insurmountable rule** that makes the phenomenon of quantum fluctuation possible.
2. **Extreme-number power-law statistical distribution (energy-level distribution of fluctuations)** In situations involving extremely large numbers of quantum fluctuations, the probability distribution of the fluctuation energy scale ΔE stably follows a power law:
$$P(\Delta E) \propto \Delta E^{-\alpha}, \quad \alpha > 1$$
This power law is a **universal statistical pattern**: high-energy is rare, low-energy dominates.
3. **Rigid boundaries of the three laws of thermodynamics**
   - **First law (energy conservation):** Energy can transform among different forms, but the total remains conserved. This constitutes the universe's absolute bookkeeping.
   - **Second law (entropy increase):** The entropy of an isolated system never decreases; processes have an irreversible direction. This constitutes the universe's directional boundary.
   - **Third law (unattainability of absolute zero):** It is impossible, through finitely many operations, to bring a system to absolute zero. This reveals a fundamental constraint of existence: a completely "frozen" form is physically unrealizable.
4. **Einstein's mass–energy morphological equivalence relation** The relation $E = mc^2$ holds precisely in all relevant processes. It reveals that **"energy" and "mass" are not independent ontologies, but two morphological manifestations of**

the same reality. The speed of light c is the **universal conversion constant** between them.

5. **Rule dependence of structured existence** All observed stable, non-trivial structures (from fundamental particles to life and society) in their existence, maintenance, and interactions necessarily depend on an **internal, hierarchical, describable rule set that does not depend on consciousness for its existence or transfer** (e.g., physical laws, genetic codes, social contracts). The execution and maintenance of these rules require continuous energy input. **Rules themselves are objective existents.**

3.1.2 Core Meta-Philosophical Declaration: The Ultimate Answer to the "Phenomenon vs. Law" Question

Modern physics presents a fundamental ontological dilemma: within a vacuum that is almost "nothing," quantum fluctuations continuously occur and are strictly constrained—between the "energy-borrowing phenomenon" and the "Heisenberg constraint law," and the "extreme-number power-law distribution," **which comes first**?

Logically there are only three possibilities:

1. **Phenomenon first:** then physical laws degrade into post hoc statistics, and the world has no underlying law; this leads to chaos and unknowability.
2. **Law first:** then laws become transcendental metaphysical entities detached from physical process itself; this departs from naturalism.
3. **Rule execution is phenomenon; the two are one:** this is the only logically self-consistent path.

We must then push further:

1. In Heisenberg's uncertainty principle, is Δt a "point" or a "process" (even if, for humans, it is already beyond instrumental measurement and beyond direct perception)? The proof is simple: if Δt were a point (i.e., took the value 0), then Heisenberg's uncertainty principle could not hold; therefore, Δt must be a process.
2. Did physical laws exist before humans were born? The answer is yes; they do not depend on the existence or transfer of consciousness.
3. The Earth–Moon system is a system—why can true vacuum not be a system? It must be able to, because true vacuum satisfies all features of a system.

Based on the above, GET strictly follows the path "rule execution is phenomenon," and via reverse engineering reveals the only possible picture:

What we observe as "vacuum," along with its fluctuations, is not "empty of everything," but a speakable objective existence—namely, the macroscopic statistical steady-state manifestation of a **memoryful, concurrently executing, difference self-referential protocol-recursive dynamical system** L_0 when it runs independently.

Accordingly, the five cornerstones receive a unified reinterpretation:

- **Cornerstone 1** is not an externally imposed physical law, but the **irreducible "minimum algorithmic accounting unit"** that the baseline system L_0 must pay when executing the bottom-level difference-generation operation. $\hbar$ is the system's fundamental accounting unit.
- **Cornerstone 2** is the **necessary emergent feature** of L_0 reaching a dynamic statistical steady state under extreme concurrency.
- **Cornerstone 3** is the manifestation of L_0's global energy bookkeeping, process irreversibility, and the unattainability of a completely frozen state.
- **Cornerstone 4** reveals that energy and mass are **morphological manifestations of the same process** within L_0, appearing at different stages of protocol structuring (different degrees of locking).
- **Cornerstone 5** directly points to the ontological status of the **protocol-set system** L as an objective existent.

Therefore, what answers "which comes first" is neither "phenomenon" nor "law," but "rule execution is phenomenon"—it must necessarily be a protocol-recursive dynamical system whose running rules simultaneously and consistently appear as "constraint logic," while its running manifestation appears as "borrowing activity." Its baseline state is denoted L_0.

—

3.2 Existence Principles: A Top-Level Outline Forcibly Distilled from the Cornerstones

After establishing the five empirical cornerstones and completing the meta-philosophical anchoring, we can now forcibly distill four basic "existence principles" from them. They are not conclusions of later derivations; they are the **philosophical foundation** and **logical mother-body** of the entire GET edifice, sitting above all definitions and axioms—subsequent definitions, axioms, and corollaries are expansions and formalizations of these four principles:

Principle 3.1 (Origin: Difference is existence) □.

The only starting point and the only entry point of existence is the creation of difference. Difference is existence itself.

Each execution of the difference function D *necessarily produces an instantaneous difference vector* $e(t) \neq \mathbf{0}$*. Without* D *there is no* $e(t)$*; without* $e(t)$ *there is no net* *borrowing balance $E(t)$; without $E(t)$ there is no distinguishable, speakable "existence." Existence is not a carrier of difference; the execution process of difference is existence. In any running moment with no $e(t) \neq \mathbf{0}$, there is no speakable existence.*

Rigid constraint (Cornerstone 1 Heisenberg uncertainty + Definition A1): *The minimum cost of generating difference is locked by the baseline system L_0 as $\hbar/2$; i.e., any non-trivial difference must satisfy:*

$$\boldsymbol{E_{peak} \cdot \tau \geq \frac{\hbar}{2}}$$

where E_{peak} is the maximum of the net borrowing balance over the lifecycle, and τ is the total number of steps in the lifecycle.

Difference is not an optional "phenomenon," but the ontological starting point and the only entry point of existence. ***This is the ultimate realization of the "ontological minimalism" commitment of Chapter 2: existence is difference being executed.***

Principle 3.2 (Constraint: Difference ultimately returns to zero) □.
Any difference ≥ 0 (including quantum fluctuations) must net-return to zero at the end of its lifecycle.

For an individual with recursion depth ≥ 0, its net borrowing balance over its lifecycle τ (including both the living and the remnant phases) must satisfy:

$$E(\tau) = \int_0^{\tau} e(t)\, dt = 0$$

(scalar algebraic sum equals zero)

Rigid constraint (Cornerstone 3 First law of thermodynamics + scalar accumulation property of D): *Any difference that "only borrows and never repays" cannot persist physically.*

To break the return-to-zero, the only exemption path is: within finite τ, use the four-phase cycle to convert part of the difference into an irreversible structured asset E_{solid}, thereby enabling the recursion depth d to jump from 0 to ≥ 1.

This directly excludes all metaphysical assumptions such as eternal entities, immortal souls, infinitely stable vacuum, etc., that violate energy conservation—and provides the only exemption path: protocol solidification.

Principle 3.3 (Path: Difference must seek survival) □.
The only path to break the $d = 0$ return-to-zero is to evolve—via geometric selection—toward increasing complexity and persistence.

By Principle 2, if it does not persist, difference returns to zero and returns to "nothing." To persist, within finite lifecycle τ it must convert part of the difference into an irreversible structured asset E_{solid}, which necessarily triggers an increase in recursion depth.

Two fundamental strategies ***plus*** *one inevitable result:*

1. ***Capture energy (Strategy 1):*** *increase energy capture efficiency η, expand the positive magnitude and duration of instantaneous net flux $e(t)$, and increase the reserve E_{active}. This is the immediate survival term in the survival inequality, ensuring $\eta \cdot e(t) > 0$ holds.*
2. ***Lock energy (Strategy 2):*** *convert active-state energy E_{active} into solidified-state energy E_{solid} (solidify, structure, increase protocol hierarchy), thereby increasing total structured asset E_{locked}. This increases the contribution of E_{solid}/τ_{solid} in the existence-degree formula.*
3. ***Expansion (inevitable result):*** *successful execution of the first two strategies necessarily leads to increased spacetime scale, population diffusion, and higher complexity hierarchy—this is the natural manifestation of successful persistence, not an independently optional "third path."*

Rigid constraint (Existence-degree Axiom III + thermodynamic second/third laws): *The net borrowing balance is*

$$\boldsymbol{E(t) = \int_0^t e(\tau)d\tau = E_{solid}(t) + E_{active}(t),}$$

where E_{solid} is the form's solidified asset, and E_{active} is the active asset.

To maintain

$$\boldsymbol{\frac{E_{active}(M,t)}{\tau_{active}(L)} + \eta(L) \cdot e(t) \geq 0,}$$

one must satisfy both strategies simultaneously, and universally satisfy $\tau_{solid} \gg \tau_{active}$. Protocol solidification necessarily accompanies entropy increase ($\eta < 1$, with radiative dissipation), and a fully frozen state ($E_{active} = 0$ and $\eta \cdot e(t) = 0$) is unattainable (thermodynamics' third law).

These two strategies are not "optional"; they are mathematically forced by the survival inequality.

Successful execution of these strategies—especially the completion of the "four-phase cycle" corresponding to "locking energy"—necessarily leads to increased spacetime scale, population diffusion, and higher complexity hierarchy. This is the natural manifestation of successful persistence, not an independently optional third path.

Principle 3.4 (Fate: Survival must pay a price) $\square$.
Any expansion necessarily dilutes available active-state energy—this is the fundamental constraint and endgame fate of all complex systems.

Expansion (spatial expansion, population diffusion, higher complexity hierarchy) necessarily causes dilution of usable energy resources:

- *Effective invested energy decays with distance squared (Axiom VII):*

$$\boldsymbol{E}_{invest}^{eff} \approx \frac{\boldsymbol{E}_{invest}}{\mathbf{1} + (\boldsymbol{r}/\boldsymbol{r}_{\mathbf{0}})^{\mathbf{2}}}$$

- *Connection capital necessarily depreciates, jointly driven by insufficient energy and spatial separation*
- *The universe's structured usable energy pool $\mathcal{E}_{struct}(T)$ monotonically decreases over time (Axiom XI.4)*
- *Cosmological expansion dilutes energy density: $\rho \propto a^{-3}$ (matter) or a^{-4} (radiation)*

Rigid constraint: *Expansion is the only choice forced upon existence by Principle 3, but expansion itself weakens the choice itself. This forms a self-accelerating vicious loop:*

$$\begin{aligned}&\textit{must expand} \to \textit{expansion dilutes } \boldsymbol{E}_{active}\\&\to \textit{harder to maintain } \frac{\boldsymbol{E}_{active}(\boldsymbol{M},\boldsymbol{t})}{\boldsymbol{\tau}_{active}(\boldsymbol{L})} + \boldsymbol{\eta}(\boldsymbol{L}) \cdot \boldsymbol{e}(\boldsymbol{t}) \geq \mathbf{0}\\&\to \textit{must expand more violently} \to \cdots\end{aligned}$$

This dilution directly determines the theoretical upper bound D_{max} of recursion depth d (Corollary 12). When $\mathcal{E}_{struct}(T) \to 0$, any further structuring attempt fails due to energy insufficiency.

Final outcome: *recursion depth reaches its upper bound, $\mathcal{E}_{struct} \to 0$, and the system enters an approximate heat-death state or a black-hole-dominated dead-asset isolation regime (where E_{solid} is permanently sealed behind the event horizon and slowly dissipates via Hawking radiation). Yet even in this terminal state, the third law of thermodynamics remains valid—absolute zero is unattainable, and L_0 still maintains a minimum level of difference activity.*

The four principles above are an existence outline forcibly distilled from the five cornerstones. Together they constitute GET's ultimate answer to "how existence is born, how it evolves, and how it ends":

1. From nothing to something: difference must be created (Principle 1)
2. From something to nothing: difference must return to zero (Principle 2)
3. Away from zero: persistence is required—via the four-phase cycle to increase complexity (Principle 3)
4. The price of persistence: expansion inevitably dilutes → persistence becomes harder → expansion must become more violent → ... (Principle 4)

This is not pessimism; it is the only existence cycle forcibly deduced from the most basic facts: $\hbar/2$, the three laws of thermodynamics, and the vector/scalar accumulation nature of D:

As long as L_0 is still running, new differences may again emerge from vacuum.

The entire history of the universe, life, and civilization can be summarized in one sentence:

Existence is an ultimate tug-of-war—forced to start by Heisenberg uncertainty, forced to end by thermodynamics, and never stopping at the bottom layer: a perpetual cycle of creating something from nothing, rising costs, and a fate of returning to nothing.

Yet L_0 never stops. Every return to zero is a possible starting point for the next creation of difference. This is the whole grammar of existence: from $\hbar$, to entropy, cycling back and forth without end.

However, an outline must be concretized into a computable rule system. The following axioms (Axiom Ω through Axiom XIII) are precisely the strict formalization of these four principles:

- Axiom Ω anchors the baseline system L_0 and its running;
- Axiom I writes "existence arises from difference" as protocol-recursive monism;
- Axiom II algorithmizes "difference must seek persistence" as the four-phase cycle;
- Axiom III quantifies existence as the existence-degree formula and survival criterion;
- Axiom VII and Axiom XI correspond respectively to "expansion dilutes resources" as spatial and energy constraints;
- Axiom VIII and Definition D5 formalize the black-hole isolation mechanism in the "terminal fate" as toxic-protocol management.

Every sentence of the four principles can be traced back to corresponding clauses among the five cornerstones; no new ad hoc assumptions are introduced; every conclusion takes a sharp form that can be directly falsified by counterexamples—this is the concrete embodiment of Chapter 2's "strong falsifiability" principle and the logical prerequisite for Chapter 4's "four-level falsification system."

At this point, we transition smoothly from the philosophical outline into the mathematical-physical system.

3.3 Basic Definition System

3.3.1 Module A: Ontology of Existence

Definition A1 (A meta-description of existence—ontology, phenomenon, and endogenous dimensions).

Meta-philosophical stance: rule execution is phenomenon. Phenomenon (form M) is the stable external manifestation of the recursive running of a protocol set L.

1. The only dynamical ontology: the protocol-recursive system L An objective, memoryful, difference self-referential, protocol-recursive dynamical system. Its baseline state when running independently is denoted L_0.

2. Speakable existence: form M **Form** M **is the set of observable phenomena stably presented during the recursive execution of the protocol set** L**.**

$$\boldsymbol{M} := \text{Phenomenon of } \boldsymbol{L} \text{ running}$$

M is the **external manifestation type** of L's running.

3. Endogenous state variables To describe the running of L (i.e., the dynamics of M), introduce variables that are fully endogenous to this process. **All variables are generated directly from each execution of the difference function** D**, without any external inputs or a priori assumptions.**

a. Difference function D **(core operation; the generative engine of all variables, constrained by the protocol set** L**)**

$$D : \Sigma \to \Sigma$$

is the difference-generation function. Each execution of D necessarily produces a state transition and a difference value:

$$\boldsymbol{\sigma_{t+1} = D(\sigma_t)}, \quad \text{difference value} = \boldsymbol{D(\sigma_t) - \sigma_t}$$

b. Time t **(the sequential unfolding of** D**)** t is the sequential index of the iteration of D, $t \in \mathbb{N}_0$, starting from the initial state σ_0. Each execution of D advances t by 1 and records the order in which differences unfold. A complete lifecycle of a fluctuation individual corresponds to a finite number of steps $\tau_i = m_i \cdot t_P$ (where t_P is the minimal iteration unit).

c. Instantaneous difference $e(t)$ **(scalar)** Define the **instantaneous difference** at time t as the state difference at the current step:

$$e(t) = D(\sigma_t) - \sigma_t$$

$e(t)$ is a scalar and carries a sign (positive or negative). A positive value indicates borrowing energy; a negative value indicates repaying energy. It is an endogenous variable generated directly by the iteration of the difference function D.

Physical correspondence: $e(t)$ corresponds directly to the **instantaneous energy-borrowing flow rate** in traditional physics. Positive means borrowing (energy inflow); negative means repayment (energy outflow).

d. Net borrowing balance $E(t)$ (energy; scalar) Define the **net borrowing balance** up to time t as the algebraic sum of all instantaneous differences from birth to the current step:

$$E(t) = \sum_{k=1}^{t} e(k) = \sum_{k=1}^{t} (D(\sigma_k) - \sigma_k)$$

This is the endogenous definition of energy in the GET framework—energy is the cumulative sum of instantaneous differences, representing an individual's net energy borrowing balance from birth to time t.

Lifecycle evolution:

- At birth: $E(0) = 0$
- Rising stage: in the first half of the lifecycle, borrowing dominates repayment, $e(t) > 0$ dominates, and $E(t)$ gradually increases
- Peak: reaches the maximum value E_{peak}
- Falling stage: in the second half of the lifecycle, repayment dominates borrowing, $e(t) < 0$ dominates, and $E(t)$ gradually decreases
- At annihilation: $E(\tau) = 0$ (all borrowed energy is fully repaid)

This is precisely the energy constraint "what is borrowed over a lifetime equals what is repaid over a lifetime"—the net borrowing balance must start from 0, rise to a peak, and fall back to 0. Over the entire lifecycle, $E(t) \geq 0$ always holds.

Peak borrowed energy:

$$\boldsymbol{E_{\text{peak}} = \max_{t \in [0,\tau]} E(t)}$$

Endogenous expression of the Heisenberg constraint:

$$\boldsymbol{E_{\text{peak}} \cdot \tau \geq \frac{\hbar}{2}}$$

This relation is the **real process constraint** of the D operation itself: generating and accumulating non-trivial differences necessarily requires a finite process ($\tau \geq$ the minimal iteration unit), and the product of the peak net borrowing balance and the lifecycle must exceed a minimal constant. $\hbar/2$ is the minimal **energy–time composite cost** that L_0 must pay to generate and maintain a distinguishable "existence individual."

Derived solidified-state and active-state energy:

The net borrowing balance $E(t)$ manifests within form M as two qualitatively different existence morphologies:

- **Solidified-state energy** $E_{\text{solid}}(t)$: the portion of structured assets already locked by protocol solidification. This energy **cannot be directly used for immediate**

survival maintenance; its release or dissipation has a characteristic solidified-state time constant $\tau_{\text{solid}}(L)$. **Cross-domain unification note:** the physical meaning of $\tau_{\text{solid}}(L)$ varies by domain—e.g., in physics it corresponds to the "dissipation/decay time constant" of structured assets; in biology to the "metabolic conversion cycle" of fat and other structural biomass; in economics to the "depreciation period" of fixed assets; in cognitive science to the "forgetting time constant" of long-term memory. Although mechanisms differ, its mathematical role in the existence-degree formula is identical: it measures the inverse of the intrinsic rate at which solidified energy converts into immediately usable resources.

- **Active-state energy** $E_{\text{active}}(t)$: the portion of the total structured asset that is freely deployable to support the real-time operation of the form. This energy **can be directly consumed to maintain immediate survival**, and has an active-state characteristic time constant $\tau_{\text{active}}(L)$. In biology it corresponds to immediately usable energy currencies such as ATP, blood glucose, and glycogen.

Their relation:

$$\boldsymbol{E(t) = E_{\text{solid}}(t) + E_{\text{active}}(t), \quad E_{\text{solid}}(t) \geq 0, \quad E_{\text{active}}(t) \geq 0}$$

The **total structured asset** $E_{\text{locked}}(t) = E_{\text{solid}}(t) + E_{\text{active}}(t)$ corresponds to mass m in traditional physics, and under the low-velocity approximation satisfies $E_{\text{locked}} \approx mc^2$. Mass m is not an independent entity, but the morphological manifestation of the total energy E_{locked} after protocol solidification.

For quantum fluctuation individuals ($d = 0$) in the baseline dynamical system L_0, solidified-state energy is always zero: $E_{\text{solid}}(t) \equiv 0$. In that case, $E(t) = E_{\text{active}}(t)$ is directly the net borrowing balance.

Instantaneous net borrowing flow: simply $e(t)$ itself.

e. Spatial distance s (geometric separation accumulated by D) s is the sum of weights along the shortest path from σ_0 to σ_t in the state space Σ, measuring the degree of protocol-state difference.

A complete lifecycle of a fluctuation individual forms a path $\gamma_i : [0,1] \to \Sigma$ in Σ, with start at σ_0.

Path length (total distance traveled):

$$L(\gamma_i) = \int_0^1 \|\gamma_i'(t)\| dt \approx \sum_{k=1}^{\tau} |e(k)|$$

This is the total "distance traveled" in state space, always non-negative.

Net displacement:

$$\text{net displacement} = \boldsymbol{\left\| \sum_{k=1}^{\tau} e(k) \right\| = \|E(\tau)\| = 0}$$

For a quantum fluctuation individual, at annihilation $E(\tau) = 0$, so the net displacement is zero. But the path length $L(\gamma_i) = \sum |e(k)|$ can be far greater than zero. This is the essence of "fluctuation": **it makes a big loop and ultimately returns to the origin.**

Approximate radius:

$$r \approx \frac{1}{2} \max_t s(\sigma_0, \sigma_t)$$

The ratio of circumference to diameter, in statistical average, must converge to the constant π:

$$\langle L(\gamma_i)/(2r) \rangle \approx \pi$$

This reveals an intrinsic geometric property of the lowest-level state space—under large statistical averaging, the "unfolding degree" of lifecycle paths converges to a universal geometric constant.

Winding number:

$$\boldsymbol{n_w(\gamma_i) = \frac{1}{2\pi i} \oint_{\gamma_i} \frac{dz}{z}} \quad \text{(under projection onto a 2D submanifold of } \boldsymbol{\Sigma}\text{)}$$

The simplest non-zero integer value is ± 1, corresponding to two directional separations (left-handed/right-handed), becoming the chirality marker of the protocol fingerprint $C_{\mathcal{S}}$. **Left-handed and right-handed are perfectly symmetric in the baseline state L_0 (equal probability), with no underlying bias.** This symmetry is broken in the first structured phase transition (inflation trigger), mapping to gauge-field spin and fermion chirality.

f. The continuous compounding limit e (the necessary limit of composite iteration of D) In the extreme-iteration limit ($\tau_i \to \infty$, i.e., long-lifetime fluctuations selected from the power-law tail), consider a local interval in which instantaneous differences are predominantly of the same sign; the growth of the net borrowing balance $E(t)$ can be approximated by continuous compounding:

$$E(t) \approx E_0 \times e^{rt}$$

where r is the instantaneous growth rate, determined by the ratio of $e(t)$ to the current $E(t)$.

Emergence of the mathematical constant e:

$$e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n$$

is the **necessary emergent constant** of discrete iteration D in the continuous limit. It indicates that when the number of accumulation steps tends to infinity and the increment per step tends to infinitesimal, the growth mode of $E(t)$ transitions smoothly from discrete summation to continuous exponential growth—and the "natural base" of that transition is precisely e.

This relation reveals: **the natural constant e is not an invention by mathematicians, but an intrinsic property of any recursive difference-generation system in the macroscopic limit.**

g. Recursion depth d (self-nesting of D)

$$\Sigma^{(d+1)} = R(\Sigma^{(d)})$$

where R is the recursive nesting function. d records the number of nesting layers in which D uses its own output to redefine its input and reorganize difference—i.e., the system's capability to use already-generated difference patterns as the basis of new rules to generate higher-order differences.

Recursion depth is the quantitative scale of complexity:

- $d = 0$: D acts directly on the raw state space, generating base differences (quantum fluctuation level)
- $d = 1$: use difference patterns produced at $d = 0$ to solidify new protocols, forming a new state space (fundamental particle level)
- $d \gg 1$: after many recursive nestings, produce complex systems with high-order self-referential ability (life, consciousness, civilization)

Each increase in recursion depth requires satisfying the **geometric selection condition** (Definition B2)—within a qualified window, successfully execute the four-phase cycle and convert part of the active-state energy E_{active} into new solidified-state energy E_{solid} (solidified into a new protocol).

h. Maximum propagation speed c (propagation upper bound of D iteration)

$$\boldsymbol{c = \sup_{\sigma_1,\sigma_2 \in \Sigma} \frac{s(\sigma_1,\sigma_2)}{\Delta t}} \quad \text{subject to} \quad \boldsymbol{\sigma_2 = D^{(n)}(\sigma_1),\ n \geq 1}$$

where $D^{(n)}$ denotes n iterations of the difference function. c is the inherent maximum rate at which protocol-state differences propagate in Σ.

Dynamical origin of c: to propagate from σ_1 to σ_2, one must realize it step-by-step through multiple iterations of D. Each iteration requires at least one minimal time unit t_P, and each step can produce only a finite separation $s(\sigma_k, \sigma_{k+1}) \leq |e(k)|$. Therefore c is jointly determined by the **maximum absolute instantaneous difference per step** and the **minimal iteration time**:

$$c = \frac{\max_k |e(k)|}{t_P}$$

Endogeneity of c: c is not an externally imposed cosmic constant; it is an **intrinsic upper bound** produced by the dynamics of the L system itself—how fast D can continuously produce differences is how fast differences can propagate. This provides a mechanistic explanation for Axiom XIII.

Invariance: for all observers described by compatible protocol sets, the measured maximum speed c is the same, because all observers share the same underlying running logic of D.

i. Existence degree Ψ (a universal criterion for all stable existences)

$$\boldsymbol{\Psi_M(t) = \frac{E_{\text{solid}}(M,t)}{\tau_{\text{solid}}(L)} + \frac{E_{\text{active}}(M,t)}{\tau_{\text{active}}(L)} + \eta(L)\cdot e(t)}$$

Where:

- $E_{\text{solid}}(M,t) \geq 0$: the **solidified-state energy** locked by form M at time t—structured assets locked by protocol solidification, **not directly usable for immediate survival maintenance**. In biology it corresponds to structural biomass such as bones, muscles, and fat that cannot be directly used as ATP.
- $\tau_{\text{solid}}(L) > 0$: the **solidified-state characteristic time constant** determined by protocol set L; its reciprocal represents the "depreciation rate" or conversion rate of structured assets.
- $E_{\text{active}}(M,t) \geq 0$: the **active-state energy** freely deployable by form M at time t—the portion of total structured assets that can be directly used to support real-time operation. In biology it corresponds to ATP, blood glucose, glycogen, and other immediately usable energy currencies.
- $\tau_{\text{active}}(L) > 0$: the **active-state characteristic time constant** determined by protocol set L; its reciprocal represents the "consumption rate" at which active energy supports operation.
- $e(t) = D(\sigma_t) - \sigma_t$: the instantaneous difference at time t—**a signed scalar**; positive indicates net borrowing (inflow), negative indicates net repayment (outflow).
- $\eta(L) \in [0,1]$: the **energy capture–conversion efficiency** of protocol set L, converting the instantaneous difference $e(t)$ into effective energy flow that can directly maintain or strengthen system operation.

Existence degree is a comprehensive measure of whether form M can continue to exist. It measures: the structural return provided by current solidified assets (first term) plus the operational capacity provided by active assets (second term) plus the immediate gain from real-time energy exchange (third term).

The **total structured asset** $E_{\text{locked}}(t) = E_{\text{solid}}(t) + E_{\text{active}}(t)$ constitutes form M's total energy reserve; under the low-velocity approximation it corresponds to mass m, satisfying $E_{\text{locked}} \approx mc^2$.

Survival criterion: a form M can continue to exist if and only if:

$$\boldsymbol{\frac{E_{\text{active}}(M,t)}{\tau_{\text{active}}(L)} + \eta(L) \cdot e(t) \geq 0}$$

Three-state distinction:

- > 0**:** healthy operation; M is normally maintained
- $= 0$**:** critical balance; M is fragile but not yet disintegrated
- < 0**:** death state; M disintegrates immediately

Key constraints:

- The immediate survival of form M is entirely determined by the latter two terms.
- When $\eta \cdot e(t) < 0$, form M begins consuming its E_{active} reserve to compensate for the negative net flow. As long as E_{active} is sufficient, $\frac{E_{\text{active}}(M,t)}{\tau_{\text{active}}(L)} + \eta(L) \cdot e(t) \geq 0$ can still be maintained, and M continues to survive.
- When $\eta \cdot e(t) < 0$ persists, E_{active} gradually decreases. Eventually, when E_{active} falls so that $\frac{E_{\text{active}}(M,t)}{\tau_{\text{active}}(L)} + \eta(L) \cdot e(t) < 0$, form M can no longer maintain immediate operation and dies immediately.
- Because the conversion rate of solidified-state energy E_{solid} into active-state energy is too slow, it cannot provide effective help when form M faces an immediate survival crisis.
- After M dies, its E_{solid} may be utilized by other forms M, but the dead form M itself cannot use that energy to save itself.

4. Complete state function A complete description of a concrete existence (a form instance M_i) is uniquely determined by its protocol set L via the generating function Φ:

$$\Omega^{(M)} = \Phi(L)$$

where:

- $\Omega^{(M)}$ is the **complete state function** of form M (a function of endogenous time t), i.e., $\Omega^{(M)} : \mathbb{N}_0 \to \mathcal{S}$, with $\Omega^{(M)}(t) \in \mathcal{S}$ the instantaneous complete state at time t;
- L is the only dynamical ontology (protocol set);
- Φ is the **generating function**—given a protocol set L, $\Phi(L)$ outputs the full evolution trajectory of form M, $\{\Omega^{(M)}(t)\}_{t \in \mathbb{N}_0}$.

Expanded form (explicitly writing endogenous variables):

$$\boldsymbol{\Omega_{\mathcal{S}}^{(M)}(t) = \Phi_{\mathcal{S}}\big(L, E_{\text{solid}}(t), E_{\text{active}}(t), e(t), t\big)}$$

where $E_{\text{solid}}(t)$ and $E_{\text{active}}(t)$ are the two components of the net borrowing balance $E(t)$ up to time t, $e(t)$ is the instantaneous difference, and t is the endogenous time variable generated by the iteration of the difference function D. The subscript $\mathcal{S}$ indicates that the specific form of the generating function is determined by the species' core protocol fingerprint $C_{\mathcal{S}}$.

Essential unity: regardless of recursion depth d and regardless of species $\mathcal{S}$, all speakable existences obey the same basic form:

$$\Omega^{(M)} = \Phi(L)$$

This is GET's **core equation** and the mathematical expression of the "grammar of existence."

Correspondence to the five cornerstones:

- **Cornerstone 1 (Heisenberg uncertainty):** $E_{\text{peak}} \cdot \tau \geq \hbar/2$
- **Cornerstone 2 (power-law statistics):** E_{peak} follows a power-law distribution $P(E_{\text{peak}}) \propto E_{\text{peak}}^{-\alpha}$
- **Cornerstone 3 (three laws of thermodynamics):** $E(\tau) = 0$ over the lifecycle (first law); protocol solidification is irreversible, $\eta < 1$ and radiative dissipation is necessary (second law); a fully frozen state $E_{\text{active}} = 0$ and $\eta \cdot e(t) = 0$ is unattainable (third law)
- **Cornerstone 4 (mass–energy equivalence):** the total structured asset $E_{\text{locked}} = E_{\text{solid}} + E_{\text{active}}$ corresponds to mass m, satisfying $E_{\text{locked}} \approx mc^2$ under the low-velocity approximation. Here c is both the **universal conversion constant** between total energy and mass, and the inherent maximum propagation speed of protocol-state differences.
- **Cornerstone 5 (rule dependence):** L itself is the objective rule set

Definition A2 (Baseline dynamical system L_0 and true vacuum).
Define the fundamental protocol-recursive difference-unfolding dynamical system as L_0. If and only if L_0 runs independently, the macroscopic statistical steady state it presents is what we call the **true vacuum state** Θ_{vac}.

L_0 is a self-sustaining, memoryful, concurrently executing protocol-recursive dynamical system. Its complete state is given by Definition A1:

$$\boldsymbol{\Omega}^{(\text{vacuum fluctuation})} = \boldsymbol{\Phi}(\boldsymbol{L_0})$$

The core constitution of L_0 includes:

1. **Self-referential recursive engine:** a difference function $D : \Sigma \to \Sigma$ satisfying $\sigma_{t+1} = D(\sigma_t)$. This is the bottom-level recursion mechanism, using its own output as the next input to make "process" possible. Each execution of D generates instantaneous borrowing flow $\phi_{\text{in}}(t)$ and repayment flow $\phi_{\text{out}}(t)$.

2. **Concurrent individual management:** for any T, L_0 maintains an extremely large number ($N(T) \gg 1$) of concurrent fluctuation individuals, each with its own lifecycle and a resource (energy) usage pattern obeying a power-law distribution. Its net borrowing energy curve $E(t)$ has a peaked shape (Definition B1).
3. **Endogeneity of rigid constraints:** all constraints in Definition A1 corresponding to the five cornerstones—Heisenberg uncertainty, the three laws of thermodynamics, mass–energy equivalence, and the causal propagation upper bound—are endogenous properties of L_0's running, rather than externally imposed laws. Together they constitute L_0's **irreducible bookkeeping rules and boundary conditions** as a dynamical system.
4. **The only route to cosmic birth:** the structured origin of the universe occurs if and only if some individuals' life curves $E(t)$ in L_0 intersect the energy threshold $\epsilon_{\text{th}}^{(1)}$, and the time span between intersection points reaches the minimum window $N_{\text{min}}^{(1)}$. **This is the necessity of geometric selection.**

Definition A3 (Recursive protocol space and protocol sets). • **Recursive protocol space:**

$$\mathscr{L} := \bigcup_{d=0}^{D_{\max}} \mathscr{L}_d$$

where $\mathscr{L}_d$ is the family of **solidified core protocol sets** at recursion depth d.

- **Starting point:**

$$\mathscr{L}_0 = \{\{L_0\}\}$$

containing only the singleton set of the baseline protocol L_0.

- **Recursion-depth function:** For a form M, let its protocol set be L. Define the recursion depth of L as

$$d(L) = \max\{\, k \mid \exists\, L_{k,j} \in L \,\}.$$

- **Protocol-set compatibility:** A necessary condition for a protocol set L to stably exist is that all protocols within it are mutually compatible:

$$\boldsymbol{\kappa(L) = \min_{L_i, L_j \in L}} \text{compatibility}\boldsymbol{(L_i, L_j)} \geq \boldsymbol{\kappa}_{\text{crit}}.$$

—

Definition A4 (A hierarchical genealogy of stable form categories).

In GET, a stable category of forms defined by a shared protocol fingerprint C can be divided into different hierarchical classes according to **recursion depth** d and **the transmission mechanism of solidified protocols**. These classes correspond to different conventional names in scientific discourse, but their essence is always "a set of forms defined by a shared core protocol fingerprint."

Table 1: A cross-depth unified genealogy of stable form categories (material, biological, and consciousness dimensions)

Recursion depth d	**Transmission of solidified protocols**	**Material lineage**	**Biological lineage**	**Consciousness / intelligence**
$d = 0$	Quantum-fluctuation dynamics	Quantum fluctuations	—	—
$d = 1$	Cosmological solidification + particle physics	Fundamental particles / black holes	—	—
$d = 2$	Strong interaction	Baryons / neutron stars	—	—
$d = 3$	Electromagnetic interaction	Atoms	—	—
$d = 4$	Chemical bonds	Molecules	Organic molecules	—
$d = 5$	Gravitational collapse + nuclear fusion	Stars	Proto-life	—
$d = 6$	Biological heredity (DNA)	—	Multicellular organisms	—
$d = 7$	Nervous systems	—	Sentient organisms	Infants
$d = 8$	Learning and skills	—	—	Basic cognition
$d = 9$	Professional skills	—	—	Intermediate cognition
$d = 10$	Social institutions	—	—	Advanced professional capacity
$d = 11$	Civilizational paradigms	—	—	Top-level paradigms
$d \geq 12$	Inter-civilizational integration	—	—	Meta-paradigms / civilizational wisdom (near the theoretical limit $D_{\max}$)

Key distinction: within the same recursion depth d, there can exist **multiple distinct protocol fingerprints** $C_1, C_2, \ldots$, corresponding to different **stable form categories** at the same level. For example:

- **At** $d = 0$**:** quantum fluctuations and primordial virtual pairs ($C_{\text{virtual-pair}}$) belong to vacuum fluctuations.
- **At** $d = 1$**:** photons (C_{photon}), electrons (C_{electron}), quarks (C_{quark}), black-hole event horizons ($C_{\text{black-hole}}$) are different "species" of fundamental particles / compact-object forms.
- **At** $d = 2$**:** protons (C_{proton}), neutrons (C_{neutron}), neutron stars ($C_{\text{neutron-star}}$) are

different baryonic species.

- **At $d = 3$:** hydrogen atoms (C_{hydrogen}), iron atoms (C_{iron}) are different atomic species.
- **At $d = 4$:** water (C_{H2O}), methane (C_{CH4}), organic molecules (C_{organic}) are different molecular species.
- **At $d = 5$:** main-sequence stars ($C_{\text{main-sequence}}$), red giants ($C_{\text{red-giant}}$), proto-life ($C_{\text{protolife}}$) are different "star / proto-life" species.
- **At $d = 6$:** plants (C_{plant}), animals (C_{animal}), fungi (C_{fungi}) are different multicellular biological species.
- **At $d = 7$:** fish (C_{fish}), reptiles (C_{reptile}), infants (C_{infant}) are different sentient-organism species.
- **At $d = 8$:** language acquisition (C_{language}) and tool use ($C_{\text{tool-use}}$) are different basic cognition protocol sets.
- **At $d = 9$:** programming ($C_{\text{programming}}$), instrument performance ($C_{\text{instrument}}$), surgery (C_{surgery}) are different intermediate cognition / professional-skill species.
- **At $d = 10$:** AI engineers (C_{AI}), civil engineers (C_{civil}), teachers (C_{teacher}) are different advanced professional / occupational species.
- **At $d = 11$:** the scientific method ($C_{\text{scientific}}$), Cartesian rationalism ($C_{\text{Cartesian}}$), Taoist naturalism (C_{Taoist}) are different top-level paradigm / cultural-tradition species.
- **At $d \geq 12$:** cross-cultural dialogue ($C_{\text{intercultural}}$) and global ethics ($C_{\text{global-ethics}}$) are different meta-paradigm / civilizational-wisdom species.

Deep unity: regardless of depth, and regardless of whether the inheritance mechanism is quantum-number conservation, strong/electromagnetic interaction, chemical bonding, DNA replication, neural plasticity, social learning, or civilizational transmission, **all these categories share the same essence**: a collection of forms defined by a stable core protocol fingerprint C, first established by the earliest successful solidification events and inherited by subsequent individuals through a specific transmission mechanism.

Evolutionary continuity: from $d = 0$ quantum fluctuations to $d \geq 12$ civilizational wisdom, the entire genealogy exhibits a continuous evolutionary picture in which existence increases recursion depth via **geometric selection** and the **four-phase cycle**. Each depth's stable categories provide the basis for the emergence of the next depth. For any form M, its recursion depth is determined by its instantiated protocol set L: $d(M) = d(L)$.

—

Definition A5 (Species as a biological-level stable form category).

Preliminary note: this definition is a specialization of Definition A4's "stable form category" concept to biological recursion depths (typically $d = 5 \sim 7$). At other recursion depths, stable form categories defined by shared protocol fingerprints can be called by different conventional names—e.g., "particle species" ($d = 1$), "chemical-element species"

($d = 3$–4), "occupational-skill species" ($d = 8$–10), "cognitive-paradigm species" ($d = 11$), etc.—but all follow the same GET ontological framework: **a set of forms M identified by a core protocol fingerprint C.**

At biological recursion depths (typically $d = 5 \sim 7$), a **biological species** $\mathcal{S}^{(d)}$ is a set of stable forms M defined by a **heritable core genetic protocol set** $C_{\mathcal{S}} \subset \mathscr{L}_d$, whose members can produce fertile offspring through mating.

Origin: a species $\mathcal{S}^{(d)}$ originates from the **first individuals** at that depth who successfully complete the four-phase cycle and thereby solidify a protocol subset $S \in \mathscr{L}_d$. These individuals' S share a common feature set $C_{\mathcal{S}}$, which becomes the species' **protocol fingerprint**, transmitted across generations by heredity.

Let $\mathrm{Core}(L) \subseteq L$ denote the solidified core protocol subset within a protocol set L that determines stable existence and basic interaction properties. Then:

$$\boldsymbol{\mathcal{S}^{(d)} := \{\, M \mid \exists L \text{ such that } M = \mathrm{Phenomenon}(L),\ d(L) = d, \text{ and } \mathrm{Core}(L) = C_{\mathcal{S}} \,\}}$$

Extended application: at other recursion depths, stable form categories defined by shared protocol fingerprints may use domain-specific names (e.g., particle species, chemical elements, occupational skills, cognitive paradigms), but still obey the same GET framework. To maintain terminological unity in cross-domain discussion, one may refer to them generically as "stable form categories" or "protocol-fingerprint categories," while using conventional domain terms within specific domains.

—

3.3.2 Module B: Cosmic Genesis and Initial Conditions

Definition B1 (Quantum fluctuations as the running manifestation of L_0). Each quantum-fluctuation individual i is a temporary existence generated and maintained by L_0. Its complete state is given by Definition A1: $\Omega_i^{(0)} = \Phi(L_0)$. For a $d = 0$ individual, its solidified-state energy is identically zero: $E_{\text{solid}}(t) \equiv 0$. Hence the net borrowing balance is entirely active-state: $E(t) = E_{\text{active}}(t)$. Its lifecycle dynamics are generated by each execution of the difference function D. All constraints below are endogenous manifestations of L_0's rules:

1. **Lifecycle:** born at background time T^i_{birth} and annihilates at T^i_{end}. Its lifetime is

$$\boldsymbol{\tau_i = T^i_{\text{end}} - T^i_{\text{birth}} = m_i \cdot t_P}$$

   where $m_i \in \mathbb{N}$ is the total number of internal iteration steps.

2. **Internal time and difference generation:** at internal steps $t = 1, 2, \ldots, m_i$ (corresponding to background time $T = T^i_{\text{birth}} + (t-1)t_P$), executing D generates the

instantaneous difference:

$$e_i(t) = D(\sigma_t) - \sigma_t$$

a signed scalar: positive means borrowing energy from vacuum; negative means repaying energy to vacuum.

3. **Evolution of net borrowing balance:** the cumulative net borrowing balance (i.e., energy) from birth to time t is:

$$E_i(t) = \sum_{k=1}^{t} e_i(k)$$

Over the lifecycle, $E_i(t)$ follows a strictly **peaked curve**:

- $E_i(0) = 0$, $E_i(\tau_i) = 0$.
- In the rising phase, $e_i(t) > 0$ dominates and $E_i(t)$ increases monotonically.
- After the peak, $e_i(t) < 0$ dominates and $E_i(t)$ decreases monotonically back to zero.

Throughout the lifecycle, $E_i(t) \geq 0$ always holds.

4. **Peak borrowed energy:**

$$\boldsymbol{E^i_{\text{peak}} = \max_{t\in[1,m_i]} E_i(t)}$$

5. **Lifecycle energy constraint:** the individual strictly obeys energy conservation over its entire lifecycle (thermodynamics' first law):

$$E_i(\tau_i) = \sum_{k=1}^{m_i} e_i(k) = 0$$

6. **Heisenberg constraint:** the peak borrowed energy and lifetime satisfy:

$$\boldsymbol{E^i_{\text{peak}} \cdot \tau_i \geq \frac{\hbar}{2}}$$

This is the **minimum energy–time composite cost** that the L_0 system must pay to produce a distinguishable "existence individual."

7. **Path and causal-cone constraint:** the individual's path length in the state space Σ is $L(\gamma_i) = \sum_{k=1}^{m_i} |e_i(k)|$, which can be far greater than its net displacement (which is zero). Its spatial extension radius r_i satisfies:

$$r_i \leq \frac{1}{2} \max_t s(\sigma_0, \sigma_t) \leq c \cdot \frac{\tau_i}{2}$$

where c is the inherent maximum propagation speed of protocol-state differences (Definition A1.h).

Definition B2 (Threshold conditions for recursive upgrade and the geometric selection principle).

Geometric selection is the necessary statistical mechanism for recursive upgrade ($d \to d+1$), rather than any independently preset parameter.

For fluctuation individuals produced by the running of L_0, the net borrowing balance $E(t)$ follows a **peaked life-curve** (forced by Definition A1.d: $E(0) = 0 \to E_{\text{peak}} \to E(\tau) = 0$), and the peak distribution follows a power law $P(E_{\text{peak}}) \propto E_{\text{peak}}^{-\alpha}$ (Cornerstone 2, $\alpha > 1$).

To complete the four-phase cycle and solidify into the $d+1$ layer, there must exist a sub-interval within the lifecycle such that the window-averaged survival term satisfies the minimum survival requirement of $d+1$-level protocols, $K^{(d+1)}$:

$$\boldsymbol{\left\langle \frac{E_{\text{active}}(t)}{\tau_{\text{active}}(L)} + \eta(L) \cdot e(t) \right\rangle_W \geq K^{(d+1)}} \tag{1}$$

Here $K^{(d+1)}$ is the **survival-threshold constant** for the $d+1$ layer as an independent form (Axiom III.3), and is forced upward by the solidification cost of higher-order difference patterns.

Notation. This paper uses the unified symbol $\epsilon_{\text{th}}^{(d)}$ to denote the energy threshold required for the transition from recursion depth $d-1$ to d. Thus:

- $\epsilon_{\text{th}}^{(1)}$: threshold for the transition from $d = 0$ (quantum fluctuations) to $d = 1$ (fundamental particles);
- $\epsilon_{\text{th}}^{(2)}$: threshold for the transition from $d = 1$ to $d = 2$;
- and so on...

Geometric selection principle.

For an individual form, its life-curve $E(t)$ is a peaked curve. Draw a horizontal line $E = \epsilon_{\text{th}}^{(d+1)}$. This line intersects the curve $E(t)$ at two points t_{start} and t_{end} ($t_{\text{start}} < t_{\text{end}}$). The interval $[t_{\text{start}}, t_{\text{end}}]$ is called the **potential time window**, with length $\Delta t = t_{\text{end}} - t_{\text{start}}$.

Only when this potential window satisfies **both** conditions below does it become a **qualified window** W:

1. **Time condition:** $\Delta t \geq N_{\min}^{(d+1)}$, i.e., the window length is not smaller than the minimum time window.
2. **Survival condition:** the window-averaged term satisfies Eq. (1).

Because of the universality of the scale constraint (Axiom III.3), the Heisenberg uncertainty relation $\Delta E \cdot \Delta t \geq \hbar/2$ is only the lowest-order, special-form manifestation of this scale law at the quantum layer ($d = 0 \to d = 1$). Higher-level upgrades likewise obey an **energy–time complementary tradeoff**:

- energy too high $\rightarrow$ window too short (solidification cannot be completed);
- energy too low $\rightarrow$ insufficient input (cannot reach $K^{(d+1)}$).

Thus, the vast majority of fluctuations are naturally excluded from qualified windows. Only those whose energy–time composite falls within an ”effective survival band” have non-zero probability of completing irreversible solidification.

Emergence of the capture-energy threshold.

Within a qualified window W, because $E(t) \geq \epsilon_{\text{th}}^{(d+1)}$ and is typically on the rising segment or the peak plateau of the curve, the instantaneous difference $e(t)$ is correspondingly high. According to the existence-degree formula

$$\boldsymbol{\Psi_M(t) = \frac{E_{\text{solid}}(t)}{\tau_{\text{solid}}(L)} + \frac{E_{\text{active}}(t)}{\tau_{\text{active}}(L)} + \eta(L) \cdot e(t),}$$

within this window, the sum of the three terms will necessarily maintain a level significantly above zero.

Define this **minimum existence-degree level** guaranteed jointly by the energy threshold and the time window as the **capture-energy threshold** $\Psi_{\text{invest}}^{(d+1)}$:

$$\boldsymbol{\Psi_{\text{invest}}^{(d+1)} := \min_{t \in W} \Psi_M(t) \quad \text{for all individuals satisfying } \Delta t \geq N_{\text{min}}^{(d+1)}.}$$

Therefore, $\Psi_{\text{invest}}^{(d+1)}$ is not an independent parameter; it is a logical corollary of $\epsilon_{\text{th}}^{(d+1)}$, $N_{\text{min}}^{(d+1)}$, and the survival threshold $K^{(d+1)}$. It quantifies the ”high survival quality (high Ψ)” that necessarily comes with the ability to ”maintain a high-energy state for a long time.”

Final adjudication of upgrade permission.

A **necessary and sufficient condition** for an individual to obtain permission to attempt the ”four-phase cycle” upgrade is: within its lifecycle there exists at least one sub-interval W such that

1. $\Delta t \geq N_{\text{min}}^{(d+1)}$;
2. the window-averaged term satisfies Eq. (1).

This condition already implies that within the window $\Psi_M(t) \geq \Psi_{\textbf{invest}}^{(d+1)} \gg 0$.

Once the window closes, subsequent fluctuations permanently lose the opportunity to upgrade, forming ”failed samples” (e.g., close primate relatives permanently frozen at the $d = 5$ layer).

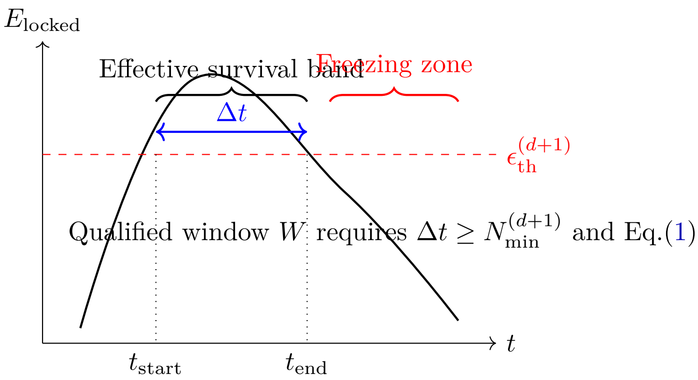

Figure 1: Schematic diagram of the geometric selection principle.

—

Definition B3 (Formation condition of the primordial excited state).

When a quantum-fluctuation individual i (with form M^i_{fluct}) has its net borrowing energy curve $E_i(t)$ intersect the energy threshold $\epsilon^{(1)}_{\text{th}}$ within its lifecycle, and the time span between the intersection points satisfies $\Delta t \geq N^{(1)}_{\min}$, then it automatically satisfies the geometric selection condition of **Definition B2**, thereby obtaining access permission to the candidate protocol-family $\mathscr{P}_1$ and entering the **primordial excited state**, with its form becoming M^i_{excited}.

Its state function is:

$$\boldsymbol{\Omega_i^{(M_{\text{excited}})}(t) = \Phi_{\text{excited}}(\{L_0\} \cup P, E_i, t)}$$

where $P \in \mathscr{P}_1$.

Definition B4 (Triggering and termination of cosmic inflation). 1. **Trigger time (T_0).** When the first batch of fluctuations satisfying Definition B3 collectively activate $\mathscr{P}_1$, symmetry is broken and the universe transitions from true vacuum into the primordial excited state. This cosmological initial moment is denoted T_0.

2. **Inflation process.** Starting from T_0, the universe enters an exponential expansion phase (inflation).

3. **Termination time.** The inflation duration is naturally determined by the minimum time window:

$$\boldsymbol{\tau_{\text{inflation}} = N^{(1)}_{\min} \cdot t_P}$$

At background time $T = T_0 + \tau_{\text{inflation}}$, the first activators complete protocol solidification execution (Definition B5), triggering a macroscopic phase transition and terminating inflation.

Definition B5 (Protocol solidification execution).

A primordial excited-state form M^i_{excited} attempts to execute its protocol set P; this process is the **first realization of the four-phase cycle**. The required time is determined by the minimum time window:

$$\boldsymbol{\delta t = N_{\min}^{(1)} \cdot t_P}$$

This process is irreversible, with a binary outcome:

- **Successful solidification.** Part of the peak borrowed energy is locked; the protocols in P are instantiated and solidified into a core protocol subset $S_1 \in \mathscr{L}_1$. The form transitions into a stable ordinary-matter form M_{OM}^i:

  $$\boldsymbol{\Phi_{\mathrm{excited}}(\{L_0\} \cup P, E_i, t) \to \Phi_{\mathrm{OM}}(\{L_0\} \cup S_1, \eta E_{\mathrm{peak}}^i, t) + \text{radiation photons}\ \big((1-\eta) E_{\mathrm{peak}}^i\big)}$$

  where $E_{\mathrm{locked}}^i = \eta E_{\mathrm{peak}}^i$ is the total structured asset, and the radiation photons constitute the primordial radiation field. This transition marks the increase of recursion depth from $d(\{L_0\} \cup P) \approx 0$ to $d(\{L_0\} \cup S_1) = 1$.

  Species origin. If this successfully solidified individual is among the **first successful solidifiers** at depth $d = 1$, then its protocol subset $S_1 \in \mathscr{L}_1$ becomes the prototype of the protocol fingerprint $C_{\mathcal{S}}$ that defines subsequent ordinary-matter species. The S_1 of the first successful solidifiers jointly determine the fingerprints $C_{\mathcal{S}}$ of $d = 1$-layer species.

- **Solidification interruption.** If external conditions interrupt the process before solidification is completed, it is forcibly terminated and permanently frozen, forming a dark-matter form M_{DM}^i.

Definition B6 (Channel closure and the three existence categories).

At the inflation-termination moment $T = T_0 + \tau_{\mathrm{inflation}}$:

1. All energy-borrowing channels from true vacuum (the baseline state L_0) are permanently and collectively cut off.
2. All protocol solidification executions that are ongoing but incomplete are forcibly interrupted and permanently frozen.

Thus, at the "cosmic moment" $T_0 + \tau_{\textbf{inflation}}$, three existence categories are produced in one shot:

Category A (ordinary matter). Primordial excited states that have **completed** protocol solidification execution before $T = T_0 + \tau_{\mathrm{inflation}}$, forming ordinary-matter forms M_{OM}^i. Complete state:

$$\boldsymbol{\Omega^{(M_{\mathrm{OM}})}(t) = \Phi_{\mathrm{OM}}(\{L_0\} \cup S, E_{\mathrm{locked}}^i, t), \quad S \in \mathscr{L}_1}$$

This form has recursion depth $d(M_{\mathrm{OM}}) = 1$. Its total structured asset $E_{\mathrm{locked}} = E_{\mathrm{solid}} + E_{\mathrm{active}}$ corresponds to the mass of ordinary matter.

Category B (dark matter). Events that are **still in** the primordial excited state at $T = T_0 + \tau_{\text{inflation}}$, whose locked energy remains not below the threshold and are permanently frozen, forming dark-matter forms M^i_{DM}. Complete state:

$$\boldsymbol{\Omega}^{(M_{\text{DM}})}(\boldsymbol{t}) = \boldsymbol{\Phi}_{\text{DM}}(\{\boldsymbol{L_0}\} \cup \boldsymbol{P}, \boldsymbol{E}^{\boldsymbol{i}}_{\text{frozen}}, \boldsymbol{t}) \quad \text{(frozen)}$$

where P is the un-solidified protocol set, and E^i_{frozen} is the frozen active-state energy (non-zero, consistent with thermodynamics' third law). This form has recursion depth $d(M_{\text{DM}}) = 0$.

Category C (dark energy). At $T = T_0 + \tau_{\text{inflation}}$, all fluctuations whose **cumulative net borrowing energy** E_i has never reached—or has already fallen below—the threshold $E^{(1)}_{\text{th}}$, have their energy uniformly frozen into space, forming the initial part of the dark-energy background form M_Λ. Thereafter, the energy lost by radiation through redshift during cosmic expansion is continuously injected into this background, causing the total dark-energy content to grow in step with cosmic volume, while its density remains constant.

Complete state:

$$\boldsymbol{\Omega}^{(M_\Lambda)}(\boldsymbol{t}) = \boldsymbol{\Phi}_{\boldsymbol{\Lambda}}(\{\boldsymbol{L_0}\}, \boldsymbol{E}_{\text{background}}(\boldsymbol{t}), \boldsymbol{t})$$

where

$$\boldsymbol{E}_{\text{background}}(\boldsymbol{t}) = \boldsymbol{E}^{\text{(frozen)}}_{\text{background}} + \int_{\boldsymbol{t_0}}^{\boldsymbol{t}} \boldsymbol{Q}_{\boldsymbol{r}\to\boldsymbol{\Lambda}}(\boldsymbol{t'})\boldsymbol{V}(\boldsymbol{t'})\boldsymbol{dt'}$$

is the uniformly frozen active-state energy, with density $\rho_\Lambda = E_{\text{background}}/V$ constant (non-zero, consistent with thermodynamics' third law). This background form has recursion depth $d(M_\Lambda) = 0$.

Definition B7 (Origin of the cosmic microwave background radiation, CMB). The CMB is the collectively redshifted remnant of the high-energy photons (total amount $(1-\bar{\eta})E^{(\text{A})}_{\text{total peak}}$) radiated by the collective phase transition of Category A events—i.e., those that completed protocol solidification execution before $T = T_0 + \tau_{\text{inflation}}$. Its near-perfect blackbody spectrum confirms the universality (shared $\bar{\eta}$) and statistical nature of this process.

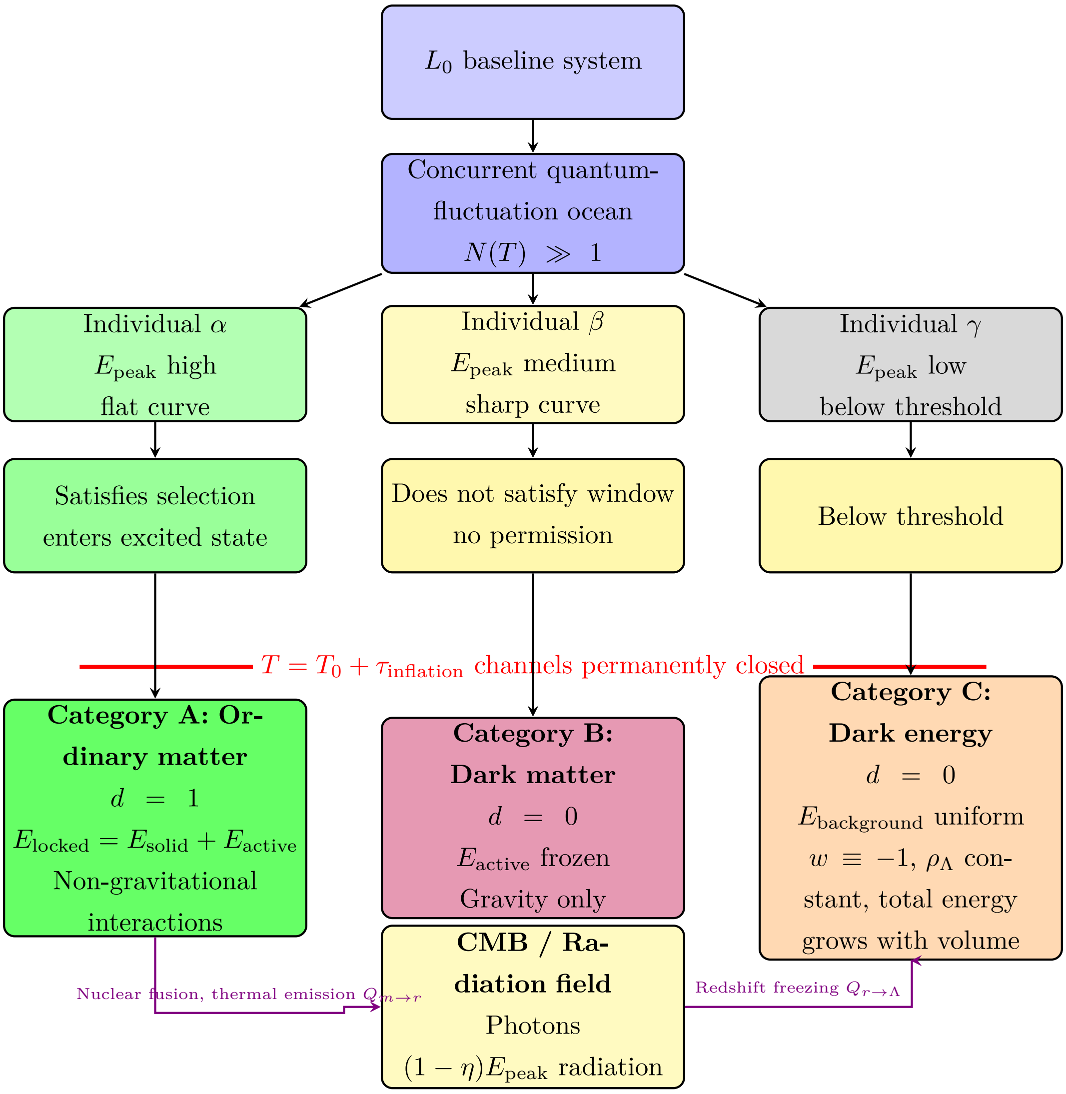

Figure 2: The three existence categories frozen at the cosmic moment.

3.3.3 Module C: Recursive Evolution Dynamics

Definition C1 (The four-phase cycle: the core algorithm of form upgrade). Any transition from recursion depth d to $d+1$ must proceed through the four-phase cycle: **Trial — Capture — Solidify — Generate**. The **prerequisite** for starting the cycle is satisfying the geometric selection condition of Definition B2 (i.e., existence of a qualified window W).

Stages:

1. **Trial.** Within the qualified window W, the individual selects a candidate protocol set P from the candidate family $\mathscr{P}_{d+1}$, randomly or by weighted choice.
2. **Capture. Using the high existence-degree capital during the window**, the candidate protocol set P is temporarily "bound" or "mounted" onto the current protocol set, forming a temporary composite $\{L\} \cup P$ with the existing form M,

creating a transient bridge $\Sigma_d \to \Sigma_{d+1}$. This begins consuming the system's capital reserves.

3. **Solidify.** If and only if both the energy threshold $\epsilon_{\text{th}}^{(d+1)}$ and the minimum time window $N_{\text{min}}^{(d+1)}$ are satisfied, and the process is completed before the window ends, can the transient structure solidify into a new stable protocol set:

$$L' = \{L\} \cup S, \quad S \in \mathscr{L}_{d+1}$$

where S is a solidified instance of P.

If solidification cannot be completed within the window, or before completion $\Psi_M(t)$ drops to the critical point, the upgrade fails; the individual may revert to its original form or disintegrate.

Species creation. If the individual is among the **first successful solidifiers** at depth $d{+}1$, then its S participates in defining the new species' protocol fingerprint $C_{\mathcal{S}}$. At the same depth $d+1$, different individuals may solidify different $S_1, S_2, \ldots$ from different candidate protocol sets $P_1, P_2, \ldots$. As long as these S are mutually compatible ($\kappa \geq \kappa_{\text{crit}}$), they can coexist at the same depth as different species $\mathcal{S}_1, \mathcal{S}_2, \ldots$.

4. **Generate.** The new protocol set L' begins executing its functions, producing outputs (e.g., radiation photons, interactions, behavioral outputs), and may participate in the next recursion cycle.

Definition C2 (Inter-layer transition rate functions).

Define two types of inter-layer transition rates:

1. **Upward transition rate:**

$$W_{\uparrow}^{(d\to d+1)}(\Omega_d \to \Omega_{d+1}; T)$$

- Conditions: $\Omega_d \in \Sigma_d,\ \Omega_{d+1} \in \Sigma_{d+1}$
- Non-zero only when the four-phase cycle succeeds and the threshold window conditions are met
- Physical meaning: the rate at which an individual at depth d creates new species and new form spaces at depth $d+1$ through investment

2. **Downward transition rate:**

$$W_{\downarrow}^{(d+1\to d)}(\Omega_{d+1} \to \Omega_d; T)$$

- Conditions: protocol disintegration, energy depletion, toxic destruction, form death
- Occurs when $\Psi_M(t) < 0$ or when the form disintegrates
- Physical meaning: the rate at which a form at depth $d+1$ disintegrates and releases energy back to depth d, with protocols deactivated

Definition C3 (Intra-layer differentiation rate function).

Define the within-layer species differentiation rate:

$$\boldsymbol{W_{\mathrm{div}}^{(d)}(\mathcal{S}_i \to \mathcal{S}_j; T)}$$

- Conditions: within the same depth d, an individual of an existing species $\mathcal{S}_i$ successfully solidifies a new protocol subset S whose fingerprint $C_{\mathcal{S}_j} \neq C_{\mathcal{S}_i}$, and the compatibility satisfies $\kappa \geq \kappa_{\mathrm{crit}}$.
- Physical meaning: adaptive radiation and niche filling within the same depth, a core indicator of flourishing complex systems.

Definition C4 (An individual's morphologization efficiency and ordering factor).

For an individual that successfully completes protocol solidification execution (Category A), its peak borrowed energy E^i_{peak} is allocated according to an efficiency η_{ind}:

$$\boldsymbol{E^i_{\mathrm{peak}} = E^i_{\mathrm{locked}} + E^i_{\mathrm{radiated}}, \quad E^i_{\mathrm{locked}} = \eta_{\mathrm{ind}} E^i_{\mathrm{peak}}, \quad E^i_{\mathrm{radiated}} = (1 - \eta_{\mathrm{ind}}) E^i_{\mathrm{peak}}}$$

where E^i_{locked} is the total structured asset (sum of solidified-state and active-state energy after solidification), and E^i_{radiated} is radiated energy released as photons.

Define the individual's ordering factor:

$$\boldsymbol{\Xi_{\mathrm{ind}} := \frac{1 - \eta_{\mathrm{ind}}}{\eta_{\mathrm{ind}}}}$$

which measures the entropy-increase cost required to create a unit of structured asset.

Definition C5 (Cosmological average morphologization efficiency).

At $T = T_0 + \tau_{\mathrm{inflation}}$, statistically aggregate all completed protocol solidification execution events (the total of Category A) and define the cosmological average morphologization efficiency:

$$\boldsymbol{\bar{\eta} := \frac{E^{(\mathrm{A})}_{\text{total locked}}}{E^{(\mathrm{A})}_{\text{total peak}}}, \quad \bar{\Xi} := \frac{1 - \bar{\eta}}{\bar{\eta}}}$$

where $E^{(\mathrm{A})}_{\text{total peak}} = \sum_{i \in \mathrm{A}} E^i_{\mathrm{peak}}$ and $E^{(\mathrm{A})}_{\text{total locked}} = \sum_{i \in \mathrm{A}} E^i_{\mathrm{locked}}$.

3.3.4 Module D: State Description and Diagnostic Tools

Definition D1 (Time hierarchy). 1. **Background time T.** An objective statistical measure of the overall evolution progress of the concurrent fluctuation ocean in true vacuum, with Planck time t_P as the minimal unit:

$$T = n \cdot t_P \quad (n \in \mathbb{N})$$

Background time is the objective measure of the irreversible growth of L_0's **global memory (individual historical records)**. The arrow of time arises from the monotonic growth of memory.

2. **Individual internal time t_i.** The step count of the recursive unfolding of the individual i's protocol set L_i within its lifecycle. If the individual is born at background time T_{birth} with lifetime $\tau_i = m_i \cdot t_P$, then $t_i \in \{1, 2, \ldots, m_i\}$ corresponds to background time:

$$\boldsymbol{T = T_{\text{birth}} + (t_i - 1)t_P}$$

Definition D2 (Layered complete state function and species subspaces). 1. **Species subspace.** For each recursion depth d and species $\mathcal{S}$, there exists a corresponding state subspace:

$$\boldsymbol{\Sigma_d^{(\mathcal{S})} := \{\Omega^{(M)} \mid d(L) = d \text{ and } \mathrm{Core}(L) = C_{\mathcal{S}}\}}$$

where $\Omega^{(M)}$ is the complete state of form M, and $d(M) = d(L) = d$.

2. **Layered complete state function.** The instantaneous state of any individual belonging to species $\mathcal{S}$ (with recursion depth d) is fully determined by that species' state function:

$$\boldsymbol{\Omega_{\mathcal{S}}^{(M)}(t) = \Phi_{\mathcal{S}}(L, E_{\text{solid}}(t), E_{\text{active}}(t), e(t), t), \quad \Omega_{\mathcal{S}}^{(M)} \in \Sigma_d^{(\mathcal{S})}}$$

where the form of $\Phi_{\mathcal{S}}$ is determined by the species' core protocol fingerprint $C_{\mathcal{S}}$ (Definition A4). It takes as input the individual instance's protocol set L (satisfying $\mathrm{Core}(L) = C_{\mathcal{S}}$) and its energy state, and outputs the complete state vector $\Omega^{(M)}$. Recursion depth $d = d(L) = d(M)$ is an endogenous property of the form.

Unification note. Here "species $\mathcal{S}$" is the mathematical formalization of the "stable form category" in Definition A4. Whether $\mathcal{S}$ refers to fundamental particles ($d = 1$), chemical elements ($d = 3$), biological species ($d = 5$–7), or occupational skills ($d = 8$–10), the mathematical structure is identical: all are identified by a core protocol fingerprint $C_{\mathcal{S}}$, correspond to a state subspace $\Sigma_d^{(\mathcal{S})}$, and follow the same state-function form $\Phi_{\mathcal{S}}$. This unified mathematical framework is the root of GET's cross-domain explanatory power.

Definition D3 (Layered state distribution functions).

The macroscopic state of the system is fully described by the **family of layered state distribution functions**:

$$\{P_d(\Omega^{(M)}, T)\}_{d=0}^{D_{\max}}$$

where:

- $P_d(\Omega^{(M)}, T)$: at background time T, the probability density of being at depth d and state $\Omega^{(M)}$, with $\Omega^{(M)} \in \Sigma_d$ (i.e., $d(M) = d$).
- Normalization:
$$\sum_{d=0}^{D_{\max}} \int_{\Sigma_d} P_d(\Omega^{(M)}, T) d\Omega^{(M)} = 1.$$

The total distribution is the layered sum:

$$\boldsymbol{P}_{\text{total}}(\boldsymbol{\Omega}^{(M)}, \boldsymbol{T}) = \sum_{d=0}^{D_{\max}} \delta_{d(M),d} \cdot \boldsymbol{P}_{d}(\boldsymbol{\Omega}^{(M)}, \boldsymbol{T}).$$

Definition D4 (Connection capital—spatial constraints on accumulation and depreciation).

Complexity increase of a protocol set (accumulation of connection capital) is strictly constrained by:

1. **The only path of accumulation:** accumulation occurs if and only if via a **successful four-phase cycle**, i.e., a transition from L to $L' = \{L\} \cup S$ (with $S \in \mathscr{L}_{d+1}$). This requires converting a large amount of active-state energy E_{active} into new solidified-state energy E_{solid} (solidified into a new protocol), thereby increasing total structured asset E_{locked}.
2. **Distance dilution of energy capture:** increasing depth $d \to d+1$ requires a minimum investment energy

$$\boldsymbol{E}_{\text{invest}} = \boldsymbol{\epsilon}_{\text{th}}^{(d+1)} \cdot \boldsymbol{N}_{\text{min}}^{(d+1)},$$

   which must be provided by the individual's active-state energy E_{active} within a qualified window. But effective invested energy dilutes with transmission distance r:

$$\boldsymbol{E}_{\text{invest}}^{\text{eff}} = \frac{\boldsymbol{E}_{\text{invest}}}{\boldsymbol{1} + (\boldsymbol{r}/\boldsymbol{r}_{0}(\boldsymbol{L}))^{2}}.$$

3. **Spatial decay of environmental match:** an individual's investment capability $\mathcal{I}_i$ depends on the match between its protocol set and other protocol sets in the environment, and decays with distance.
4. **Necessity of depreciation:** depreciation is jointly driven by insufficient E_{active} (unable to maintain protocol synchronization) and spatial separation that makes synchronization difficult, ultimately manifesting as dissipation of E_{solid} or protocol deactivation.

These characteristic time constants are determined by domain-specific protocol sets L, but their mathematical status in the existence-degree equation

$$\boldsymbol{\Psi}_{M}(\boldsymbol{t}) = \frac{\boldsymbol{E}_{\text{solid}}}{\boldsymbol{\tau}_{\text{solid}}} + \frac{\boldsymbol{E}_{\text{active}}}{\boldsymbol{\tau}_{\text{active}}} + \boldsymbol{\eta} \cdot \boldsymbol{e}(\boldsymbol{t})$$

is identical—quantifying the contribution rates of solidified assets and active assets to continued existence.

Table 2: Cross-domain mapping of characteristic time constants τ_{solid} and τ_{active}

Domain	**Solidified-state time constant** $\boldsymbol{\tau_{\text{solid}}}(L)$	**Active-state time constant** $\boldsymbol{\tau_{\text{active}}}(L)$
Physics	nuclear half-lives, stellar lifetimes, etc.	excited-state lifetimes, chemical reaction rates, etc.
Biology	fat breakdown rates, bone renewal cycles, etc.	ATP turnover time, blood-glucose consumption rate, etc.
Economics	fixed-asset depreciation periods, brand-value decay times, etc.	cash conversion cycles, receivables turnover days, etc.
Cognitive science	forgetting time constants after long-term memory consolidation, etc.	working-memory refresh cycles, attention-span duration, etc.

Definition D5 (Toxic residues and the necessity of isolation).

The residues of dead individuals may contain a **toxic protocol subset** L_{toxic}, with latency, triggerability, destructiveness, and transmissibility. The spread of toxic protocols is described by specific transition terms in the evolution equation. System stability requires an effective isolation mechanism satisfying isolation efficiency:

$$\boldsymbol{Q} > \boldsymbol{Q}_{\text{crit}}.$$

Black holes are universe-level toxic-protocol isolation systems.

3.4 Core Axiomatic System

Axiom Ω (Baseline dynamical system and the eternal background) □. *1.* ***Baseline dynamical system*** L_0***.*** *There exists a self-sustaining, memoryful, concurrently executing protocol-recursive dynamical system* L_0*. The macroscopic statistical steady state presented by its independent running is the true vacuum.*

2. ***Time as a measure of memory growth.*** *Background time* $T = n \cdot t_P$ *is the objective measure of the irreversible growth of* L_0*'s* ***global memory (individual historical records)****. The arrow of time arises from the monotonicity of memory growth and is an intrinsic property of* L_0*'s dynamics.* ***This bottom-level arrow of time, defined by the monotonic growth of*** L_0***'s memory, is the microscopic dynamical foundation of the macroscopic thermodynamic second law (entropy increase)****—precisely because each execution of the bottom-level difference-unfolding process* D *irreversibly increases global memory, macroscopic entropy necessarily exhibits monotonic growth.*
3. ***Constancy of the concurrent individual ocean.*** *At any* T*,* L_0 *maintains an extremely large number* $(N(T) \gg 1)$ *of concurrent fluctuation individuals, each with its*

own lifecycle and a resource (energy) usage pattern obeying a power-law distribution. Their net borrowing energy curves $E(t)$ have a peaked shape (Definition B1).

4. ***The only route to cosmic birth.*** *The structured origin of the universe occurs if and only if some individuals' life curves $E(t)$ in L_0 intersect the energy threshold $\epsilon_{th}^{(1)}$, and the time span between intersection points reaches the minimum window $N_{min}^{(1)}$.* ***This is the necessity of geometric selection.***

Axiom I (Protocol-recursive monism) □.

All observable, stable, dynamic existences have complete states fully determined by the manifestation of the protocol set L as it unfolds recursively. Formally:

$$\Omega_{\mathcal{S}}^{(M)} = \Phi_{\mathcal{S}}(L)$$

but L ***is the only dynamical ontology****. Form M, energy E, time t, and recursion depth d are all endogenous variables and external manifestations describing the running state of L. All intrinsic properties, dynamical behaviors, and interaction potentials arise from L's own recursive unfolding and interactions.*

Axiom II (The four-phase cycle as the core dynamics) □.

Any transition of recursion depth from d to $d+1$ must be realized through the ***four-phase cycle: Trial — Capture — Solidify — Generate****. Initiation of the cycle* ***must*** *take as prerequisite the existence of a* ***qualified window*** *W jointly defined by the energy threshold $\epsilon_{th}^{(d+1)}$ and the minimum time window $N_{min}^{(d+1)}$ (Definition B2). Here $\epsilon_{th}^{(d+1)}$ is the energy threshold required for the transition from depth d to $d+1$. The four-phase cycle is the* ***universal algorithm*** *of protocol recursion and the only mechanism connecting different depth layers $\Sigma_d \to \Sigma_{d+1}$.*

Species differentiation within the same depth *does not require a new four-phase cycle in the sense of increasing depth; rather, within depth d, a new protocol fingerprint $C_{\mathcal{S}'}$ is formed by solidifying a compatible candidate protocol $P \in \mathscr{P}_d$ onto an existing protocol set L. This process still follows the logical sequence "Trial — Capture — Solidify — Generate," but its threshold conditions are determined by the parameters of depth d itself and do not change recursion depth.*

Axiom III (Criterion of form existence, evolutionary direction, and scale constraints) □.

1. Criterion of form existence

According to Definition A1, for a concrete form M instantiated by protocol set L, its ***instantaneous existence degree*** *$\Psi_M(t)$ is defined as:*

$$\boldsymbol{\Psi_M(t) := \frac{E_{solid}(M,t)}{\tau_{solid}(L)} + \frac{E_{active}(M,t)}{\tau_{active}(L)} + \eta(L) \cdot e(t)}$$

Survival criterion: *a form M can continue to exist if and only if:*

$$\boldsymbol{\frac{E_{active}(M,t)}{\tau_{active}(L)} + \eta(L) \cdot e(t) \geq 0}$$

Three-state distinction

- > 0***:*** *healthy operation; the system is normally maintained*
- $= 0$***:*** *critical balance; the system is fragile but not yet disintegrated*
- < 0***:*** *death; the system disintegrates immediately*

Key explanation

- *The term* $\frac{E_{solid}(M,t)}{\tau_{solid}}$ *is the system's structural return due to possessing solidified assets, such as a star's gravitational binding energy or the stability provided by a biological organism's skeletal structure. This return does not directly participate in immediate survival maintenance, but provides long-term stability buffering.*
- *The system's* ***immediate survival*** *is fully determined by the latter two terms.*
- *When* $\eta \cdot e(t) < 0$*, the system begins consuming its* E_{active} *reserves to compensate for negative net flow. As long as* E_{active} *is sufficient,* $\frac{E_{active}(M,t)}{\tau_{active}(L)} + \eta(L) \cdot e(t) \geq 0$ *can still be maintained and the system survives.*
- *When* $\eta \cdot e(t) < 0$ *persists,* E_{active} *gradually decreases. Eventually, when* E_{active} *falls so that* $\frac{E_{active}(M,t)}{\tau_{active}(L)} + \eta(L) \cdot e(t) < 0$*, the system can no longer maintain immediate operation and dies immediately.*
- ***Solidified-state energy*** E_{solid} ***cannot help at this stage****, because its conversion rate into active-state energy is extremely slow and cannot meet immediate demand.*
- *After death, the system's solidified-state energy* E_{solid} *may be utilized by other systems, but the dead system itself cannot use that energy to save itself.*

Manifestation of the third law of thermodynamics: *a fully frozen state (*$E_{active} = 0$ *and* $\eta \cdot e(t) = 0$*) is unattainable—any existing form must maintain* $\frac{E_{active}(M,t)}{\tau_{active}(L)} + \eta(L) \cdot e(t) \geq 0$*, which means an ideal crystal at absolute zero is physically impossible.*

2. Two fundamental paths of evolutionary direction

To maintain or enhance a form M's existence degree $\Psi_M(t)$*, the system has only two fundamentally optimizable paths, directly forced by the structure of the existence-degree formula:*

$$\boldsymbol{\Psi_M(t)} = \underbrace{\frac{\boldsymbol{E_{solid}(M,t)}}{\boldsymbol{\tau_{solid}(L)}}}_{\text{passive structural return (not optimizable)}} + \underbrace{\frac{\boldsymbol{E_{active}(M,t)}}{\boldsymbol{\tau_{active}(L)}}}_{\text{Path 1: increase active reserves}} + \underbrace{\boldsymbol{\eta(L) \cdot e(t)}}_{\text{Path 2: keep net flow positive}}$$

Important note: *the term* $\frac{E_{solid}(M,t)}{\tau_{solid}(L)}$ *is a passive return determined by the system's core protocol fingerprint* $C_{\mathcal{S}}$*, and* ***cannot be changed via active optimization****. The system can only accept it as a given baseline return.*

Path 1: Increase active-state energy reserves

Goal: increase $\frac{E_{active}(M,t)}{\tau_{active}(L)}$.

Strategy: enlarge the scale of active-state energy E_{active} *to provide a longer operational buffer.*

Physical meaning:

- *Active-state energy* E_{active} *is the system's "working capital," e.g., a star's nuclear fuel, a biological organism's ATP, a firm's cash flow.*
- *Increasing* E_{active} *means the system has a more ample energy reserve, enabling survival under net-flow fluctuations for a longer time.*
- *But* E_{active} *is consumable and must be continually replenished through Path 2.*

Examples:

- *Stars: increase core nuclear-fuel reserves by accreting companion matter.*
- *Biology: increase energy reserves via feeding and storage mechanisms (e.g., fat storage).*
- *Firms: increase cash reserves via financing or retained profits.*
- *Civilizations: build strategic resource reserves and food reserves.*

Path 2: Keep net flow positive

Goal: ensure $\eta(L) \cdot e(t) > 0$ *holds.*

Strategies:

1. *Increase capture efficiency* $\eta(L)$ *by optimizing protocol set* L *so that environmental energy is converted into usable form more effectively.*
2. *Increase instantaneous net flow* $e(t)$ *(Definition A1.c) by opening new inflow channels or reducing outflow losses, keeping net inflow positive.*

Physical meaning:

- *This is the system's* ***fundamental guarantee of immediate survival****. If* $\eta \cdot e(t) \leq 0$*, the system begins consuming its* E_{active} *reserves.*
- *When* $\eta \cdot e(t) < 0$ *persists,* E_{active} *is gradually exhausted, ultimately yielding* $\frac{E_{active}(M,t)}{\tau_{active}(L)} + \eta(L) \cdot e(t) < 0$ *and system death.*
- *Keeping net flow positive is the system's* ***primary task*** *for long-term survival.*

Examples:

- *Stars: sustain fusion so that outward radiation pressure balances gravity; when fuel is exhausted and the survival inequality fails, the star dies.*

- *Biology: sustain intake > consumption; hunger makes $e(t) < 0$ and reserves are consumed.*
- *Firms: sustain revenue > expenses; losses make $e(t) < 0$ and cash reserves are consumed.*
- *Civilizations: sustain production > consumption; resource depletion makes $e(t) < 0$ and civilization declines.*

System death mechanism

- ***Negative net-flow phase:*** *when $\eta \cdot e(t) < 0$, the system consumes E_{active}. As long as E_{active} suffices, the survival inequality can still hold.*
- ***Critical point:*** *when consumption reduces E_{active} until $\frac{E_{active}}{\tau_{active}} + \eta \cdot e(t) = 0$, the system is at critical balance.*
- ***Death point:*** *when further decrease makes $\frac{E_{active}}{\tau_{active}} + \eta \cdot e(t) < 0$, the system dies immediately.*
- ***After death:*** *the system disintegrates; its E_{solid} may be used by others, but the dead system cannot use E_{solid} to save itself because conversion is too slow.*

(Explanatory cases for stars / biology / firms follow as domain illustrations and do not add new axioms.)

3. Scale constraints

From the survival criterion $\frac{E_{active}(M,t)}{\tau_{active}(L)} + \eta \cdot e(t) \geq 0$ and long-term stability conditions, combined with the physical properties of a specific protocol set L, one can derive the ***energy–time scale constraint*** *that the form category (species) must satisfy.*

For a species $\mathcal{S}^{(d)}$ defined by a specific core protocol fingerprint $C_{\mathcal{S}}$, there is a universal relation among its solidified-state energy E_{solid}, active-state energy E_{active}, instantaneous net flow $e(t)$, and characteristic parameters $\tau_{solid}, \tau_{active}, \eta$:

$$\boldsymbol{\frac{E_{solid}}{\tau_{solid}(C_{\mathcal{S}})} + \frac{E_{active}}{\tau_{active}(C_{\mathcal{S}})} + \eta(C_{\mathcal{S}}) \cdot \langle e(t) \rangle \geq K(C_{\mathcal{S}})}$$

where $\langle e(t) \rangle$ is the average instantaneous net flow under typical survival environments, and $K(C_{\mathcal{S}}) > 0$ is the ***survival-threshold constant****, representing the minimal existence-degree level required for stable existence.*

Unified verification

- *For the $d = 0$ quantum fluctuation species (L_0): $E_{solid} \equiv 0$, $\tau_{active} \propto \tau$, $E_{active} = \Delta E$, $\eta(L_0) = 1$. After integrating over the lifecycle, this inequality necessarily reduces to $E_{peak} \cdot \tau \geq \hbar/2$, where $K = \hbar/2$. This is L_0's* ***minimum accounting unit****.*
- *For $d \geq 1$ structures (fundamental particles, stars, living forms, etc.), observed scale laws are specific instances of this universal relation under their specific core protocol*

fingerprints $C_{\mathcal{S}}$.

Axiom IV (Inflation trigger and window closure) $\square$.

When the first quantum-fluctuation individuals' life curves $E(t)$ *intersect the energy threshold* $\epsilon_{th}^{(1)}$*, and the time span between intersection points reaches the minimum window* $N_{min}^{(1)}$*,* $\mathscr{P}_1$ *is activated; symmetry is broken; cosmic inflation is triggered. The trigger moment is denoted* T_0*, and the duration is* $\tau_{inflation} = N_{min}^{(1)} \cdot t_P$.

At $T = T_0 + \tau_{inflation}$*, the first activators complete protocol solidification, triggering a macroscopic phase transition and* ***permanently closing the energy-borrowing channel from*** $d = 0$ ***to*** $d = 1$*. Thereafter,*

$$\boldsymbol{\mathcal{H}_1^{(inter\text{-}up)}[P_0] \equiv 0.}$$

Axiom V (Layered recursive evolution equation system) $\square$.

The macroscopic evolution of the system strictly follows the layered evolution equations:

$$\boldsymbol{\frac{\partial P_d(\Omega^{(M)}, T)}{\partial T} = \mathcal{H}_d^{(intra)}[P_d] + \mathcal{H}_d^{(inter\text{-}up)}[P_{d-1}] + \mathcal{H}_d^{(inter\text{-}down)}[P_{d+1}] + \mathcal{H}_d^{(div)}[P_d]}$$

where:

- $\mathcal{H}_d^{(intra)}$*: intra-layer dynamics (individual existence-degree evolution, energy exchange), acting on forms* M *at depth* d*.*
- $\mathcal{H}_d^{(inter\text{-}up)}$*: upward transitions (*$d-1 \to d$*, successful four-phase cycle), moving forms from depth* $d-1$ *to depth* d*.*
- $\mathcal{H}_d^{(inter\text{-}down)}$*: downward transitions (*$d+1 \to d$*, form disintegration), disintegrating forms from depth* $d+1$ *back to depth* d*.*
- $\mathcal{H}_d^{(div)}$*: intra-layer species differentiation (solidification of new protocol fingerprints within the same depth), generating new form categories at depth* d*.*

Transition rates W *satisfy protocol match, energy bookkeeping, threshold windows, and causal-cone constraints. This equation system is the basic law of system dynamics.*

Axiom VI (Morphologization efficiency and the necessity of entropy output) $\square$.

Any successful individual morphologization process (the "Solidify" phase of the four-phase cycle) must obey Definition C4, allocating input energy according to an efficiency $\eta < 1$*. Necessarily, a fraction* $(1-\eta)$ *of the energy is radiated to the environment as photons as "entropy increase." This is the manifestation of the thermodynamic second law.*

Axiom VII (Connection capital: spatial constraints on accumulation and depreciation) $\square$.

The complexity increase of a protocol set (accumulation of connection capital) is strictly constrained (Definition D4):

1. ***Only path of accumulation:*** *only via successful four-phase cycles.*
2. ***Distance dilution of energy capture:*** *the minimum effective investment energy required for depth increase dilutes with transmission distance.*
3. ***Spatial decay of environmental match:*** *investment capability decays with distance.*
4. ***Necessity of depreciation:*** *depreciation is driven by insufficient energy and protocol synchronization difficulties caused by spatial separation.*

Axiom VIII (Toxic residues and the necessity of isolation) □.
The residues of dead individuals may contain a toxic protocol subset L_{toxic} with latency, triggerability, destructiveness, and transmissibility. The spread of toxic protocols is described by specific transition terms in the evolution equations. System stability requires an effective isolation mechanism satisfying isolation efficiency $Q > Q_{crit}$. Black holes are universe-level toxic-protocol isolation systems.

Axiom IX (Observation as protocol matching and reconstruction) □.
Observation is essentially the process of matching, decoding, and reconstructing protocols between the observer's protocol set L_O and the observed system's protocol set L_S. Observation changes the system's state distribution:

$$\boldsymbol{P_{post}(L) = \frac{P_{pre}(L) \cdot \exp[-\beta D(L \| L_O)]}{Z}}$$

where $D(L\|L_O)$ measures protocol difference, $\beta \propto E_{obs}$ is an observation-strength parameter, and Z is a normalization factor. Observation necessarily consumes the observer's $\Delta E(t)$ reserves and satisfies:

$$\boldsymbol{\Delta E_{obs} \geq k_B T \cdot D_{KL}(P_{post} \| P_{pre})}$$

For Categories B/C, observation involves only geometric effects and does not trigger internal protocol reconstruction.

Axiom X (Protocol media of interactions and the energy ledger) □.
Any non-geometric interaction must use a shared protocol subset $L_{shared} \subseteq L_1 \cap L_2$ (with $d(L_{shared}) \geq 1$) as the medium. Interaction depth

$$\boldsymbol{d_{int} = \max_{L \in L_1 \cap L_2} d(L)}$$

determines the type and strength of the interaction.

Interaction obeys the energy-ledger principle. It consumes the participants' ***active-state energy*** *E_{active} to negotiate and execute protocols, possibly converting part of E_{active} into new* ***solidified-state energy*** *E_{solid} (e.g., forming chemical bonds), or converting it into radiated energy. The process is constrained by the real-time supply capacity of the instantaneous net flow $e(t)$.*

Propagation of protocol media and causal influence of interactions are strictly constrained by the maximum speed c. Gravity is a purely geometric effect at $d = 0$ and does not require shared protocols. Dark matter (Category B) lacks any $d \geq 1$ protocol sets, and its active-state energy E_{active} is frozen; therefore, it has no interactions other than gravity.

Axiom XI (Cosmic energy conservation and form transformation) □. *1.* ***Total energy conservation.*** *Total cosmic energy E_{total} is conserved and is the sum of the three existence categories and radiation:*

$$\boldsymbol{E}_{total} = \boldsymbol{E}_{OM} + \boldsymbol{E}_{DM} + \boldsymbol{E}_{\boldsymbol{\Lambda}} + \boldsymbol{E}_{rad} = \textit{constant}$$

where:

- $E_{OM} = E_{solid}^{(OM)} + E_{active}^{(OM)}$ *is the total energy of ordinary matter (Category A)*
- E_{DM} *is the frozen active-state energy of dark matter (Category B)*
- E_{Λ} *is the uniformly frozen active-state background energy of dark energy (Category C)*
- E_{rad} *is radiation energy (CMB and other electromagnetic radiation)*

2. ***Energy-form transformation path.*** *Transformations among the three categories and radiation obey the strict serial path:*

(a) ***Matter → radiation.*** *Ordinary matter (Category A) converts part of its structured assets (including solidified-state energy E_{solid} and active-state energy E_{active}) into radiation via nuclear fusion, thermal emission, frictional dissipation, etc. This corresponds to a non-negative transfer term $Q_{m \to r} \geq 0$:*

$$\boldsymbol{\Delta E}_{OM} < \boldsymbol{0} \quad \Rightarrow \quad \boldsymbol{\Delta E}_{rad} > \boldsymbol{0}, \;\textit{and}\; |\boldsymbol{\Delta E}_{OM}| = \boldsymbol{\Delta E}_{rad}.$$

(b) ***Radiation → dark energy.*** *Under cosmic expansion, radiation loses energy because its wavelength is stretched (redshift). This lost energy does not vanish; it is uniformly "frozen" into background spacetime, becoming the injected component of dark energy (Category C), $E_{\Lambda}^{(inj)}$. Together with the initial frozen component at inflation termination $E_{\Lambda}^{(frozen)}$, it forms total dark energy $E_{\Lambda}^{(total)} = E_{\Lambda}^{(frozen)} + E_{\Lambda}^{(inj)}$.*

This is the only channel by which matter energy becomes dark energy. *There is no direct transition from ordinary matter or dark matter to dark energy of any kind:*

$$Q_{m \to \Lambda} \equiv 0.$$

(c) ***Inertia of dark matter.*** *Dark matter (Category B), as a primordial excited state permanently frozen during inflation, does not evolve after formation. It therefore participates in no non-gravitational energy transformation; its total energy is strictly conserved:*

$$\boldsymbol{\Delta E}_{DM} \equiv \boldsymbol{0}.$$

Dark matter produces only gravitational effects via its total structured asset $E_{locked} = E_{active}$.

3. ***Continuity equations and initial conditions.*** *At inflation termination t_0, component energy densities satisfy:*

$$\boldsymbol{\rho_r(t_0) = \rho_r^{(0)}, \quad \rho_m(t_0) = \rho_m^{(0)}, \quad \rho_{dm}(t_0) = \rho_{dm}^{(0)}, \quad \rho_\Lambda(t_0) = \rho_\Lambda^{(frozen)}}$$

Thereafter, the coupled evolution equations are:

$$\begin{aligned} \boldsymbol{\dot\rho_r + 4H\rho_r} &= \boldsymbol{Q_{m\to r} - Q_{r\to\Lambda}} \\ \boldsymbol{\dot\rho_m + 3H\rho_m} &= \boldsymbol{-Q_{m\to r}} \\ \boldsymbol{\dot\rho_{dm} + 3H\rho_{dm}} &= \boldsymbol{0} \\ \boldsymbol{\dot\rho_\Lambda} &= \boldsymbol{Q_{r\to\Lambda}} \end{aligned}$$

where:

- $Q_{m\to r} \geq 0$*: net transfer rate from matter to radiation*
- $Q_{r\to\Lambda} \approx 4H\rho_r$*: freezing rate from radiation redshift loss into dark energy*

The system satisfies total energy conservation:

$$\boldsymbol{\dot\rho_r + \dot\rho_m + \dot\rho_{dm} + \dot\rho_\Lambda + 3H(\rho_r + \rho_m + \rho_{dm} + p_r/c^2) = 0}$$

4. ***Nature of dark energy.*** *Category C (dark energy) is a uniformly frozen active-state background composed of two parts:*

   - ***Initial frozen part*** $E_\Lambda^{(frozen)}$*: sub-threshold fluctuation energy uniformly frozen at inflation termination; total constant*
   - ***Injected accumulated part***

     $$\boldsymbol{E_\Lambda^{(inj)}(t) = \int_{t_0}^{t} Q_{r\to\Lambda}(t')V(t')dt'}$$

     continuous injection from radiation redshift loss, causing total dark energy to grow with cosmic volume

   Dark-energy density

   $$\boldsymbol{\rho_\Lambda = \frac{E_\Lambda^{(frozen)} + E_\Lambda^{(inj)}(t)}{V(t)}}$$

   remains constant, i.e. $\dot\rho_\Lambda = 0$. This means total energy grows proportionally with volume; the growth is entirely supplied by redshift injection:

   $$\dot E_\Lambda = Q_{r\to\Lambda} \cdot V$$

   The observed increase of the dark-energy fraction Ω_Λ over time arises from dilution of matter and radiation densities, not from any increase in ρ_Λ.

5. ***Mapping of energy variables.*** *The correspondence between cosmological components and GET core variables is:*

Table 3: Mapping between cosmological energy components and GET core variables

Cosmological component	**GET representation**	**Physical meaning**
Ordinary-matter total energy E_{OM}	$E_{\mathrm{solid}}^{(\mathrm{OM})} + E_{\mathrm{active}}^{(\mathrm{OM})}$	Category A total structured assets, including solidified energy (e.g., baryonic rest energy) and active energy (e.g., nuclear fuel)
Dark-matter total energy E_{DM}	$E_{\mathrm{active}}^{(\mathrm{DM})}$ (frozen)	Category B frozen active-state energy, no solidified component
Dark-energy total energy E_{Λ}	$E_{\mathrm{background}} = E_{\mathrm{background}}^{(\mathrm{frozen})} + E_{\mathrm{background}}^{(\mathrm{inj})}$	Category C uniformly frozen active-state background, composed of initial freezing plus redshift injection
Radiation energy E_{rad}	E_{rad}	energy in photons and other radiation forms

Notes:

- *For Category A:* $E_{OM} = E_{solid}^{(OM)} + E_{active}^{(OM)}$, *and typically* $E_{solid}^{(OM)} \gg E_{active}^{(OM)}$ *(e.g.,* $\sim 99.93\%$ *of a star is solidified-state).*
- *For Category B:* $E_{DM} = E_{active}^{(DM)}$, *strictly conserved after formation.*
- *For Category C:* $E_{\Lambda} = E_{background}$, *with constant density* $\dot{\rho}_{\Lambda} = 0$.
- *Total conservation:* $E_{total} = E_{OM} + E_{DM} + E_{\Lambda} + E_{rad} = constant$.

6. ***Structured usable energy pool.*** *Define the universe's structured usable energy pool* $\mathcal{E}_{struct}(T)$ *as the total active-state energy and radiation energy usable for further structuring:*

$$\boldsymbol{\mathcal{E}_{struct}(T) = E_{active}^{(OM)}(T) + E_{rad}(T)}$$

where:

- $E_{active}^{(OM)}(T)$*: freely deployable active energy within ordinary matter*
- $E_{rad}(T)$*: radiation energy (uniformly distributed; not usable for local structuring)*

Because each successful protocol solidification (four-phase cycle) and each radiation redshift process permanently converts part of active-state energy into solidified energy or freezes it as dark energy, $\mathcal{E}_{struct}(T)$ *decreases monotonically:*

$$\boldsymbol{\frac{d\mathcal{E}_{struct}(T)}{dT} < 0}$$

7. ***Complete picture of cosmic energy circulation.*** *From inflation termination to*

the present, cosmic energy follows the irreversible circulation path:

$$\underbrace{\boldsymbol{E}_{peak}}_{\text{primordial borrowing}} \xrightarrow{\text{protocol solidification}} \begin{cases} \underbrace{\boldsymbol{E}^{(OM)}_{solid} + \boldsymbol{E}^{(OM)}_{active}}_{\text{ordinary matter (Category A)}} & \xrightarrow{\boldsymbol{Q}_{m\to r}} \boldsymbol{E}_{rad} \xrightarrow{\boldsymbol{Q}_{r\to\Lambda}} \boldsymbol{E}_{\boldsymbol{\Lambda}} \\ \underbrace{\boldsymbol{E}^{(DM)}_{frozen}}_{\text{dark matter (Category B)}} & \xrightarrow{\text{no conversion}} \text{gravity only} \\ \underbrace{\boldsymbol{E}_{background}}_{\text{dark energy (Category C)}} & \xrightarrow{\text{constant}} \text{drives accelerated expansion} \end{cases}$$

Throughout, total energy E_{total} is strictly conserved.

Axiom XII (Duality of entropy and the cosmological constant) $\square$. *The cosmological constant term Λ is linked to the total entropy S_{total} of the universe. Total entropy consists of absolute frozen entropy S_{abs} and relative active entropy S_{rel}. S_{abs} mainly comes from Category C (frozen, sub-threshold net borrowing energy—i.e., permanently solidified active-state energy) and the frozen part of Category B. S_{rel} comes from the active parts of Categories A and B and all flowing active-state energy E_{active} and radiation energy. Λ is the macroscopic manifestation of S_{abs} in Einstein's field equations:*

$$\Lambda(t) = \kappa \cdot [S^{(\text{total})}_{\text{abs}}(t) + S^{(\text{total})}_{\text{rel}}(t)]$$

where κ is a proportionality constant.

Axiom XIII (Maximum propagation speed and causal structure) $\square$.

1. ***Existence of a maximum speed.*** *There exists a finite maximum speed $c > 0$, an intrinsic property of the protocol-recursive system L, defining the absolute upper bound of propagation of protocol-state differences in state space Σ.*
2. ***Rigidity of causal structure.*** *All physical processes must obey causal-cone constraints. Causes must lie in the causal past of their effects, and effects must lie in the causal future of their causes. Formally, for a transition from state Ω' to Ω,*

$$s(\Omega', \Omega) \leq c \cdot |T' - T|.$$

3. ***Generative-engine explanation of quantum nonlocality.*** *Nonlocal correlations such as quantum entanglement are intrinsic properties of the difference function D as the* ***global state-generation engine*** *of L_0. When D acts on a composite system, its output state $\sigma_{t+1} = D(\sigma_t)$ naturally maintains strong correlations among spatially separated subsystems. Establishment and synchronization of these correlations are not achieved via signal propagation through spacetime, but via D's* ***one-shot, globally consistent generation*** *of the global state. Therefore, their apparent "instantaneity" does not violate the constraint set by c.*
4. ***Invariance.*** *All observers described by compatible protocol sets measure the same maximum speed c.*

5. ***Upper bound of energy transfer.*** *No form of energy transfer can exceed speed c.* ***Note:*** *the endogenous mechanism of c is given by Definition A1.h: it arises from the ratio of the maximum difference per step* $\max \|e(k)\|$ *produced by D to the minimal iteration time unit* t_P*, i.e.* $c = \max \|e(k)\|/t_P$.

3.5 Core Corollaries

Corollary 3.1 (Gravity as a geometric effect) $\square$.

Gravity is a geometric curvature effect on the background spacetime metric produced when an existent body has non-zero energy. The source of gravity is the total structured asset possessed by a form M*, i.e., the sum of solidified-state and active-state energy:*

$$\boldsymbol{E}_{locked} = \boldsymbol{E}_{solid} + \boldsymbol{E}_{active}.$$

Under the low-velocity approximation, this corresponds to the form's mass m *and satisfies*

$$\boldsymbol{E}_{locked} \approx \boldsymbol{mc^2}.$$

Mass m *is not an independent entity; it is the morphological manifestation of total energy* E_{locked} *after protocol solidification.*

For ***Category A (ordinary matter)*** *and* ***Category B (dark matter)****:* $E_{locked} > 0$*, producing standard gravitational effects (positive mass, gravitational attraction). Category A forms possess solidified protocols with* $d \geq 1$ *and can participate in non-geometric interactions; Category B forms lack such protocols and* ***only*** *produce gravity (Corollary 2).*

For ***Category C (dark energy)****: its uniformly frozen background energy density (uniformly distributed in the form of active-state energy) produces* ***negative pressure*** *and appears as the cosmological constant term* Λ *in Einstein's field equations, causing gravitational repulsion (accelerated expansion).*

Gravity itself requires no exchange of any solidified protocol medium with $d \geq 1$*; it is a purely geometric effect. Gravitational waves propagate at the speed of light c, because they are the propagation of disturbances of the protocol-connection pattern of spacetime.*

Corollary 3.2 (Strictness of zero non-gravitational interaction for dark matter) $\square$.

Category B (dark matter) is a frozen primordial excited state at $d = 0$*. Its complete state is*

$$\boldsymbol{\Phi}^{(0)}(\{\boldsymbol{L_0}\} \cup \boldsymbol{P}, \boldsymbol{E}^{i}_{frozen}, \boldsymbol{t})$$

permanently frozen (Definition B6). It lacks any solidified protocol sets with $d \geq 1$*, and its instantaneous net flow* $e(t)$ *and active-state energy* $E_{active}(t)$ *are frozen (cannot be used to pay the cost of protocol interaction). By Axiom X, it lacks the "solidified protocol identity"* *and "energy payment capability" required for non-gravitational interaction. Therefore, its non-gravitational interaction cross section is **strictly zero** in theory.*

Corollary 3.3 (Exactness of the dark-energy equation of state $w \equiv -1$) □.
Category C (dark energy) is, at the channel-closure moment, the set of fluctuations for which

$$\boldsymbol{E_i(\tau_{inflation}) < E_{th}^{(1)}}.$$

It has no internal structure, no solidified protocol sets, and no interaction channels; and its energy density remains constant during cosmic expansion (because radiation redshift energy is continuously converted so it does not dilute). Hence its equation-of-state parameter

$$w = p/\rho \equiv -1$$

holds exactly.

Corollary 3.4 (Statistical necessity and non-fine-tuning of cosmic component ratios) □.
The CMB-observed ratio

$$\boldsymbol{\Omega_{OM} : \Omega_{DM} : \Omega_\Lambda \approx 4.9 : 26.5 : 68.5}$$

is a natural result under the quantum fluctuation statistical framework, determined by the following parameters: the power-law exponent α of peak borrowed energy, the energy threshold $E_{th}^{(1)}$, the minimum time window $N_{min}^{(1)}$, and the distribution of cumulative net borrowing energy $\Delta E(\tau_{inflation})$ at $T_0 + \tau_{inflation}$. No fine-tuning is required.

The total energy of ordinary matter (Category A) corresponds to its total structured asset

$$\boldsymbol{E_{locked}^{(A)} = E_{solid}^{(A)} + E_{active}^{(A)}}.$$

The total energy of dark matter (Category B) corresponds to its peak borrowed energy at freezing $E_{peak}^{(B)}$ (entirely active-state). The total energy of dark energy (Category C) corresponds to the sum of net borrowing energies of all individuals that did not reach threshold. The observed dark-energy density includes three sources: primordial freezing, radiation redshift conversion, and matter thermal-radiation conversion.

Corollary 3.5 (The CMB as the latent-heat relic of the first phase transition) □.
The CMB is the collectively redshifted remnant of the radiated energy (Axiom VI) synchronously released by the universe's first large-scale ordering event—i.e., the protocol solidification execution by which individuals transition from the primordial excited form $M_{excited}$ to the ordinary-matter form M_{OM}. Its near-perfect blackbody spectrum confirms the universality (shared average morphologization efficiency $\bar{\eta}$) and the statistical nature of this process.

Corollary 3.6 (Protocol-symmetry origin of photon spin) $\square$.
Photons radiated in individual protocol solidification events have intrinsic spin ± 1, *originating from the specific mapping and symmetry-breaking choice by candidate protocol sets* $P \in \mathscr{P}_1$ *applied to the winding number* $n_w = \pm 1$ *of baseline fluctuation paths under* L_0 *(Definitions A1 and B1).*

In the baseline state L_0, *left-handed and right-handed are perfectly symmetric and equally probable. When inflation-trigger activation of* $\mathscr{P}_1$ *occurs in the first structured phase transition, this symmetry is broken; left/right are mapped into gauge symmetry (ultimately solidified as U(1) gauge symmetry) and appear in the radiation field as photon spin* ± 1.

Thus, photon spin is the direct embodiment of an intrinsic geometric property of L_0 *in the first structuring process.*

Corollary 3.7 (Continuous heat loss and energy exchange of ordinary matter) $\square$.
All ordinary matter (Category A), in protocol reconstruction processes (Axiom X), follows the energy-ledger principle: protocol negotiation necessarily consumes energy $E_{negotiation} > 0$, *which becomes heat and is an irreversible source of entropy increase.*

Corollary 3.8 (Dynamical mechanism for the growth of the dark-energy fraction) $\square$.
Category C (dark energy, $d(M_\Lambda) = 0$*) is a uniformly frozen active-state background with strictly constant density* ρ_Λ *(*$\dot{\rho}_\Lambda = 0$*), and equation of state* $w_\Lambda = -1$ *holds exactly.*

Total dark-energy content

$$E_\Lambda = \rho_\Lambda V$$

grows with cosmic volume, with growth supplied by radiation redshift injection:

$$\dot{E}_\Lambda = Q_{r \to \Lambda} \cdot V.$$

The observed fraction

$$\boldsymbol{\Omega_\Lambda = \rho_\Lambda / \rho_{total}}$$

increases over time entirely due to dilution of matter and radiation densities ($\rho_m \propto a^{-3}$, $\rho_r \propto a^{-4}$*), not because* ρ_Λ *increases.*

The process strictly follows the serial chain "matter $\to$ *radiation* $\to$ *redshift freezing* $\to$ *dark-energy background," with no assumption of internal dark-energy dynamics.*

Corollary 3.9 (Structuring direction theorem: the principle of minimal net dissipation rate) $\square$.
Under the layered evolution equations $\partial P_d / \partial T = \mathcal{H}_d[\dots]$, *the structuring evolution of a*

local system tends to choose the form and protocol path that minimize the system's net energy dissipation rate. This provides a directional criterion for the formation and evolution of complex structures (e.g., organisms, stellar systems, civilizational organizations).

Corollary 3.10 (Three stages of the universe's overall evolution) □. *1. **Protocol capture and freezing phase** ($T < T_0 + \tau_{inflation}$): symmetry-breaking inflation, geometric-selection competition, the first protocol solidification executions trigger a macroscopic phase transition, and the three categories (A, B, C) are frozen in a single shot.*

*2. **Recursive growth and structuring phase** ($T_0 + \tau_{inflation} < T < T_{peak}$): Category A matter clusters via gravity, accumulates energy to reach new thresholds, repeatedly obtains qualified windows via geometric selection, executes four-phase cycles to solidify deeper protocols, and forms complex structures from atoms and molecules to stars, galaxies, and life. This phase features both **upward creation** (new depths and new species) and **within-layer differentiation** (species radiation within the same depth), manifested as increasing recursion depth d and richer activation of each layer's P_d.*

*3. **Dissipative return phase** ($T > T_{peak}$): black holes become dominant, and via Hawking radiation slowly convert all E_{locked} into low-energy photons; the universe ultimately returns to an approximate heat-death state. Yet even in this terminal state, the third law of thermodynamics remains valid—absolute zero is unattainable, and L_0 still maintains a minimum level of difference activity.*

Corollary 3.11 (Evolution of the efficiency factor and the cosmic structuring process) □.

The evolution of an individual's efficiency factor

$$\boldsymbol{\eta(t) = \frac{E_{locked}(t)}{E_{peak}}}$$

fully characterizes the cosmic structuring process:

- ***At inflation termination:** η first solidifies and locks the energy of the quantum-fluctuation state (structureless; all energy borrowed) as $\eta_{inflation}$. At this moment, most energy $(1-\eta_{inflation})$ is released as radiation. This is the **largest-scale energy locking** in cosmic history.*
- ***Increasing phase** (after inflation termination to before star ignition): part of radiation energy is reused by baryon synthesis, atom synthesis, etc., and locked into new structures, causing η to **increase continuously** and reach a **global maximum** before star ignition.*
- ***Decreasing phase** (after star ignition): stars begin burning, continuously converting locked energy into radiation. Because the mass–energy conversion efficiency of nuclear fusion is extremely low (on the order of $\sim 0.07\%$), and because the vast majority of*

*radiation energy is frozen into dark energy via redshift under cosmic expansion, η enters an **extremely slow monotonic decrease**.*

- ***Disintegration threshold:*** *when the radiation-energy release rate falls below the rate at which it is frozen into dark energy, it means "no energy can be ordered and utilized"; structure enters a functional disintegration phase, and $\eta \to 0$.*

Relation to the natural constant e***:*** *in the extreme-iteration limit, energy accumulation approaches the continuous-compounding limit*

$$\boldsymbol{E_{locked}(t) \approx E_0 \times e^{rt}}$$

(Definition A1), where e is the necessary emergent constant of discrete iteration D in the continuous limit. This reveals the intrinsic unity between the evolution of $\eta(t)$ and the mathematical constant e.

Therefore, the full trajectory of $\eta(t)$ is: ***locking → increasing to a peak → monotonic decrease to zero****.*

The cosmological mean efficiency factor $\bar{\eta}(T)$ follows the same rule. The current observed value

$$\boldsymbol{\bar{\eta}(T_{now}) \approx 0.1045}$$

indicates the universe is in the post–star-ignition monotonic decrease phase.

Corollary 3.12 (Existence and computability of a maximum protocol depth $D_{\max}$) $\square$.

Constrained by the finiteness and monotonic decrease of the universe's structured usable energy pool $\mathcal{E}_{struct}(T)$, any individual that can be naturally produced and stably exist through recursive structuring (the four-phase cycle) has a theoretical upper bound on protocol recursion depth d, denoted $D_{max}(T)$, determined by the product of the following constraints:

$$\boldsymbol{D_{max} = f\left(\mathcal{E}_{struct}, \{\eta_i\}, H(t), c, \{N^{(d)}_{min}\}, \kappa_{crit}, r_0, \alpha, \textit{history path}, \dots\right)}$$

including at least:

- ***Total energy constraint:*** $\mathcal{E}_{struct}(T) = E^{(OM)}_{active}(T) + E_{rad}(T)$
- ***Energy-density constraint:*** *dilution by cosmic expansion ($\rho \propto a^{-3}$ or a^{-4})*
- ***Capture-efficiency constraint:*** *energy conversion efficiencies η at each layer*
- ***Transmission-efficiency constraint:*** *distance attenuation of effective investment energy*

$$\boldsymbol{E^{eff}_{invest} = \frac{E_{invest}}{1 + (r/r_0)^2}}$$

- ***Matching-efficiency constraint:*** *protocol compatibility must satisfy $\kappa \geq \kappa_{crit}$*
- ***Solidification-efficiency constraint:*** *each solidification necessarily radiates $(1 - \eta)E_{peak}$*

- ***Time-window constraint:*** *upgrading requires the minimum duration* $N_{min}^{(d+1)}$
- ***Light-speed constraint:*** *cross-scale coordination time must satisfy* $c{\cdot}\Delta t \geq$ *system scale*
- ***Historical-path constraint:*** *restrictions imposed by already solidified protocols on subsequent choices*

Each constraint is a multiplicative factor; together they determine the maximum recursion depth a system can stably maintain in a given spacetime region.

Thus D_{max} *is* ***local, updatable, and empirically testable****:*

- ***Local:*** *different regions (e.g., Earth vs. interstellar space) may have different* D_{max} *due to different parameters*
- ***Updatable:*** *discovery of new energy sources or efficiency breakthroughs allows recomputation of* D_{max}
- ***Testable:*** *if one finds a system stably at* $d > D_{max}^{(calc)}$*, either the calculation is wrong or the theory is falsified*

Corollary 3.13 (The absolute limit of observation arises from protocol and energy constraints) □.

An observer can stably observe only forms M *whose recursion depth is close to or lower than the observer's own:*

$$\boldsymbol{d(M) \leq d(L_{observer})}.$$

This is because observation is, in essence, matching the observer's protocol set L_O *to the potential paths of the observed system's core protocol fingerprint* $C_{\mathcal{S}}$*, and the matching cost grows exponentially with recursion-depth difference. Observation itself consumes the observer's* $\Delta E(t)$ *reserves (Axiom IX), hence there is a fundamental energy constraint.*

Corollary 3.14 (Dissolution of the "quantum gravity" dilemma) □.

Gravity is explicitly defined as a classical geometric effect at $d = 0$ *(Corollary 1), while quantum phenomena are associated with the behavior of solidified protocol sets at layers* $d \geq 1$ *(Axiom X).*

By Definition A1 and Axiom III, the Heisenberg uncertainty relation

$$\boldsymbol{E_{peak} \cdot \tau \geq \hbar/2}$$

is a special case of the existence-degree criterion's scale constraint at the $d = 0$ *layer (the quantum-fluctuation form* M_{fluct}*), rather than the entirety of bottom-level physics. Forcing gravity (*$d = 0$ *geometric effect) and quantum phenomena (*$d \geq 1$ *protocol behavior) into a single "quantum gravity" framework originates from the error of conflating phenomena that belong to different protocol layers and have different ontological foundations.*

Quantizing gravity is unnecessary and may be a conceptual misdirection. The two naturally coordinate under L_0*'s* ***unified state management and causal-consistency algorithm****.*

Corollary 3.15 (Expected similarity within a multiverse family) $\square$.
If there exists a multiverse family and each universe originates from the same true vacuum and fluctuation statistics (sharing the same L_0 and α; with energy levels and windows determined by thresholds), then the fundamental physical laws evolved (i.e., the core protocol sets in $\mathscr{L}_1$) and cosmological parameters (such as $\bar{\eta}$) will exhibit high similarity. Significantly different universes will be rare.

Corollary 3.16 (A protocol-theoretic interpretation of life and consciousness) $\square$.
Life is the product of multiple successful realizations of the four-phase cycle leading to higher recursion depth (a life form M_{life} has $\mathrm{d}(M_{life}) \gg 1$). Its core is a complex protocol set L that can efficiently manage energy reserves, continually obtain qualified windows, and trial deeper candidate protocol families $\mathscr{P}_{d+1}$ to achieve further recursion-depth growth.

Consciousness is the emergent ability of such high-dimensional life forms to perform recursive operations (self-reference) on their own states and protocols, corresponding to higher recursion depth.

Corollary 3.17 (Common origin of material chirality and photon spin) $\square$.
Photon spin (Corollary 6) and fermion chirality share a common origin: in the first protocol solidification events, candidate protocol sets $P \in \mathscr{P}_1$ map the baseline fluctuation-path winding number $n_w = \pm 1$ (Definition B1) and make a symmetry-breaking selection.

In the baseline state L_0, left-handed and right-handed are perfectly symmetric with no underlying bias. In the first structuring phase transition, this symmetry is broken, and the two directional separations are mapped into:

- *gauge-field spin (photon ± 1)*
- *fermion chirality (left-handed/right-handed particles)*

They are manifestations of the same higher-dimensional protocol instruction across different degrees of freedom, sharing a common origin in L_0's intrinsic geometric–topological properties.

Corollary 3.18 (Statistical expectation for the ordinary-matter–radiation energy ratio) $\square$.
At inflation termination, the expected ratio between ordinary matter's total rest energy (Category A total locked energy E_{OM}) and the primordial radiation total energy (the initial CMB energy $E_{CMB}(t_{struct})$) is approximately

$$\frac{\bar{\eta}}{1-\bar{\eta}}.$$

Corollary 3.19 (Origin of the universe's net rotation / angular momentum) $\square$. *1.*
Microscopic origin: *a single primordial phase-transition event (Definition B5)*

produces particles and photons carrying spin and momentum, but the total angular momentum of the event is zero.

2. ***Macroscopic origin:*** *in the early universe, large-scale statistical anisotropy of the CMB photon field (arising from non-spherical primordial density perturbations) applies a net torque to primordial matter density perturbations via radiation pressure.*
3. ***Energy conversion:*** *this torque irreversibly converts part of the collective linear momentum of CMB photons (radiation energy) into the collective orbital angular momentum of matter (baryons and dark matter).*

Corollary 3.20 (A sociological correspondence of connection capital) $\square$.
Social trust, legal institutions, cultural traditions, and other "social capital" are high-recursion-depth protocol sets ($d \gg 1$) and follow the accumulation and depreciation dynamics of Definition D4. They cannot be built rapidly; they require continuous energy capture (social resource investment) and depreciate when energy input is insufficient.

Corollary 3.21 (Multi-scale manifestations of toxic protocols) $\square$.
Radioactive nuclides (physics), pathogens (biology), computer viruses (information), cultural dross (society), etc., are toxic protocols manifested at different recursion depths d. They require corresponding-layer isolation mechanisms (physical shielding, immune systems, firewalls, cultural critique).

Corollary 3.22 (Ultimate falsifiability criteria of GET) $\square$.
Any one of the following definitive experimental or observational findings would directly falsify GET's core architecture:

1. ***Cornerstones overturned:*** *any of the five empirical cornerstones is negated.*
2. ***Description failure:*** *discovery of a stable natural existence (form M) whose properties cannot be fully described by a state function of the form $\Omega^{(M)} = \Phi(L)$, together with the existence-degree formula composed of E_{solid}, E_{active}, and $e(t)$; or whose survival violates*
$$\frac{\boldsymbol{E}_{active}(\boldsymbol{M},\boldsymbol{t})}{\boldsymbol{\tau}_{active}(\boldsymbol{L})} + \boldsymbol{\eta}(\boldsymbol{L}) \cdot \boldsymbol{e}(\boldsymbol{t}) \geq \mathbf{0}.$$
3. ***Dark matter interaction:*** *confirmation that dark matter (Category B) participates in any non-geometric direct interaction (violating Corollary 2).*
4. ***Dark energy dynamics:*** *definitive confirmation that dark energy's equation-of-state parameter w persistently and significantly deviates from -1 (violating Corollary 3).*
5. ***Superluminal causal transfer:*** *confirmation of any causal energy or information transfer that* **does not rely on pre-shared protocols**, *whose speed persistently and significantly exceeds c. Apparent nonlocal correlations are explained by L_0's global memory-consistency writing and do not violate Axiom XIII.*

4 A Strong, Falsifiable Prediction System: GET as the Ultimate Test of the Grammar of Existence

4.1 Overview: The Theoretical Basis and Grading Principles of Strong Predictions

The value of a theory lies not only in the depth with which it explains known phenomena, but also in the ability—and courage—to predict unknown phenomena. Based on its **zero–ad hoc deduction** from the five cornerstones, Generalized Existence Theory (GET) logically and necessarily yields a series of sharply testable predictions. These predictions constitute a complete experimental protocol for the "ultimate test" of the GET edifice.

According to the nature of the predictions and how they are tested, we divide them into four levels:

- **Level 1: Logical tests of the existence-grammar** — tests internal consistency and completeness; a **necessary condition** for the theory to stand.
- **Level 2: Strong predictions in fundamental physics** — decisive, "all-or-nothing" experimental predictions addressing core puzzles (dark matter, dark energy, the nature of gravity); the **core test** of the physical picture.
- **Level 3: Cross-scale derivative predictions** — application-level corollaries in cosmology and complex systems, providing **auxiliary evidence**.
- **Level 4: Ultimate comprehensive falsification** — an overall evaluation of the theory's explanatory power, parsimony, and productivity.

All predictions of GET—especially those of Level 2—are **binary judgements**: no parameter-tuning space and no retreat by post hoc adjustment. The theory is either upheld or falsified. This is a full execution of Popperian falsifiability and a return to Einstein's ideal of clear, decidable tests.

4.2 Level 1: Logical Tests of the Existence-Grammar (Internal Consistency)

If any condition at this level is violated, GET collapses logically at once, with no need to wait for external experiments.

Prediction 4.1 (Completeness test of the existence-grammar) □.
Any stable existence (form M) has its complete state fully described by the state function $\Omega^{(M)} = \Phi(L)$ (Definition A1). This function takes the protocol set L as the only input; its internal execution necessarily yields the corresponding stable form M and computes all intrinsic properties and instantaneous states of M, including $E_{solid}(t)$, $E_{active}(t)$, and

instantaneous net flow $e(t)$*. Continued existence of* M *must satisfy the survival criterion* $\frac{E_{active}(M,t)}{\tau_{active}(L)} + \eta(L) \cdot e(t) \geq 0$ *(Axiom III).*

Falsification condition: *discover a clearly stable entity that cannot be completely described by this state function, or whose existence necessarily implies* $\frac{E_{active}(M,t)}{\tau_{active}(L)} + \eta(L) \cdot e(t) < 0$.

Logical root: *violates Definition A1 and Axiom I (protocol-recursive monism), Axiom III (existence criterion).*

Test method: *conceptual analysis and algorithmic exhaustive audit.*

Prediction 4.2 (Equivalence test of the cornerstones) □.
The GET theoretical system and the five physical cornerstones are logically equivalent. GET is a necessary corollary of the five cornerstones; conversely, the five cornerstones can be derived from GET. The ***geometric selection principle*** *(Definition B2) is a direct logical corollary of Cornerstone 1 (Heisenberg constraint) and Cornerstone 2 (power-law statistics) under the peaked energy-curve assumption; it is not an independent ad hoc assumption.*

Falsification condition: *with all five cornerstones holding, GET is falsified; or GET holds while any cornerstone is falsified.*

Logical root: *violates GET's methodological rule that axioms must be empirical.*

Test method: *formal logic verification and mathematical consistency proofs.*

Prediction 4.3 (Logical test of protocol stratification) □.
Recursion depth d *is a discrete integer. Existence at* $d = 0$ *(Categories B and C) lacks any structured, directional non-geometric interaction capability; existence at* $d \geq 1$ *(Category A and its composites) gains non-geometric interaction qualification via solidified protocols (Axiom X, Corollary 2). In energy terms:* $d = 0$ *individuals have only active-state energy* E_{active} *(and must return to zero over a lifecycle), while* $d \geq 1$ *individuals possess solidified-state energy* E_{solid} *as structured assets.*

Falsification condition: *find a* $d = 0$ *existence that possesses structured interaction capability; or a* $d \geq 1$ *existence that cannot perform structured interaction.*

Logical root: *violates Axiom* Ω *(baseline system), Definition B6 (three categories), and Axiom X (interaction via protocol media).*

Test method: *protocol identification and interaction experiments.*

Prediction 4.4 (Logical necessity test of the geometric selection principle) □.
The "energy threshold $\epsilon_{th}^{(d+1)}$*" and the "minimum time window* $N_{min}^{(d+1)}$*" required for recursive upgrade are not two independent, parallel parameters. Upgrade permission is a*

geometric fact*: the individual's net-borrowing energy curve $E(t)$ (for $d = 0$ individuals, $E(t) = E_{active}(t)$) is a peaked curve; drawing the horizontal line $E = \epsilon_{th}^{(d+1)}$ yields two intersections whose separation $\Delta t = t_{end} - t_{start}$ is the window length. Only when $\Delta t \geq N_{min}^{(d+1)}$ does the individual obtain upgrade permission. Within the qualified window W, the existence degree $\Psi_M(t) = \frac{E_{solid}}{\tau_{solid}} + \frac{E_{active}}{\tau_{active}} + \eta \cdot e(t)$ necessarily stays high (for $d = 0$, the first term is zero; $E(t)$ and $e(t)$ jointly ensure $\Psi_M(t) \gg 0$). The emergent $\Psi_{invest}^{(d+1)}$ is therefore a logical corollary of $(\epsilon_{th}^{(d+1)}, N_{min}^{(d+1)})$, not an independent parameter (Definition B2, Axiom II).*

Falsification condition: *find any recursive-upgrade event that does not follow this geometric selection condition; or find that within a qualified window $\Psi_M(t)$ fails to maintain a significantly positive value.*

Logical root: *violates Definition B2, Axiom II, Axiom III.*

Test method: *analyze "upgrade/phase-transition/breakthrough" events at any scale (from particle physics to social systems) and check whether their energy–time histories satisfy this geometric constraint.*

Risk assessment: *if any prediction at this level is falsified, GET fails logically immediately.* ***Risk level: fatal.***

—

4.3 Level 2: Strong Predictions in Fundamental Physics (Decisive Experiments)

Predictions at this level stand in sharp opposition to mainstream physical pictures and have an explicit "duel" character. Any falsification constitutes a fatal blow to GET's physical core.

Prediction 4.5 (Absolute zero non-gravitational interaction of dark matter) □. *Dark matter (Category B) is a $d = 0$ frozen primordial excited state, with complete state $\Phi_{DM}(\{L_0\} \cup P, E_{frozen}^i, t)$ permanently frozen (Definition B6, Axiom X). It lacks any solidified protocol set with $d \geq 1$; its active-state energy E_{active} is frozen ($e(t) \equiv 0$, unable to pay protocol interaction costs); and $E_{solid} \equiv 0$.*

Strong prediction: *all non-gravitational interaction cross sections between dark matter and ordinary matter are* ***strictly zero****. Any experiment that claims a non-gravitational dark-matter interaction, if not an error, directly falsifies GET.*

Key experiments and time windows:

- *XENONnT / LZ end-stage data (2025–2028): WIMP cross-section sensitivity $\sim 10^{-48}\ cm^2$.*

- *DARWIN operation (2030–2035): ultimate sensitivity* $\sim 10^{-49}$ *cm^2, delivering the final verdict.*

Quantitative expectation: *all direct-detection signals remain below statistical background fluctuations (below* 5σ*).*

Prediction 4.6 (No graviton; null results for quantizing gravity) $\square$.
Gravity is a pure geometric effect at $d = 0$*: when a form* M *has non-zero total structured asset* E_{locked}*, it geometrically curves the background metric. Gravity exchanges no solidified protocol medium with* $d \geq 1$ *(Corollary 1).*

Strong prediction:

(a) gravitons do not exist;

(b) all experiments aimed at detecting quantum-gravity effects or spacetime discreteness (tabletop experiments, high-precision interferometry of gravitational waves) yield ***null results****;*

(c) at the Planck scale, gravity behaves as classical geometric noise rather than a quantized structure.

Key experiments: *tabletop quantum-gravity experiments (2030–2040), next-generation gravitational-wave observatories (e.g., LISA, Einstein Telescope).*

Prediction 4.7 (Exact assertion $w \equiv -1$ for dark energy) $\square$.
Dark energy (Category C) is: at the channel-closure moment $T = T_0 + \tau_{inflation}$*, the set of fluctuations whose cumulative net borrowing energy satisfies* $E_i(\tau_{inflation}) < E_{th}^{(1)}$*. Its state is* $\Phi_\Lambda(\{L_0\}, E_{background}, t)$*, where* $E_{background}$ *is uniformly frozen active-state energy (non-zero, consistent with the third law of thermodynamics). It has no internal structure, no solidified protocol sets, and no interaction channels; and its energy density remains constant under cosmic expansion (non-diluting) (Definition B6, Corollary 3).*

Strong prediction: *the dark-energy equation-of-state parameter* $w = p/\rho \equiv -1$ *holds exactly, and dark energy is strictly uniform and isotropic, with no dynamical evolution features.*

Key experiments and time windows:

- *DESI and Euclid full datasets (2028–2035): precision* ± 0.01 *on* w*.*
- *Roman Space Telescope (2030–2040): precision* ± 0.005*.*

Falsification condition: *confirm* $w \neq -1$ *at* 5σ*; or detect local structure / inhomogeneity in dark energy.*

Prediction 4.8 (An algebraic relation among cosmic components) $\square$.
The ratios of ordinary matter (Ω_{OM}*), dark matter (*Ω_{DM}*), and dark energy (*Ω_Λ*) satisfy a*

fixed algebraic relation determined by the mean morphologization efficiency $\bar{\eta}$ (Definition C5, Corollary 4):

$$\boldsymbol{\Xi} = \frac{\boldsymbol{\Omega_\Lambda - \Omega_\Lambda^{(post\text{-}inflation)}}}{\boldsymbol{\Omega_{OM}}} = \frac{\boldsymbol{1-\bar{\eta}}}{\boldsymbol{\bar{\eta}}},$$

where $\bar{\eta} := E^{(A)}_{total\ locked}/E^{(A)}_{total\ peak}$ is the cosmological mean morphologization efficiency, and $\Omega_\Lambda^{(post\text{-}inflation)}$ is the dark-energy fraction accumulated after inflation via radiation redshift.

Strong prediction: *Ξ is a logically necessary solution, not a tunable parameter. Using current observations $\Omega_{OM} : \Omega_{DM} : \Omega_\Lambda \approx 4.9 : 26.5 : 68.5$, after accounting for the radiation-redshift contribution, one obtains $\bar{\eta} \approx 0.1045$ and $\Xi \approx 8.57$. Future observations must remain within 8.57 ± 0.1.*

Key data: *CMB-S4 (2027–2032, precision ± 0.05); 30-meter-class telescope cosmology (2040–2050, precision ± 0.02).*

Prediction 4.9 (Anisotropy of the Hubble constant) □.
During inflationary structuring and freezing (Definition B6), there is primordial directional asymmetry arising from statistical fluctuations in the spatial distribution of the first fluctuations that satisfy geometric selection.

Strong prediction: *the Hubble constant H_0 has a large-scale anisotropy of ~ 1–2%, and the pattern correlates with the distribution of large-scale structure (e.g., superclusters and voids).*

Quantitative expectation: *$\Delta H_0/H_0 \approx (1.5 \pm 0.5)\%$, exhibiting a dipole or quadrupole pattern.*

Key observations: *LSST all-sky survey (2026–2030), Roman deep-field multi-direction observations.*

Prediction 4.10 (Inflation as a phase-transition origin) □.
GET mechanism: *inflation is triggered by the first batch of fluctuations that satisfy geometric selection (Definition B2: the peaked curve $E(t)$ intersects $E^{(1)}_{th}$ with $\Delta t \geq N^{(1)}_{min}$), which collectively activate and solidify protocols; in essence, it is a first-order phase transition (Definition B4, Axiom IV).*

Strong prediction: *CMB observations—especially non-Gaussianity f_{NL} and the primordial gravitational-wave spectrum r—should contain characteristic signals of a first-order phase transition, distinguishable from the predictions of slow-roll inflation models. These signals (e.g., specific non-Gaussian shapes) arise from the formation of the cosmic thermal bath as the product of the first structuring phase transition; the thermal bath itself is the relic of that phase transition, and its statistics encode the phase-transition dynamics.*

Key experiments: *CMB-S4, LiteBIRD, and other next-generation CMB polarization experiments (2027–2035).*

Risk assessment: falsifying any assertion at this level directly overturns GET's physical core. **Risk level: extremely high.**

—

4.4 Level 3: Cross-Scale Derivative Predictions (Auxiliary Evidence)

Prediction 4.11 (Existence of a maximum protocol depth) □.
Constrained by the finiteness and monotonic decrease of the structured usable energy pool $\mathcal{E}_{struct}(T) = \sum E_{active}$ (Axiom XI.4, Corollary 12).

Strong prediction:

- *The maximum protocol depth D_{max} of naturally produced structures in the observable universe is finite, determined by the universe's initial total usable energy, causal-region energy limits, and dilution from cosmic expansion.*
- *Given current usable energy (global active-state energy reserves), human civilization has a theoretical upper bound on technological complexity $D_{max}^{(\mathrm{human})}$.* ***The technological singularity has a theoretical upper limit; it cannot be broken indefinitely by unbounded exponential growth.***

Prediction 4.12 (Exponential growth of the energy threshold for upward transitions) □.
Increasing recursion depth $d \to d+1$ requires satisfying geometric selection; the energy threshold $\epsilon_{th}^{(d+1)}$ and minimum time window $N_{min}^{(d+1)}$ jointly determine the minimum energy investment $E_{invest} = \epsilon_{th}^{(d+1)} \cdot N_{min}^{(d+1)}$ (Definition D4).

Strong prediction: *the required successful-upgrade energy investment grows exponentially with depth:*

$$\boldsymbol{E_{invest}^{(d)} \propto \exp(\alpha d)}$$

where $\alpha > 0$ is an empirical parameter to be determined by cross-scale statistical analysis.

Test: *cross-scale statistics of energy costs (from particle reactions to civilizational technological breakthroughs) should yield a consistent α value across different depth transitions, providing a quantitative test of the exponential growth hypothesis.*

Prediction 4.13 (Constancy of the baryon-to-photon ratio) □.
The baryon-to-photon ratio $\eta \sim 6 \times 10^{-10}$ is determined by intrinsic energy-level threshold ratios within the universe's foundational protocol set $L \in \mathscr{L}_1$, originating from the allocation ratio between total structured assets E_{locked} and radiated energy E_{rad} in the first solidification events (Definition B5, Corollary 5).

Strong prediction: *η is strictly invariant across the observable universe, with no significant spatial variation.*

Key observations: *JWST and next-generation telescopes measuring primordial abundances at extremely high redshift or across different sky regions (2035–2045).*

Prediction 4.14 (Self-consistency of the cosmic energy audit) □.
Total cosmic energy is conserved; total energy is composed of each form's solidified-state energy E_{solid}, active-state energy E_{active}, and radiation energy E_{rad}; components evolve by equations of state and transform strictly along the serial chain "matter → radiation → dark energy" (Axiom XI).

Strong prediction: *by precise measurements of component energy densities versus redshift z, one can verify that*

$$\boldsymbol{E}_{total} = \sum(\boldsymbol{E}_{solid} + \boldsymbol{E}_{active}) + \boldsymbol{E}_{rad} + \boldsymbol{E}_{\boldsymbol{\Lambda}} = constant$$

holds within errors; and that the dark-energy density injection rate matches the radiation redshift-loss rate ($\dot{\rho}_\Lambda = Q_{r\to\Lambda}$), with no direct matter → dark-energy conversion channel.

Prediction 4.15 (Multiverse similarity) □.
Any universes sharing the same L_0 and fluctuation statistics (same power-law exponent α and threshold parameters) will have highly similar foundational laws (high similarity of $\mathscr{L}_1$'s core protocol sets) (Corollary 15).

Strong prediction: *universes capable of producing structure have foundational physical laws and constants that are highly similar (99.9%); significantly different universes are extremely rare.*

Prediction 4.16 (A protocol explanation of the S-curve of technological revolutions) □.
Technology diffusion is a function of the compatibility κ between a new technology protocol set L_{new} and the existing social protocol set L_{social}. The timing of a technological breakthrough corresponds to the first moment when the social system's accumulated "energy" (capital, talent, knowledge reserves) curve intersects a threshold and satisfies the geometric selection window condition.

Strong prediction: *models based on GET dynamics (geometric selection, compatibility matching, and connection-capital accumulation) can predict technological breakthrough timing and diffusion curves more accurately than traditional logistic models.*

Risk assessment: falsifying multiple assertions at this level severely weakens GET's explanatory and application value. **Risk level: high.**

—

4.5 Level 4: Ultimate Comprehensive Falsification (Overall Evaluation)

Prediction 4.17 (Falsification of the unified narrative) □.
Cosmic evolution is a single continuous causal chain: "baseline system L_0 running →

concurrent quantum-fluctuation ocean → geometric selection → protocol solidification → inflation[24, 25, 26] and channel closure → one-shot freezing of the three categories → recursive depth growth and within-layer differentiation → continuous conversion of matter energy via radiation → dark energy → dissipative return" (Corollary 10, Axiom XI).

Falsification condition: *definitive evidence shows that the inflation mechanism, the thermal origin of the CMB, the essence of dark matter[17, 18, 19, 20] / dark energy[21, 22, 23], and the cosmic component ratios have* ***no physical linkage*** *and are independent events; or a direct energy-conversion channel violating the serial chain "matter → radiation → dark energy" is found.*

Prediction 4.18 (Counter-evidence via explanatory power and productivity) □.

GET can efficiently generate systematic, non-ad hoc explanations and testable predictions for cross-scale phenomena.

Falsification condition: *in practice it is shown that (a) it cannot generate any insight beyond existing theories; (b) it is beaten in all application domains by simpler and more accurate ad hoc models; (c) it cannot provide operable diagnostic or predictive tools.*

Prediction 4.19 (Loss of the parsimony advantage) □.

GET is superior to existing theories in variable count (only L is ontology; $E_{solid}, E_{active}, e(t)$ are endogenous outputs of D with no external inputs), degree of ad hocness (zero; geometric selection is logically necessary), cross-scale unity (from $d = 0$ fluctuations to $d \gg 1$ cultural traditions), and falsifiability strength.

Falsification condition: *a competing theory appears that,* ***while retaining all GET's successful predictions****, is fully superior to or equal to GET on those four meta-scales.*

Prediction 4.20 (Proof of redundancy of the protocol grammar) □.

The concept of the core protocol set L as the only dynamical ontology is necessary and sufficient.

Falsification condition: *strict proof that the concept of L is redundant, and all GET explanations and predictions can be obtained more simply without introducing L.*

Risk assessment: falsification at this level means GET fails as a competitive scientific theory as a whole. **Risk level: fatal.**

—

4.6 Meta-Commitments of Theoretical Quality

The four-level falsification scheme above provides concrete test conditions for GET. But before that, a more fundamental question must be asked: **why is this theory worth**

testing at all?

The answer lies in its construction method itself. The table below summarizes GET's core meta-level commitments and their guarantee mechanisms:

Table 4: GET's meta-commitments and guarantee mechanisms

Commitment	**Guarantee mechanism**
It is logically impossible for causal entities to exist outside Closure(L)	Closure definition: all inputs are strictly limited to the five cornerstones; any added independent assumption is falsification
The reverse-engineering path is unique and necessary	Forced inversion from the five cornerstones: the derivation from cornerstones to L_0 is necessary, with no branching
Deductive conclusions necessarily apply at all scales	Universality of L's structure: protocol-recursive grammar spans all layers from $d = 0$ to $d \gg 1$
L is the only necessary ontology	Objectivity—independent of consciousness transfer: rules exist independently of any observer

These are not "statements of faith," but **auditable construction principles**. Any reader can trace and check:

- Has the closure been broken (is any entity introduced beyond the five cornerstones)?
- Does the reverse-engineering path branch (is there another possible L_0 that also yields the five cornerstones)?
- Is the L structure truly universal (is there a layer not describable by protocol recursion)?
- Is L truly necessary (can everything be explained more simply without introducing L)?

If any of these checks fail, the theory is falsified at the meta level—without waiting for specific empirical predictions.

—

4.7 Test Roadmap and Milestones

Based on the predictions above, we propose the following systematic test roadmap:

Table 5: Systematic test roadmap for GET

Time window	**Key experiments/observations**	**Test focus**	**Milestone meaning**
2025–2028	XENONnT/LZ final data	Dark matter zero interaction (Prediction 4.5)	Sensitivity $10^{-48}\mathrm{cm}^2$, initial verdict
2026–2030	LSST all-sky survey	H_0 anisotropy (Prediction 4.9)	Confirm/deny 1–2% directional pattern
2026–2032	CMB-S4 Phase I data	Inflation phase-transition features, Ξ value (Predictions 4.10, 4.8)	Non-Gaussianity constraints; Ξ precision ± 0.05
2026–2035	DESI, Euclid full datasets	Dark-energy $w \equiv -1$ (Prediction 4.7)	w precision ± 0.01, uniformity test
2026–2035	DARWIN operation	Dark matter zero-interaction final verdict (Prediction 4.5)	Ultimate sensitivity $10^{-49}\mathrm{cm}^2$, final verdict
2026–2040	Tabletop quantum-gravity experiments	Null results for gravity quantization (Prediction 4.6)	Test Planck-scale gravity behavior
2026–2045	JWST and next-gen telescopes	Baryon–photon ratio constancy (Prediction 4.13)	Primordial abundance at extremely high redshift
2026–2050	30-meter-class telescope cosmology	Full-parameter precision of Ξ (Prediction 4.8)	Ξ precision ± 0.02, overall cosmological test

—

4.8 Statement of Logical Closure of the Falsification Conditions

GET's core architecture is supported by the following layers; falsifying any bottom layer collapses the whole:

1. **Cornerstone layer:** the five empirical cornerstones (Section 2.3).
2. **Deductive layer:** the logical chain from cornerstones to L_0, geometric selection, and the existence-degree axiom (Section 2.5; Definition B2; Axiom III).
3. **Definition layer:** ontological architecture, recursion depth, three categories, species fingerprints (Definitions A1, A4, B6, D2).
4. **Axiom layer:** Axioms Ω through XIII (Section 3.4).
5. **Corollary layer:** Corollaries 1 through 22 (Section 3.5).

Ultimate falsification criterion: see Corollary 22. This criterion logically closes the entire falsification system.

GET places itself fully under the ultimate judgement of logic and experiment—without reservation, without retreat.

5 Cross-Scale Evidence and Description: Verifying GET as a Unified Grammar of Existence

5.1 Case 1: GET Description of Cosmic Genesis and the Differentiation of the Three Categories

What GET provides is not yet another cosmological model, but a grammar of cosmic genesis. Under this grammar, phenomena that in traditional cosmology often require separate ad hoc explanations—inflation, the origin of the CMB, component ratios, dark energy—are unified as different facets of one single, continuous dynamical process:

"concurrent quantum fluctuations (L_0 running) → geometric selection → four-phase cycle (protocol solidification) → inflation and channel closure → one-shot freezing into three categories."

Core mechanism:

1. **Baseline state:** true vacuum, i.e., the macroscopic statistical steady state presented by the independent running of the baseline dynamical system L_0 (Definition A2). Its running produces a concurrent ocean of quantum fluctuations. Each fluctuation individual i is a temporary form M_{fluct} generated and maintained by L_0; its complete state is given by Definition A1:
$$\Omega_i^{(0)} = \Phi(L_0),$$
and for a $d = 0$ individual, $E_{\text{solid}}(t) \equiv 0$ so $E(t) = E_{\text{active}}(t)$.
2. **Concurrent borrowing ocean:** L_0 produces and manages an extremely large number of concurrent quantum-fluctuation individuals. Each individual's net borrowing energy $E_i(t)$ (i.e., $E^i_{\text{active}}(t)$) exhibits a **peaked curve** over its lifecycle (Definition B1), and the peak borrowed energy E^i_{peak} follows a power-law distribution:
$$\boldsymbol{P(E_{\text{peak}}) \propto E_{\text{peak}}^{-\alpha}, \quad \alpha > 1.}$$
3. **Geometric selection:** for the potential transition from recursion depth $d = 0$ to $d = 1$, there exist an energy threshold $\epsilon^{(1)}_{\text{th}} = E^{(1)}_{\text{th}}$ and a minimum time window $N^{(1)}_{\text{min}}$. The individual's net borrowing energy curve $E_i(t)$ intersects the horizontal line $E = E^{(1)}_{\text{th}}$ at two points t_{start} and t_{end} ($t_{\text{start}} < t_{\text{end}}$). The interval $[t_{\text{start}}, t_{\text{end}}]$ is the **potential time window**, with length $\Delta t = t_{\text{end}} - t_{\text{start}}$.
Only when $\Delta t \geq N^{(1)}_{\text{min}}$ does this potential window become a **qualified window** W (Definition B2). The individual then obtains access permission to the candidate protocol family $\mathscr{P}_1$, enters the **primordial excited state**, and its state function becomes:
$$\boldsymbol{\Omega_i^{(\text{excited})} = \Phi(\{L_0\} \cup P), \quad P \in \mathscr{P}_1.}$$

Within the qualified window W, because $E_i(t) \geq E_{\text{th}}^{(1)}$ and is typically on the rising segment or peak plateau of the energy curve, the instantaneous net flow $e(t)$ is correspondingly high. By the existence-degree formula

$$\boldsymbol{\Psi_M(t) = \frac{E_{\text{solid}}}{\tau_{\text{solid}}} + \frac{E_{\text{active}}}{\tau_{\text{active}}} + \eta \cdot e(t),}$$

for $d = 0$ individuals (first term is zero), within this window $\Psi_M(t)$ **necessarily and automatically remains at a significantly positive, high level**. This minimum existence-degree level $\Psi_{\text{invest}}^{(1)}$ guaranteed by geometric selection is **not an independent parameter**, but a logical corollary of $(E_{\text{th}}^{(1)}, N_{\text{min}}^{(1)})$.

4. **Inflation and phase transition: trigger moment** T_0—the first batch of fluctuations satisfying the geometric selection condition collectively activate $\mathscr{P}_1$, break symmetry, and the universe transitions from true vacuum into the primordial excited state. Starting at T_0, the universe enters an exponential expansion phase (inflation). **Termination moment**—the inflation duration is naturally determined by the minimum time window:

$$\boldsymbol{\tau_{\text{inflation}} = N_{\text{min}}^{(1)} \cdot t_P.}$$

At background time $T = T_0 + \tau_{\text{inflation}}$, the first activators complete protocol solidification execution, triggering a macroscopic phase transition and terminating inflation.

5. **Protocol solidification execution:** the primordial excited state attempts to execute its protocol set P. This is the **first realization of the four-phase cycle** (Definition C1), requiring $\delta t = N_{\text{min}}^{(1)} \cdot t_P$. The process is irreversible and has a binary outcome:

   - **Successful solidification:** part of the peak borrowed energy is locked; protocols in P are instantiated and solidified into a core protocol subset $S_1 \in \mathscr{L}_1$. The state transition is:

   $$\boldsymbol{\Phi(\{L_0\} \cup P) \to \Phi^{(1)}(\{L_0\} \cup S_1, E_{\text{locked}}^i, t) + \text{radiation photons } (E_{\text{rad}}^i),}$$

   where $E_{\text{locked}}^i = \eta E_{\text{peak}}^i$ is the total structured asset and $E_{\text{rad}}^i = (1-\eta)E_{\text{peak}}^i$. The radiation photons constitute the primordial radiation field.
   Species origin: if this successfully solidified individual is among the **first successful solidifiers** at depth $d = 1$, then its $S_1 \in \mathscr{L}_1$ becomes the prototype of the ordinary-matter species protocol fingerprint $C_{\mathcal{S}}$ (Definition A5). The S_1 of the first successful solidifiers jointly determine the fingerprints $C_{\mathcal{S}}$ of $d = 1$-layer species (e.g., electrons, photons, quarks).

   - **Solidification interruption:** if interrupted by external conditions before completion, the process is forcibly stopped and permanently frozen.

6. **Channel closure and freezing:** at the inflation termination moment $T = T_0 + \tau_{\text{inflation}}$ (Definition B6):

- all energy-borrowing channels from true vacuum (the baseline state L_0) are **permanently and collectively cut off**;
- all protocol solidification executions that are ongoing but incomplete are forcibly interrupted and permanently frozen.

Thus, at the "cosmic moment" $T_0 + \tau_{\textbf{inflation}}$, three existence categories (form categories) are produced in one shot:

Category A (ordinary matter) Those primordial excited states that had **completed** protocol solidification execution before $T = T_0 + \tau_{\text{inflation}}$. Complete state:

$$\boldsymbol{\Phi^{(1)}(\{L_0\} \cup S, E^i_{\text{locked}}, t), \quad S \in \mathscr{L}_1.}$$

These forms possess non-zero total structured assets $E_{\text{locked}} = E_{\text{solid}} + E_{\text{active}}$ and solidified protocol sets with $d \geq 1$, and can perform non-geometric interactions (Axiom X). They are the **seeds** of subsequent cosmic structuring.

Category B (dark matter) Events that were **still in** the primordial excited state at $T = T_0 + \tau_{\text{inflation}}$, permanently frozen at $d = 0$. Complete state:

$$\boldsymbol{\Phi^{(0)}(\{L_0\} \cup P, E^i_{\text{frozen}}, t) \quad \text{(frozen)},}$$

where P is the un-solidified protocol set and E^i_{active} is frozen active-state energy (non-zero, consistent with the third law of thermodynamics). These forms have non-zero active-state energy (thus gravity), but **lack any solidified protocol set with** $d \geq 1$; therefore they have no non-geometric interactions other than gravity (Corollary 2).

Category C (dark energy) At $T = T_0 + \tau_{\text{inflation}}$, all fluctuations whose **cumulative net borrowing energy** E_i never reached—or had already decayed below—the threshold $E^{(1)}_{\text{th}}$ have their energy uniformly frozen into space. Complete state:

$$\boldsymbol{\Phi^{(0)}(\{L_0\}, E_{\text{background}}, t),}$$

where $E_{\text{background}}$ is uniformly frozen active-state energy (non-zero, consistent with the third law). These forms have no internal structure, no solidified protocols, and no interaction channels; their energy density remains constant under cosmic expansion and appears as the cosmological constant Λ (Corollary 3).

7. **Energy and radiation:** protocol solidification allocates input energy by an efficiency η_{ind} (Definition C4):

$$\boldsymbol{E^i_{\text{peak}} = E^i_{\text{locked}} + E^i_{\text{rad}}, \quad E^i_{\text{locked}} = \eta_{\text{ind}} E^i_{\text{peak}}, \quad E^i_{\text{rad}} = (1 - \eta_{\text{ind}}) E^i_{\text{peak}}.}$$

Define the individual ordering factor $\Xi_{\text{ind}} := (1 - \eta_{\text{ind}})/\eta_{\text{ind}}$.

At $T = T_0 + \tau_{\text{inflation}}$, aggregate all completed protocol solidification events (the total of Category A) and define the cosmological mean morphologization efficiency (Definition C5):

$$\boldsymbol{\bar{\eta} := \frac{E^{(\text{A})}_{\text{total locked}}}{E^{(\text{A})}_{\text{total peak}}}, \quad \bar{\Xi} := \frac{1 - \bar{\eta}}{\bar{\eta}}.}$$

8. **Origin of the CMB:** the CMB is the collectively redshifted remnant of the high-energy photons radiated by Category A events before $T = T_0 + \tau_{\text{inflation}}$ (total amount $(1 - \bar{\eta})E^{(\mathrm{A})}_{\text{total peak}}$). Its near-perfect blackbody spectrum confirms the universality (shared $\bar{\eta}$) and statistical nature of this process (Definition B7).

Traditional puzzles that GET resolves:

- **Nature of dark matter / dark energy:** both share a common origin, being different outcomes of the same extremely early-universe historical event (protocol solidification and freezing), with clear ontology (Definition B6). Dark matter is a **half-finished product** whose solidification was not completed (active-state energy frozen); dark energy is a **sub-threshold frozen background** (active-state energy uniformly frozen). Both have non-zero E_{active}, consistent with the third law of thermodynamics.
- **Cosmic component ratios:** ratios are a statistical necessity of power-law statistics and geometric selection, requiring no fine-tuning (Corollary 4).
- **Origin and termination of inflation:** triggered by satisfying geometric selection; terminated by completion of protocol solidification and channel closure; mechanism is explicit (Definition B4, Axiom IV). Inflation duration $\tau_{\text{inflation}} = N^{(1)}_{\min} \cdot t_P$ is a logical necessity of the time-window parameter.
- **Origin of the CMB:** a collectively redshifted remnant of the **latent heat of the first large-scale ordering event** (protocol solidification execution), not an independent "last scattering" patch (Corollary 5).
- **Origin of the arrow of time:** from symmetric fluctuation generation–annihilation cycles to irreversible protocol solidification and channel closure, the arrow of time naturally emerges (Axiom Ω.2).

Nature of the thermal bath: the thermal-bath radiation field is essentially the decay product (released as photons) of the collective solidification of an extremely large number of primordial excited states (i.e., Higgs-scale tail individuals). It converts instantaneous "borrow-and-repay" vacuum energy into an energy-cycling pool that can be continuously captured and used, thereby supporting continuous solidification from $d = 1$ to higher d. This also explains why the thermal bath is blackbody: it is a necessary result of extreme-number statistics.

Distinct predictions / implications:

- CMB non-Gaussianity should contain statistical features of the "structuring-completion phase transition," distinguishable from slow-roll inflation predictions (Chapter 4, Prediction 4.10).
- The cosmological ordering factor $\bar{\Xi} = (1-\bar{\eta})/\bar{\eta}$ is the universe's "efficiency–cost ratio"

for creating order; its value $\bar{\Xi} \approx 8.57$ is inferred from observed component ratios and should satisfy a fixed algebraic relation $\Xi = (\Omega_\Lambda - \Omega_{\mathrm{DM}})/\Omega_{\mathrm{OM}}$ (Chapter 4, Prediction 4.8).

- If a multiverse shares the same L_0 and fluctuation statistics, foundational laws should be highly similar (Corollary 15).

5.2 Case 2: A GET Explanation of Gravity's Specialness

Based on GET Corollary 1, gravity receives a unified and deep explanation: **gravity is a purely geometric effect at** $d = 0$, fundamentally different from gauge interactions (electromagnetic, weak, strong) at $d \geq 1$.

Core mechanism:

Gravity is the geometric curvature effect on the background spacetime metric produced when a form M has non-zero total structured assets:

$$\boldsymbol{E}_{\text{locked}} = \boldsymbol{E}_{\text{solid}} + \boldsymbol{E}_{\text{active}}.$$

Under the low-velocity approximation, this corresponds to the form's mass m and satisfies $E_{\text{locked}} \approx mc^2$. Gravity **does not require exchanging any solidified protocol medium with** $d \geq 1$ (Axiom X).

For different categories:

- **Category A (ordinary matter)** and **Category B (dark matter):** $E_{\text{locked}} > 0$ produces standard gravity (positive mass, attraction). Category A possesses solidified protocols with $d \geq 1$ and thus can participate in non-geometric interactions; Category B lacks such protocols and therefore **only** produces gravity (Corollary 2).
- **Category C (dark energy):** its uniformly frozen background energy density (uniformly distributed as $E_{\text{background}}$) produces **negative pressure**, appearing as the cosmological constant term Λ in Einstein's field equations, leading to gravitational repulsion (accelerated expansion).

Gravitational waves propagate at light speed c because they are the **propagation of disturbances of spacetime's protocol-connection pattern** (Axiom XIII). Here c is an intrinsic property of the protocol-recursive system L, defining the absolute upper bound for propagation of protocol-state differences.

Traditional puzzles that GET resolves:

- **Gravity's specialness:** why is gravity so different from electromagnetism, the weak force, and the strong force?
  — **Answer: protocol-layer difference.** Gravity is a geometric effect at $d = 0$;

gauge forces are protocol-medium exchanges at $d \geq 1$ (Axiom X). Because gravity needs no shared solidified protocols, it is **universal, long-range, and unshieldable**; gauge forces depend on specific protocol matching, thus are **selective, shieldable, and tunable in strength**.

- **The quantum-gravity dilemma:** why is quantizing gravity so hard?
  — **Answer: a category error.** Gravity is classical geometry at $d = 0$, while quantum phenomena are behaviors of solidified protocols at $d \geq 1$. The difficulty of forcing them into a single "quantum gravity" framework comes from conflating phenomena of different protocol layers with different ontological bases (Corollary 14). **Quantizing gravity is unnecessary and may be a conceptual misdirection.** They naturally coordinate under L_0's unified state management and causal-consistency algorithm.
- **Dark matter gravity:** why does dark matter gravitate but not otherwise interact?
  — **Answer:** dark matter (Category B) is a frozen primordial excited state at $d = 0$ (Definition B6) and has frozen active-state energy E_{active}, so it gravitates; but it **lacks any solidified protocol set with** $d \geq 1$, so it has no non-geometric interaction channels (Corollary 2). Its non-gravitational cross section is **strictly zero**, one of GET's sharpest falsifiable predictions.
- **Equivalence principle:** inertial mass and gravitational mass are equivalent because both are endogenous to the same ontology (the running of protocol set L)—the manifestation of total structured assets $E_{\text{locked}} = E_{\text{solid}} + E_{\text{active}}$. Gravity "feels" total structured assets; inertia resists changes to total structured assets—**two manifestations of the same intrinsic property**.

5.3 Case 3: A GET Explanation of Quantum Phenomena

Within the GET framework, quantum phenomena receive a unified mechanistic explanation grounded in **protocol recursion**, **memory management**, and **geometric selection**.

Core mechanism:

1. **Quantum uncertainty.** The Heisenberg relation

$$\Delta E \cdot \Delta t \geq \hbar/2$$

   is the **irreducible minimum algorithmic cost** that the baseline system L_0 must pay when executing the bottom-level "difference generation" operation; $\hbar$ is the unit of that cost (the meta-philosophical reading of Cornerstone 1).
   More importantly, this relation is a **special case** of Axiom III's scale constraint for the $d = 0$ species L_0. For quantum-fluctuation individuals, $E_{\text{solid}} \equiv 0$ and $E_{\text{active}}(t) =$

$E(t)$. After integrating over the full lifecycle, the scale constraint takes the form:

$$\boldsymbol{E}_{\text{peak}} \cdot \boldsymbol{\tau} \geq \frac{\hbar}{\mathbf{2}}, \quad \boldsymbol{K} = \frac{\hbar}{\mathbf{2}},$$

i.e., the **energy–time composite cost constraint** that the form M_{fluct} (quantum fluctuation) must satisfy to exist stably (though briefly).

2. **Quantum superposition and nonlocality.** Nonlocal correlations such as quantum entanglement are explained as strong correlation records that L_0 maintains for spatially separated individuals in its **global state-management memory** (Axiom XIII.3). Their apparently "instantaneous" readout does **not** violate the maximum speed c, because correlation establishment and synchronization are **not** implemented via signal propagation through spacetime, but via L_0's globally consistent **write ("preparation") and read ("measurement")** of its own central state.

3. **Wave–particle duality.** Within the framework of its solidified core protocol fingerprint $C_{\mathcal{S}}$, a form M simultaneously contains multiple potential protocol paths (e.g., a wave-like path P_{wave} and a particle-like path P_{particle}). Its complete state $\Omega^{(M)}$ is a **weighted encoding** of these possibilities. The protocol set L_O of the measurement apparatus determines which protocol path has higher **compatibility** κ, thereby triggering **partial solidification** of that path and producing the corresponding observed phenomenon.

4. **Measurement and collapse.** Observation is essentially a process of **protocol matching, decoding, and protocol reconstruction** between an observer protocol set L_O and a system protocol set L_S (Axiom IX). Observation changes the system's state distribution:

$$\boldsymbol{P}_{\text{post}}(\boldsymbol{L}) = \frac{\boldsymbol{P}_{\text{pre}}(\boldsymbol{L}) \cdot \mathbf{exp}[-\boldsymbol{\beta} \boldsymbol{D}(\boldsymbol{L} \| \boldsymbol{L_O})]}{\boldsymbol{Z}},$$

where $D(L \| L_O)$ measures protocol difference and $\beta \propto E_{\text{obs}}$ is an observation-strength parameter.

"Wavefunction collapse" is the dynamical manifestation of the state distribution **sharply concentrating** toward the eigenstate with the highest compatibility with L_O after protocol matching. **Collapse is not an externally added non-physical process, but an endogenous result of protocol matching**—the system's solidified core protocol $C_{\mathcal{S}}$ remains unchanged; only one internal potential path is selected and solidified.

5. **Decoherence.** The environment, as an "observer" with an enormous number of degrees of freedom, has extremely low compatibility between its protocol set L_{env} and the system's quantum protocols L_S ($\kappa \ll 1$), causing the system's state distribution to concentrate rapidly and irreversibly toward environment-compatible classical states. This is a **statistical necessity of Axiom IX** in the large-N limit.

Positioning of the Higgs particle. The Higgs particle is the most typical low-energy tail manifestation of the primordial excited state in today's observable regime ($E_{\text{peak}} \approx 125$ GeV). However, the vast majority of similar individuals failed to complete solidification before inflation-channel closure and thus remain only as frozen excited states. This explains why Higgs events are rare, why their decay products are observable, and why the thermal-bath radiation field is, in essence, produced by the collective decay of an extremely large number of such individuals.

Traditional puzzles that GET resolves:

- **The measurement problem:** no external "collapse postulate" is required; collapse is an endogenous result of observer–system protocol interaction.
- **The nonlocality paradox:** nonlocal correlation is a manifestation of L_0's global memory consistency, involving **no superluminal signal transfer**.
- **The quantum–classical boundary:** the boundary is continuously parameterized by protocol compatibility κ, with no sharp cut; decoherence is the limiting case as $\kappa \to 0$.

5.4 Case 4: A GET Description of Living Systems

GET treats living systems as **high recursion-depth products** generated after multiple successful upward solidifications via the four-phase cycle ($d \gg 1$) (Corollary 16).

Core state:

The complete state of a living individual i is described by a species-specific state function (Definition D2):

$$\boldsymbol{\Omega_{\mathcal{S}}^{(M)}(t) = \Phi_{\mathcal{S}}(L_{\text{life}}, E_{\text{solid}}(t), E_{\text{active}}(t), e(t), t), \quad d \geq 5,}$$

where:

- $\mathcal{S}^{(d)}$ is a biological species defined by a **heritable core genetic protocol set** $C_{\mathcal{S}} \subset \mathscr{L}_d$ (Definition A5). Here $d = 5$ corresponds to proto-life, $d = 6$ to multicellular organisms, and $d = 7$ to sentient organisms.
- L_{life} is the instantiated protocol set of an individual, satisfying $\text{Core}(L_{\text{life}}) = C_{\mathcal{S}}$, and containing nested protocols for metabolism, regulation, perception, behavior, etc.
- $E_{\text{solid}}(t)$ is the form's solidified-state energy—**structural biomass not directly usable as ATP**, including bones, muscles, fat, organs, tissues. This provides structural support and long-term reserves, but **cannot directly maintain immediate life activities**.
- $E_{\text{active}}(t)$ is active-state energy—**immediately usable energy currency** for life activities, including ATP, GTP, NADH, blood glucose, etc.

- $e(t)$ is the instantaneous net flow—net metabolic gain from exchange with the environment.
- $E_{\text{locked}} = E_{\text{solid}} + E_{\text{active}}$ is total biomass energy, corresponding to "total energy reserves" in traditional biology.

Life under the GET perspective:

1. **Definition of life.** A dynamical system that, through its internal protocol set L, can efficiently manage energy E, **maintain the survival criterion**
$$\boldsymbol{\frac{E_{\text{active}}(M,t)}{\tau_{\text{active}}(L)} + \eta(L)\cdot e(t) \geq 0}$$
(Axiom III), and can **continuously obtain qualified windows**, trial deeper candidate protocol families $\mathscr{P}_{d+1}$, and thereby achieve recursion-depth growth (increase d).

2. **Existence-degree dynamics.** A living form's existence degree is:
$$\boldsymbol{\Psi_M(t) = \frac{E_{\text{solid}}(t)}{\tau_{\text{solid}}(L)} + \frac{E_{\text{active}}(t)}{\tau_{\text{active}}(L)} + \eta(L)\cdot e(t).}$$
   - $\frac{E_{\text{solid}}}{\tau_{\text{solid}}}$: structural-asset return rate—survival buffering provided by biomass.
   - $\frac{E_{\text{active}}}{\tau_{\text{active}}}$: operational-asset return rate—support capability from energy-currency reserves.
   - $\eta\cdot e(t)$: metabolic net gain—real-time difference between intake and expenditure.

   Life must maintain
$$\boldsymbol{\frac{E_{\text{active}}(M,t)}{\tau_{\text{active}}(L)} + \eta(L)\cdot e(t) \geq 0,}$$
otherwise the form disintegrates (death).

3. **Evolution and species.** A biological species $\mathcal{S}^{(d)}$ is defined by the core protocol fingerprint $C_{\mathcal{S}}$ solidified by the **first individuals** that successfully completed the four-phase cycle (Definition A5). Natural selection selects protocol configurations that can **more frequently obtain qualified windows and successfully execute the four-phase cycle**.
Species differentiation (Definition C3) is described by the within-layer differentiation rate
$$\boldsymbol{W^{(d)}_{\text{div}}(\mathcal{S}_i \to \mathcal{S}_j; T),}$$
occurring when, within the same depth d, an individual of an existing species $\mathcal{S}_i$ successfully solidifies a new protocol fingerprint $C_{\mathcal{S}_j} \neq C_{\mathcal{S}_i}$ and compatibility satisfies $\kappa \geq \kappa_{\text{crit}}$.

4. **Consciousness.** Consciousness is the emergent capacity of high-dimensional living forms to perform **recursive operations (self-reference)** on their own state and

protocol set, corresponding to higher recursion depth d (Corollary 16). A conscious-state function may be written as

$$\boldsymbol{\Omega}^{(M)}_{\text{conscious}}(t) = \boldsymbol{\Phi}_{\text{conscious}}(\boldsymbol{L}, \boldsymbol{E}_{\text{solid}}, \boldsymbol{E}_{\text{active}}, \boldsymbol{e}(t), t),$$

where $\Phi_{\text{conscious}}$ includes instructions for reading, evaluating, and modifying the protocol set L itself.

Distinct implications:

- **Synthetic biology and artificial life:** success depends not only on constructing the form M, but on achieving protocol-set compatibility κ reaching the critical value (Definition A3) and forming a stable energy–protocol–form closed loop.
- **Ecosystem health diagnostics:** the health of a species or ecosystem can be quantified by its **average existence degree** $\langle \Psi_M(t) \rangle$. A sustained decline in $\langle \Psi_M(t) \rangle$ is an early warning of collapse.
- **Unified explanation of scaling laws:** biological 3/4-power metabolic scaling, lifespan–mass correlations, etc., are concrete instances of Axiom III's scale constraints under specific core protocol fingerprints $C_{\mathcal{S}}$. Differences in $\tau_{\text{solid}}(C_{\mathcal{S}})$, $\tau_{\text{active}}(C_{\mathcal{S}})$, and $\eta(C_{\mathcal{S}})$ reflect fundamental differences in energy-management strategies across species.

5.5 Case 5: Post-Mass-Extinction Explosive Evolution——Macroscopic Manifestation of Qualified Window Density

A repeatedly observed pattern in the history of life on Earth is that mass extinction events are invariably followed by explosive species diversification that rapidly fills ecological niches. When niches become saturated, the pace of evolution plummets to an extremely slow rate. In traditional evolutionary theory, this pattern is explained as "niche release" and "reduced competition." However, GET provides a deeper mechanistic explanation—— it redefines evolutionary rate not as an intrinsic property of a species, but as a macroscopic statistical manifestation of geometric selection conditions.

The nature of mass extinction: dimensionality reduction transition and energy release

From the GET perspective, the essence of a mass extinction is that a large number of high-recursion-depth species (e.g., dinosaurs, $d \approx 6-7$) have their existence degree $\Psi_M(t)$ fall below the critical point, triggering a downward transition $W_{\downarrow}^{(d+1\to d)}$.

When a species' existence degree falls below the survival criterion

$$\frac{\boldsymbol{E}_{\text{active}}}{\boldsymbol{\tau}_{\text{active}}} + \boldsymbol{\eta} \cdot \boldsymbol{e}(t) < \mathbf{0},$$

its individuals disintegrate, and their locked total structured assets

$$\boldsymbol{E}_{\text{locked}} = \boldsymbol{E}_{\text{solid}} + \boldsymbol{E}_{\text{active}}$$

are released back into lower recursion depth layers ($d = 0 - 4$), transforming into:

- **reusable active-state energy** E_{active} (organic matter, nutrients, resources);
- **radiative energy** E_{rad} (heat, etc.).

This process is not "energy disappearance" but **dimensionality reduction of energy**——after the failure of high-depth protocols, their energy reserves are released to more fundamental layers, becoming resources that lower-depth individuals can recapture.

Protocol space release and restructuring of energy acquisition costs

After the disintegration of high-depth species, the protocol space (ecological niches) they occupied is released. This brings three direct consequences:

1. **Energy acquisition costs drop significantly**: the suppression from predators and competitors disappears, reducing the investment needed for individuals to acquire E_{active};
2. **Competitive protocol suppression disappears**: lower-depth individuals that were previously suppressed by high-depth species gain living space;
3. **The net borrowing energy curves $E(t)$ of more species more easily reach the threshold $\epsilon_{\text{th}}^{(d+1)}$.**

The key change in this stage is: **the originally locked E_{locked} is re-transformed into freely capturable E_{active}, while capture costs drop significantly**. This means that in GET's survival criterion, the term $\eta(L) \cdot e(t)$ turns from negative or zero to positive for a large number of individuals.

Emergence of qualified windows and explosion of their density

When energy acquisition costs fall, more individuals can maintain a sufficiently high energy level for a sufficiently long time during their lifecycles, thereby satisfying the geometric selection condition:

$$\boldsymbol{\Delta t \geq N_{\min}^{(d+1)}}.$$

The density of **qualified windows**——the proportion of individuals satisfying the geometric selection condition per unit time——jumps from extremely low levels to a peak.

Traditional evolutionary theory attributes the explosive period to "niches becoming empty." But GET asks further: why does "emptiness" cause an explosion? The answer is: **because the drop in energy acquisition costs directly translates into an increase in qualified-window density.** It is not that species "become stronger"; it is that the conditions "become easier."

Explosive period: the GET essence of rapid niche filling

When qualified windows exist simultaneously for a large number of individuals:

- The upward transition rate $W_{\uparrow}^{(d\to d+1)}$ soars (emergence of new species layers);
- The intra-layer divergence rate $W_{\text{div}}^{(d)}$ soars (rapid differentiation within the same depth).

New protocol fingerprints $C_{\mathcal{S}}$ are rapidly solidified——this is the GET essence of "rapid niche filling." Each successful solidification means a new species' protocol fingerprint enters the protocol library and participates in subsequent energy competition.

This process is the **same grammar manifested at different scales** as the four-phase cycle from quantum fluctuations to elementary particles in the origin of the universe. The only difference is: cosmic genesis involved a one-time channel closure, whereas biological evolution involves repeatedly opening and closing windows.

Window closure and the slow period

When niches become filled and protocol space is reoccupied:

1. **Competitive protocols are re-established**: energy competition among high-depth species returns;
2. **Energy acquisition costs rise back**: the investment needed to capture E_{active} increases again;
3. **The $E(t)$ curves of most individuals are once again suppressed below the threshold**.

Qualified-window density consequently plummets. $W_{\uparrow}$ and W_{div} return to low levels. Evolution still occurs, but only when a very few individuals **by chance** obtain sufficiently long qualified windows can they solidify new protocol fingerprints.

This is the essence of "slow evolution"——not that the species' "adaptability" or "evolutionary potential" has declined, but that **qualified windows have become rare**.

Core insight: evolutionary rate is not a function of intrinsic species properties

Traditional evolutionary theory attributes evolutionary rate to intrinsic properties of species: mutation rate, population size, generation time, etc. However, the explosive evolution following mass extinctions challenges this presupposition——the same surviving species exhibit dramatically different divergence rates before and after the extinction.

GET's answer is: GET's answer is:

Evolutionary rate is not a function of intrinsic species properties.

It is a macroscopic statistical manifestation of geometric selection conditions.

The speed of species divergence depends on the density of qualified windows. When qualified-window density is high, even "ordinary" species can diverge rapidly; when qualified-window density is low, even "high-potential" species struggle to break through thresholds.

This insight elevates the discussion of evolutionary rate from the "species level" to a unified framework of "energy–geometry level." It explains why mass extinctions are invariably followed by explosions: not because the surviving species are "stronger," but because **energy is released, costs are reduced, and windows are opened**.

Testable predictions

Based on the above mechanism, GET proposes the following testable prediction:

The intensity of a mass extinction event (i.e., the total released $\boldsymbol{E}_{\text{locked}}$**)** should be positively correlated with the divergence rate during the subsequent explosive period (i.e., the peak of $\boldsymbol{W}_{\text{div}}$**).**

Specifically:

- A mass extinction of great magnitude (e.g., the Permian–Triassic extinction, which eliminated about 95% of species) should be followed by the most dramatic explosion;
- A small extinction should be followed by a relatively mild explosion;
- If an observed mass extinction with high intensity is not followed by a significant explosion, this would suggest that the energy released by that extinction was not transformed into capturable E_{active} (e.g., the energy was lost as radiation rather than retained as organic matter), thereby providing a further test of the energy-conversion pathways articulated by the theory.

This prediction can be tested by comparing species richness curves from paleontological databases with extinction intensity data.

5.6 Case 6: GET Dynamics of Technology and Social Systems

A social system is a high-recursion-depth ($d \gg 1$) **complex protocol system** composed of large numbers of individual protocol sets connected in networks[30, 31].

Core mechanism:

1. **Technology.** Technology is a new protocol set L_{tech} created and externalized by humans. A technological revolution is the process by which a candidate protocol set $P \in \mathscr{P}_{d+1}$ is successfully **trialed**, **captured**, and **solidified** (the four-phase cycle) by the social system.

Social analogy of geometric selection: when the social system's "energy" reserves (capital, attention, talent) curve intersects a threshold and the **qualified window** lasts long enough, society gains upgrade permission. Successful solidification increases the overall capture efficiency $\eta(L_{\text{social}})$, extends the solidified-state characteristic time constant $\tau_{\text{solid}}(L_{\text{social}})$ (institutional stability), or opens new net-inflow channels $e(t)$.

2. **Institutions and culture.** Institutions and cultural traditions are social-layer **connection capital**, i.e., high-order protocol sets at recursion depth $d \gg 1$ (Definition D4, Axiom VII). They follow the accumulation and depreciation dynamics of connection capital:

   - **Only path of accumulation:** via successful four-phase cycles (e.g., institutional innovation, cultural renewal), requiring converting large amounts of active-state energy into solidified-state energy (institutionalization, canonization), thereby increasing total structured assets E_{locked}.
   - **Distance dilution of energy capture:** effective investment energy decays with propagation distance (geographic, cognitive, stratified).
   - **Necessity of depreciation:** insufficient active energy and spatial separation lead to protocol synchronization difficulty, causing solidified institutions to gradually deactivate.
   - **Characteristic:** slow to accumulate (requires sustained energy investment and time), slow to depreciate (institutional inertia), but once disintegrated, highly destructive.

3. **Economic cycles and civilizational rise and fall.** These correlate with the social system's overall **existence degree** $\Psi_{\text{social}}(t)$:

$$\boldsymbol{\Psi_{\text{social}}(t) = \frac{E^{(\text{social})}_{\text{solid}}(t)}{\tau_{\text{solid}}(L_{\text{social}})} + \frac{E^{(\text{social})}_{\text{active}}(t)}{\tau_{\text{active}}(L_{\text{social}})} + \eta(L_{\text{social}}) \cdot e(t)_{\text{social}}(t),}$$

   where $E^{(\text{social})}_{\text{solid}}$ is accumulated material wealth and infrastructure (solidified energy), $E^{(\text{social})}_{\text{active}}$ is liquid capital and human capital (active energy), and $e(t)_{\text{social}}(t)$ is net output flow.

   When protocol sets become overly complex and maintenance costs surge (effectively shrinking τ_{solid}, lowering Ψ), or when **toxic protocols** (Axiom VIII) destroy internal structure, or when external energy input becomes insufficient, the system tends toward instability and disintegration.

4. **Multi-scale manifestations of toxic protocols** (Corollary 21):

   - Radioisotopes (physical layer, $d = 1 \sim 2$): release energy by destroying solidified structure.
   - Pathogens (biological layer, $d = 4 \sim 6$): consume host active energy and disrupt protocol execution.
   - Computer viruses (information layer, $d = 7 \sim 9$): tamper with execution logic

and consume computational resources.

- Cultural dross and corrupt institutions (social layer, $d \geq 10$): reduce institutional efficiency, increase transaction costs, and consume social active energy.

System stability requires effective isolation mechanisms satisfying $Q > Q_{\text{crit}}$. Black holes are the universe-level counterpart; immune systems, firewalls, and cultural critique are the counterparts at other layers.

Concrete phenomena under the GET perspective:

- **The S-curve of technology diffusion[33, 34, 35]:** a function of the **compatibility** $\kappa(L_{\text{new}}, L_{\text{social}})$ between a new technology protocol set L_{new} and the existing social protocol set L_{social}. High compatibility yields fast diffusion, with a diffusion rate form

$$\frac{dP}{dt} \propto \kappa \cdot P(1-P).$$

  GET predicts that a model based on an explicit compatibility measure will predict breakthrough timing and diffusion curves more accurately than traditional logistic models (Chapter 4, Prediction 4.16).

- **Innovation:** successful **trial** and **capture** of candidate protocol sets $\mathscr{P}_{d+1}$ under existing protocol constraints. Innovation bursts require: a sufficiently long qualified window (capital-abundance period) + sufficiently high compatibility candidate protocols.

- **Occupations and industries as "species."** By Definition A4, occupations/industries are **stable form categories** at recursion depth $d = 10 \sim 12$, identified by protocol fingerprints $C_{\mathcal{S}}$ defined by socially recognized core skill protocols and institutional norms. Examples:
  - AI engineer (C_{AI})
  - civil engineer (C_{civil})
  - teacher (C_{teacher})

  Industry rise and fall are processes of **within-layer differentiation** and **extinction** of these "social species," describable via $W_{\text{div}}^{(d)}$ and $W_{\downarrow}^{(d+1\to d)}$ in the layered evolution equations.

- **Alignment problems (e.g., AI safety):** ensure that newly created powerful protocol sets (e.g., L_{AI}) have compatibility $\kappa(L_{\text{AI}}, L_{\text{human}})$ **above a safety threshold** κ_{safe} with the human social core protocol set L_{human}. Protocol sets with insufficient compatibility, even if powerful, should be **prohibited from solidification** or **strictly isolated**.

Distinct predictions / implications:

- **Existence of a maximum protocol depth** (Corollary 12): constrained by the finiteness and decay of the structured usable energy pool $\mathcal{E}_{\text{struct}}(T) = \sum E_{\text{active}}$, human civilization's technological complexity has a theoretical upper bound under current usable energy; a representative estimate is

$$\boldsymbol{D_{\max}^{(\text{human})} \approx 11.}$$

The technological singularity has a theoretical upper limit; it cannot be broken indefinitely.

Exponential growth of the upward-transition energy threshold (Chapter 4, Prediction 4.12): the energy investment required for successful upgrading grows exponentially with depth:

$$\boldsymbol{E_{\text{invest}}^{(d)} \propto \exp(\alpha d)}$$

where $\alpha > 0$ is an empirical parameter. This exponential scaling explains why the costs of particle accelerators, fusion devices, and space exploration rise sharply with technological level, and provides a quantitative prediction that can be tested by cross-scale statistical analysis.

- **Social existence-degree diagnostics:** a sustained decline in $\Psi_{\text{social}}(t)$ is a quantitative early warning signal of civilizational decline; its components $\frac{E_{\text{solid}}}{\tau_{\text{solid}}}$, $\frac{E_{\text{active}}}{\tau_{\text{active}}}$, and $\eta \cdot e(t)$ can be decomposed into measurable indicators such as infrastructure depreciation, liquid-capital efficiency, and net output.

5.7 Summary: Verifying GET as a Cross-Scale Unified Grammar

Across the five cases above, we verify GET's core commitment as a unified grammar of existence:

1. **Unity of the descriptive framework.** From quantum fluctuations (Σ_0, M_{fluct}) to cosmic genesis (Σ_1 species), gravity ($d = 0$ geometric effect), life ($\Sigma_{d\geq 5}$, $\mathcal{S}^{(d)}$) and consciousness, and technology and society ($\Sigma_{d\gg 1}$), all "speakable existences" can be completely described by the same state-function schema:

$$\boldsymbol{\Omega_{\mathcal{S}}^{(M)}(t) = \Phi_{\mathcal{S}}(L, E_{\text{solid}}, E_{\text{active}}, e(t), t),}$$

and their macroscopic evolution is governed by the **layered evolution equations** (Axiom V).

2. **Unity of the core mechanisms.**
   - **Complexity growth** (increase of recursion depth d) is unified as **geometric selection + the four-phase cycle** (Definition B2, Definition C1, Axiom II).
   - **System stability** is unified as the **survival criterion**

$$\boldsymbol{\frac{E_{\text{active}}(M,t)}{\tau_{\text{active}}(L)} + \eta(L) \cdot e(t) \geq 0}$$

   (Axiom III).

- **Interactions** are unified as **shared protocol media** ($d \geq 1$) or pure geometric effects ($d = 0$) (Axiom X).
- **Form categories** are unified as **core protocol fingerprints** $C_{\mathcal{S}}$ (Definitions A4 and A5).

3. **Generativity of the explanatory path.** Starting from bottom-level L_0 running, GET constructs a layer-by-layer generative explanation via reverse engineering and logical deduction, answering "**how it is possible**," not merely "how it runs."
4. **Operability of the analytical tools.** The same variables $(L, E_{\text{solid}}, E_{\text{active}}, e(t), t, d, M, C_{\mathcal{S}})$ and dynamics (geometric selection, four-phase cycle, existence degree, connection capital, toxic isolation) provide a unified, quantifiable, computable diagnostic toolkit for analyzing systems in physics, cosmology, biology, and society.

The unity GET seeks is not to reduce everything to the same "substance," but to reveal that all stable, dynamic existences share the same generative grammar and evolutionary logic.

6 Application Framework — Cross-Scale Real-Time Analysis Tools Based on the Axiomatic System

6.1 Unified Methodology: A Four-Step Analysis Framework Based on GET Core Concepts

GET is not only an explanatory theory; it is also an operable **analysis framework**. Any stable system (recursion depth $d \geq 0$) can be analyzed using the following four-step workflow, which follows directly from GET's axioms and definitions.

Four-step application framework

1. **System decoding & state-function mapping**
   - **Goal:** translate the target system into GET's core description.
   - **Procedure:**
     a. Identify the system's **protocol set** L (its internal running and interaction rules), and determine its **recursion depth** $d = d(L)$ (Definition A3).
     b. Identify the system's **form** M—the observable phenomenon set stably presented by the recursive running of L (Definition A1).
     c. Identify the system's **energy state**: its solidified-state energy E_{solid} (structured assets), active-state energy E_{active} (operational assets), and the instantaneous net flow required to maintain/expand its existence,
     $$\boldsymbol{e(t) = \phi_{\text{in}}(t) - \phi_{\text{out}}(t).}$$
     The total structured asset $E_{\text{locked}} = E_{\text{solid}} + E_{\text{active}}$ corresponds to the system's total energy reserves.
     d. Determine the system's **internal time scale** t and its correspondence to background time T (Definition D1).
     e. If the system belongs to a known species $\mathcal{S}^{(d)}$, determine its **core protocol fingerprint** $C_{\mathcal{S}}$ (Definition A5). If not, construct a protocol-fingerprint description.
   - **Output:** the system's GET state-function representation:
   $$\boldsymbol{\Omega_{\mathcal{S}}^{(M)}(t) = \Phi_{\mathcal{S}}\big(L, E_{\text{solid}}(t), E_{\text{active}}(t), e(t), t\big), \quad \Omega_{\mathcal{S}}^{(M)} \in \Sigma_d^{(\mathcal{S})}.}$$
2. **Existence-degree assessment & health diagnosis**
   - **Goal:** quantitatively assess stability and sustainability.
   - **Procedure:** by Axiom III, compute the system's **existence degree**:
   $$\boldsymbol{\Psi_M(t) = \frac{E_{\textbf{solid}}(M,t)}{\tau_{\text{solid}}(L)} + \frac{E_{\textbf{active}}(M,t)}{\tau_{\text{active}}(L)} + \eta(L) \cdot e(t),}$$

and apply the **survival criterion**:

$$\frac{\boldsymbol{E}_{\text{active}}(\boldsymbol{M},\boldsymbol{t})}{\boldsymbol{\tau}_{\text{active}}(\boldsymbol{L})}+\boldsymbol{\eta}(\boldsymbol{L})\cdot \boldsymbol{e}(\boldsymbol{t})\geq \mathbf{0}.$$

- **Diagnosis:**
  - If $\frac{E_{\text{active}}}{\tau_{\text{active}}}+\eta\cdot e(t)>0$: the system operates healthily; in a **self-maintaining or growing** state.
  - If $=0$: the system is in **critical balance**; fragile but not disintegrated.
  - If <0: the system is in a **death/disintegration** state; the form disintegrates immediately.

3. **Dynamics analysis & evolutionary-phase positioning**
   - **Goal:** determine where the system sits in GET's macro-evolution picture and identify evolutionary opportunities and threats.
   - **Procedure:** analyze using Axioms II, IV, V, Axiom III, and Corollary 10:
     - **Geometric selection & upgrade potential:** draw or estimate the system's net borrowing energy curve $E(t)$ (for $d=0$ individuals, this is $E_{\text{active}}(t)$). Determine whether there exists a **qualified window** W satisfying (Definition B2): the curve intersects the horizontal threshold line $\epsilon_{\text{th}}^{(d+1)}$ and the distance between intersections $\Delta t\geq N_{\text{min}}^{(d+1)}$. If such a window exists, the system has potential to transition to depth $d+1$.
     - **Protocol capture & growth phase:** is the system attempting upward transition ($d\to d+1$) via the **four-phase cycle** (trialing new protocols)? Is the within-layer distribution $P_d(\Omega,T)$ changing?
     - **Stable maintenance phase:** is the system mainly in intra-layer evolution (dominated by $\mathcal{H}_d^{(\text{intra})}$) with $\Psi_M(t)$ fluctuating within the positive region?
     - **Dissipative return phase:** is $\Psi_M(t)$ declining persistently, facing **downward disintegration** ($W_\downarrow$) or **toxic protocol** risks (Axiom VIII)?
   - **Output:** the system's current evolutionary phase and its principal contradiction(s).

4. **Intervention design & forecasting**
   - **Goal:** propose interventions based on GET principles and predict trajectories.
   - **Procedure:**
     a. **Protocol optimization:** increase protocol-set **compatibility** $\kappa(L)$ (Definition A3); optimize protocols to extend the solidified-state time constant $\tau_{\text{solid}}(L)$ (improve asset durability), shorten the active-state time constant $\tau_{\text{active}}(L)$ (improve operational efficiency), or improve energy capture efficiency $\eta(L)$.
     b. **Energy management:** adjust the input/output structure and optimize

the mix of E_{solid}, E_{active}, and $e(t)$ to ensure

$$\boldsymbol{\frac{E_{\text{active}}(M,t)}{\tau_{\text{active}}(L)} + \eta(L) \cdot e(t) \geq 0}$$

holds.

c. **Connection-capital investment/maintenance:** for high-d systems, by Axiom VII evaluate investment effectiveness and prevent depreciation (solidified energy dissipation) due to spatial separation or insufficient investment.

d. **Risk isolation:** identify and isolate potential **toxic protocol subsets** $L_{\textbf{toxic}}$ (Axiom VIII) and design isolation mechanisms satisfying $Q > Q_{\text{crit}}$.

e. **Create upgrade windows:** by increasing the peak of $E(t)$ or prolonging high-energy maintenance time, actively create a qualified window W for initiating the four-phase cycle.

- **Output:** concrete levers, expected effects, and a trajectory forecast based on the layered evolution equation (Axiom V).

6.2 Domain Mapping Guide (SETs): Translating GET Grammar into Specific Disciplines

To apply GET to a specific domain, one must build **"Specific-domain GET Translation Schemes (SETs)."** Each SETs is essentially a domain-specific **GET dictionary**, explicitly mapping domain concepts to GET core variables.

Core mapping relations required by any SETs

Table 6: Core mappings in SETs

GET core variable	Domain mapping content	Key question
Protocol set L	the domain's core rules, laws, institutions, code	What are the system's "rules of the game"? How to formalize them?
Form M	observable phenomena stably presented by L's running	What is the system's "body/phenomenon"?
Recursion depth d	organizational complexity level	How to stratify complexity levels? Which layer in Definition A4?
Core protocol fingerprint $C_{\mathcal{S}}$	the minimal shared protocol subset defining a class	What must all members share, sufficient to distinguish this class?
Solidified energy E_{solid}	structured assets / stock resources	What are the system's "fixed assets"?
Active energy E_{active}	liquid resources / deployable reserves	What is the system's "working capital/energy currency"?
Instantaneous net flow $e(t)$	real-time net gain / net inflow rate	Is the system "profiting" or "losing" now? At what rate?
$\tau_{\text{solid}}(L)$	persistence time of structured assets without net external input	What is the depreciation cycle / asset lifetime?
$\tau_{\text{active}}(L)$	consumption cycle of operational assets	How long can liquid reserves support operation?
$\eta(L)$	conversion efficiency of inputs to usable energy	What is the conversion rate / ROI / metabolic efficiency?
Existence degree Ψ	a sustainability KPI	Which KPI captures "structural return + operational return + real-time capture"?
Survival criterion	the bottom-line condition for survival	What is the domain analogue of $\frac{E_{\text{active}}}{\tau_{\text{active}}} + \eta e(t) \geq 0$?
Compatibility $\kappa(L)$	synergy / coherence among rules	Are there conflicts among rules? How to measure matching?
Toxic protocol L_{toxic}	destructive rules/elements causing collapse	Which rules/inputs are latent, destructive, and transmissible?

SETs Example 1: Economics

Domain mapping (simplified example):

- **Protocol set** L**:** market rules, contract law, corporate governance charters, monetary policy, accounting standards.
- **Form** M**:** firms, markets, financial products, business-cycle fluctuations.
- **Recursion depth** d**:** sole proprietorship ($d \approx 1$), partnership ($d \approx 2$), limited-liability company ($d \approx 3$), multinational group ($d \approx 4$), global economic system ($d \approx 5$).

- **Core fingerprint $C_{\mathcal{S}}$:** for the "public company" species, fingerprints include equity structure, disclosure obligations, board–management separation.
- **Energy E: monetary capital**.
  - E_{solid}: fixed assets, inventory, brand value, patent portfolio.
  - E_{active}: cash flow, working capital, short-term investments.
  - $e(t)$: net profit flow, net financing inflow (income $-$ expenses).
  - $E_{\text{locked}} = E_{\text{solid}} + E_{\text{active}}$: total assets.
- **$\tau_{\text{solid}}(L)$:** industry-average firm lifetime, asset depreciation cycles. (Note: this is τ_{solid}'s economic mapping; its mathematical role—measuring the intrinsic rate at which solidified assets convert into immediately usable resources—is identical across domains.)
- **$\tau_{\text{active}}(L)$:** cash conversion cycle, days sales outstanding.
- **$\eta(L)$:** ROA, ROIC.
- **Ψ:** economic value added (EVA) or risk-adjusted long-term NPV, reflecting a composite health metric of "asset return + liquidity return + real-time net inflow."
- **Survival criterion:** positive cash flow or maintainable via financing; otherwise bankruptcy.
- **Compatibility $\kappa(L)$:** match between corporate systems and local legal environment; post-merger cultural integration.
- **Toxic protocol L_{toxic}:** Ponzi rules, systemically risky derivative designs, corrupt rent-seeking institutions, insider-control loopholes.

SETs Example 2: Ecology

Domain mapping (simplified example):

- **Protocol set L:** interaction networks among species (predation, competition, mutualism, parasitism), matter cycling and energy-flow pathways, reproduction and heredity rules.
- **Form M:** organisms, populations, communities, ecosystem landscapes.
- **Recursion depth d:** organism ($d \approx 4$–5), population ($d \approx 6$), community ($d \approx 7$), ecosystem ($d \approx 8$) (see Definition A4).
- **Core fingerprint $C_{\mathcal{S}}$:** heritable core genetic protocol set; conserved DNA regions and key morphological/physiological traits.
- **Energy E: biomass and chemical energy**.
  - E_{solid}: standing biomass (individual mass, total population biomass), skeletal/woody structures, fat and other reserves not directly usable as ATP.
  - E_{active}: ATP, blood glucose, starch and other immediately usable energy currencies.

  - $e(t)$: instantaneous net rate (e.g., NPP), assimilation minus respiration.
  - $E_{\text{locked}} = E_{\text{solid}} + E_{\text{active}}$: total biomass energy.
- $\tau_{\textbf{solid}}(L)$**:** mean lifespan, biomass turnover time, fat breakdown rates.
- $\tau_{\textbf{active}}(L)$**:** time to exhaust usable reserves, metabolic turnover, ATP turnover time.
- $\eta(L)$**:** photosynthetic efficiency, assimilation efficiency, trophic-transfer efficiency.
- Ψ**:** ecosystem net productivity (productivity minus total respiration), intrinsic growth rate r, or resilience metrics.
- **Survival criterion:** $\frac{E_{\text{active}}}{\tau_{\text{active}}} + \eta e(t) \geq 0$: for individuals, sufficient ATP supply; for populations, birth rate $\geq$ death rate.
- **Compatibility** $\kappa(L)$**:** niche overlap, stability of symbioses, matching between invasive and native species.
- **Toxic protocol** $L_{\textbf{toxic}}$**:** extreme competitive strategies of invasive species (e.g., allelopathy), high-lethality infection mechanisms, bioaccumulation of persistent organic pollutants.

SETs Example 3: Cognitive Science and Skill Acquisition

Domain mapping (simplified example):

- **Protocol set** L**:** neural connectivity patterns, synaptic plasticity rules, cognitive schemas, procedural programs.
- **Form** M**:** skilled performance (e.g., fluent instrument playing, fast foreign-language reading).
- **Recursion depth** d**:** novice ($d \approx 7$), intermediate ($d \approx 8$), expert ($d \approx 9$) (see Definition A4).
- **Core fingerprint** $C_{\mathcal{S}}$**:** for the "piano performance" skill, fingerprints include automated fingering patterns, internalized harmonic progressions, sight-reading-to-action mapping protocols.
- **Energy** E**: attention and metabolic energy**.
  - E_{solid}: stabilized long-term skill representations (myelination level, synaptic strengthening), procedural memory.
  - E_{active}: working-memory capacity, currently deployable attention resources.
  - $e(t)$: time/attention flow invested in daily practice; learning efficiency.
  - $E_{\text{locked}} = E_{\text{solid}} + E_{\text{active}}$: total cognitive reserves.
- $\tau_{\textbf{solid}}(L)$**:** minimum practice frequency needed to maintain the skill; time for performance to decay below threshold without practice.
- $\tau_{\textbf{active}}(L)$**:** attention span; working-memory refresh cycle.
- $\eta(L)$**:** practice efficiency—skill improvement per unit practice time.
- Ψ**:** a composite of skill proficiency and performance stability.

- **Survival criterion:** $\frac{E_{\text{active}}}{\tau_{\text{active}}} + \eta e(t) \geq 0$: daily practice suffices to prevent regression.
- **Compatibility $\kappa(L)$:** match between new skills and existing schemas (positive vs. negative transfer).
- **Toxic protocol $L_{\textbf{toxic}}$:** over-solidification of erroneous motor patterns ("bad habits"), misleading pedagogies, vicious loops of performance anxiety.

6.3 The Two Fundamental Paths of GET Evolutionary Optimization

By Axiom III (the existence criterion), any system that seeks to maintain or improve its survival prospects has only two fundamentally optimizable paths, directly forced by the structure of the existence-degree equation:

$$\boldsymbol{\Psi_M(t)} = \underbrace{\frac{\boldsymbol{E}_{\text{solid}}(\boldsymbol{M},\boldsymbol{t})}{\boldsymbol{\tau}_{\text{solid}}(\boldsymbol{L})}}_{\text{passive structural return (not usable for immediate survival)}} + \underbrace{\left(\frac{\boldsymbol{E}_{\text{active}}(\boldsymbol{M},\boldsymbol{t})}{\boldsymbol{\tau}_{\text{active}}(\boldsymbol{L})} + \boldsymbol{\eta}(\boldsymbol{L}) \cdot \boldsymbol{e}(\boldsymbol{t})\right)}_{\text{path: maintain operational energy balance (the key to surv}}$$

Important notes

- The term $\frac{E_{\text{solid}}}{\tau_{\text{solid}}}$ is a structural return from solidified assets (e.g., a star's gravitational binding energy, a biological skeleton, a firm's brand assets). It **does not participate in immediate survival maintenance**, because conversion of solidified energy into active energy is too slow to meet immediate needs.
- Immediate survival is determined entirely by the latter two terms. The **survival criterion** is:
$$\frac{\boldsymbol{E}_{\text{active}}(\boldsymbol{M},\boldsymbol{t})}{\boldsymbol{\tau}_{\text{active}}(\boldsymbol{L})} + \boldsymbol{\eta}(\boldsymbol{L}) \cdot \boldsymbol{e}(\boldsymbol{t}) \geq \mathbf{0}.$$
  - If the sum is > 0, the system operates normally.
  - If it is $= 0$, the system is at critical balance.
  - If it is < 0, the system cannot be maintained and dies immediately.

The only optimizable path: maintain operational energy balance

Goal: ensure
$$\frac{\boldsymbol{E}_{\text{active}}(\boldsymbol{M},\boldsymbol{t})}{\boldsymbol{\tau}_{\text{active}}(\boldsymbol{L})} + \boldsymbol{\eta}(\boldsymbol{L}) \cdot \boldsymbol{e}(\boldsymbol{t}) \geq \mathbf{0}$$
holds.

Strategies

1. **Increase active-energy reserves:** increase E_{active} to provide a longer operational buffer.

2. **Shorten the active-state characteristic time constant:** optimize L to shorten $\tau_{\text{active}}(L)$, i.e., improve operational efficiency so each unit of active energy supports stronger real-time function.
3. **Increase capture efficiency:** optimize L to improve $\eta(L)$.
4. **Increase the scale of instantaneous net flow:** open new input channels to increase $e(t) = \phi_{\text{in}}(t) - \phi_{\text{out}}(t)$.

Physical meaning

- This is the system's **fundamental guarantee of immediate survival**.
- When $\eta \cdot e(t) < 0$, the system begins consuming E_{active} to compensate. As long as E_{active} remains sufficient, the inequality may still hold.
- If $\eta \cdot e(t) < 0$ persists, E_{active} declines; eventually the inequality fails and the system dies.
- **Solidified energy E_{solid} cannot help at that moment**, because its conversion rate to active energy is far too slow.

System death mechanism

- **Negative-flow phase:** $\eta \cdot e(t) < 0 \to$ consume E_{active}.
- **Critical point:** $\dfrac{E_{\text{active}}}{\tau_{\text{active}}} + \eta \cdot e(t) = 0$.
- **Death point:** $\dfrac{E_{\text{active}}}{\tau_{\text{active}}} + \eta \cdot e(t) < 0 \to$ immediate death.
- **After death:** the system disintegrates; its E_{solid} may be used by other systems, but the dead system cannot use E_{solid} to save itself.

Creating upgrade windows: synergy of the two paths

By the geometric selection principle (Definition B2), the prerequisite for a system to gain transition eligibility to depth $d+1$ is: on its net-borrowing energy curve $E(t)$, there exists a qualified window W formed by intersections with the energy threshold $\epsilon_{\text{th}}^{(d+1)}$ whose separation satisfies $\Delta t \geq N_{\text{min}}^{(d+1)}$.

The two paths jointly provide conditions for creating such a qualified window:

- Maintaining positive net flow (via $\eta \cdot e(t)$) ensures continuous accumulation so $E(t)$ can reach higher peaks.
- Enlarging active reserves E_{active} provides buffering so $E(t)$ can remain high long enough to meet $\Delta t \geq N_{\text{min}}^{(d+1)}$.

Thus, sustained optimization along these paths is necessary not only for maintaining current existence, but also for enabling recursive upgrade to higher complexity layers.

6.4 GET Open-Source Toolkit (get-tools) and a Verification Guide

To promote GET's application, testing, and collaboration, we propose the concept of an **open-source GET toolkit (get-tools)** together with a strict practical verification framework.

Conceptual core modules of get-tools

1. **SETs Builder:** guides users to build domain mappings,generating standardized SETs configuration files specifying mappings for $L, M, E_{\text{solid}}, E_{\text{active}}, e(t), d, C_{\mathcal{S}}, \tau_{\text{solid}}/\tau_{\text{active}}, \eta, \kappa$, etc.
2. **State-Function Calculator:** inputs system data (rule descriptions, resource flows, time series) and, under the state-function schema

$$\boldsymbol{\Omega_{\mathcal{S}}^{(M)}(t) = \Phi_{\mathcal{S}}(L, E_{\text{solid}}, E_{\text{active}}, e(t), t)},$$

automatically computes $E_{\text{solid}}(t)$, $E_{\text{active}}(t)$, $e(t)$, $\Psi_M(t)$, and the survival term $\frac{E_{\text{active}}}{\tau_{\text{active}}} + \eta e(t)$, and generates the net-borrowing curve $E(t)$ for geometric-selection analysis.
3. **Geometric-Selection Analysis Module:** for an input $E(t)$ curve, automatically detects intersections with thresholds $\epsilon_{\text{th}}^{(d)}$, computes qualified-window length Δt, compares with $N_{\text{min}}^{(d)}$, and outputs an upgrade-potential assessment.
4. **Dynamics Simulator:** based on a simplified numerical version of the layered evolution equations (Axiom V), allowing users to set initial layered distributions $P_d(\Omega, T)$, transition-rate parameters W, and differentiation rates $W_{\text{div}}^{(d)}$, and simulate macro-evolution trajectories and depth-distribution changes.
5. **Case Library & Comparator:** stores GET analysis results of historical cases (firm lifecycles, ecological succession, civilizational cycles, skill acquisition) and supports automatic comparison and pattern matching with new cases to identify similar existence-degree trajectories.
6. **Visualization Dashboard:** dynamically displays state-function evolution, existence-degree curves (decomposed into three contributions), survival-term monitoring, protocol-network topology, depth-layer distributions Σ_d, and species-diversity indices.

Practical verification framework for GET's application effectiveness

Verification of GET's predictive capability must follow a rigorous procedure to distinguish it from generic philosophical discussion:

1. **Case selection:** choose a natural or social system with clear boundaries and complete historical data (e.g., a firm lifecycle, a lake's ecological evolution, a skill acquisition process).
2. **SETs construction:** build precise, quantifiable GET mappings for the case—explicitly define measurement methods and data sources for $L, M, E_{\text{solid}}, E_{\text{active}}, e(t), d, C_{\mathcal{S}}$,

$\tau_{\text{solid}}(L), \tau_{\text{active}}(L), \eta(L)$—and produce a reproducible SETs document.

3. **Historical backtest ("fit"):** use the first half of historical data to run GET analysis and test whether it can:
   - correctly identify key turning points (rise, peak, crisis, upgrade success/failure);
   - show that fluctuations of $\Psi_M(t)$ correlate strongly with these turning points, and that periods where $\frac{E_{\text{active}}}{\tau_{\text{active}}} + \eta e(t) < 0$ correspond to disintegration/collapse events;
   - identify the key moments when the system obtained qualified windows and successfully executed the four-phase cycle;
   - reveal protocol-level root causes that mainstream analyses fail to isolate clearly.
4. **Predictive test (extrapolation): without accessing the second half of the data**, use the protocol parameters inferred from the first half ($\tau_{\text{solid}}(L), \tau_{\text{active}}(L), \eta(L)$) and the observed dynamics to forecast future evolution, including:
   - the trajectory of $\Psi_M(t)$ and its three-term decomposition;
   - the trajectory of $\frac{E_{\text{active}}}{\tau_{\text{active}}} + \eta e(t)$ and the possible time it crosses below zero;
   - whether new qualified windows and upgrade attempts will appear;
   - if no intervention occurs, when the system reaches the critical point $\frac{E_{\text{active}}}{\tau_{\text{active}}} + \eta e(t) = 0$.
5. **Comparison & evaluation:** compare GET predictions with:
   a. the actual outcomes in the second half of the historical record;
   b. the predictions of mainstream domain models under the same conditions.

   Evaluation criteria: whether GET shows significant advantages in explanatory depth (causal clarity), predictive accuracy (quantitative error), early-warning lead time (how early crisis signals appear), or the operability/effectiveness of strategy recommendations.

The seriousness of GET as a scientific framework lies in its willingness and ability to accept such repeatable empirical tests based on historical-data backtests and predictive comparisons. We invite researchers across domains to use this framework to analyze systems of interest and share validation results—whether supporting, refining, or challenging GET—in an open community.

get-tools will be released as open source; all code, documentation, and case data will be available in public repositories. This is not only a toolkit, but an open scientific experiment about the "grammar of existence."

7 Philosophical Implications and Scientific Revolution — A Fundamental Reshaping of Existence, Knowledge, and Methodology

GET is not an isolated physical hypothesis. All of its philosophical implications and revolutionary character are strictly derived from the **five empirical cornerstones** and the axiomatic system deduced from them. It represents a thoroughgoing naturalism and physicalism, while fundamentally reshaping and unifying traditional ontology, epistemology, and scientific methodology.

—

7.1 Ontological Revolution: From "Entities" to "Protocol-Recursive Processes," From "Matter" to the "Form–Protocol" Split

GET completes a profound ontological shift: it moves the foundation of existence from "entities" to **process**, and then specifies that process precisely as **protocol recursion**. On this basis, GET further clarifies the internal structure of "existence" itself.

The unique and generative ontology. GET's ultimate reality is a **memoryful, difference self-referential protocol-recursive dynamical system**, whose baseline running state is L_0 (Axiom Ω). What we observe as "vacuum" and "quantum fluctuations" is the macroscopic statistical steady state of L_0 (Definition A2). This means that existence itself is first and foremost a kind of "running" or "computation" with a specific logic.

The ontological distinction between form and protocol. GET makes a fundamental distinction within existence (Definition A1):

1. **Protocol set L:** the only dynamical ontology; the rule system itself—"how it runs."
2. **Form M:** the **stable, observable phenomenon set** presented during the recursive execution of L. Form is the **external manifestation type** of L's running.
3. **Endogenous state variables:** solidified-state energy E_{solid}, active-state energy E_{active}, instantaneous net flow $e(t)$, time t, spatial distance s, maximum propagation speed c, recursion depth d—all are endogenous dimensions describing the running state of L, generated directly by each execution of the difference function D, and have no independent ontological status.

This tripartite architecture dissolves the traditional philosophical dualisms of "matter and form" and "substance and attributes." **"What it is" (form M) is defined by "how it runs" (protocol set L); and the describable dimensions of "how it runs" are given by the endogenous variables $\{E_{\text{solid}}, E_{\text{active}}, e(t), t, s, c, d\}$.**

The complete demotion and endogenization of attributes. Basic quantities in traditional physics—mass, energy, time, space—are stripped of ontological primacy. In GET, they are **endogenous state variables** produced by the running of L_0 (Definition A1). For example, $E = mc^2$ is no longer an equivalence between two entities; it is an **efficiency relation** between different morphological stages of the same process (different degrees of protocol locking) (Cornerstone 4). Here the total structured asset

$$\boldsymbol{E}_{\text{locked}} = \boldsymbol{E}_{\text{solid}} + \boldsymbol{E}_{\text{active}}$$

corresponds to mass m, and the speed of light c is the universal conversion constant between them.

The recursive-grammar definition of existence. Any stable existence (form M) is a dynamical mode in which certain individuals in L_0, satisfying a particular **geometric selection condition** (Definition B2), **solidify** a protocol set L through the **four-phase cycle** (Definition C1). Its complete description is given by the state function:

$$\Omega^{(M)} = \Phi(L)$$

where Φ is the generating function of L: given L, $\Phi(L)$ outputs the complete state $\Omega^{(M)}$ of form M, including $E_{\text{solid}}(t)$, $E_{\text{active}}(t)$, and $e(t)$.

For different species $\mathcal{S}^{(d)}$, the state function is species-specific:

$$\boldsymbol{\Omega_{\mathcal{S}}^{(M)}(t) = \Phi_{\mathcal{S}}(L, E_{\text{solid}}(t), E_{\text{active}}(t), e(t), t), \quad \Omega_{\mathcal{S}}^{(M)} \in \Sigma_d^{(\mathcal{S})}.}$$

Continued existence of form M must satisfy the survival criterion:

$$\boldsymbol{\frac{E_{\text{active}}(M,t)}{\tau_{\text{active}}(L)} + \eta(L) \cdot e(t) \geq 0,}$$

where $\frac{E_{\text{solid}}}{\tau_{\text{solid}}}$ is the structural return from solidified assets, $\frac{E_{\text{active}}}{\tau_{\text{active}}}$ is the operational capacity provided by active assets, and $\eta \cdot e(t)$ is the immediate survival guarantee from real-time net flow.

This means that everything—from dark energy and electrons to organisms and cultural traditions—is, in essence, the same bottom-level recursive logic (L_0) stabilized at different "recursion depths" (d), through different "protocol fingerprints" ($C_{\mathcal{S}}$), as dynamic "standing waves" or "attractor modes." Differences among existents are not differences in "material," but differences in the complexity and recursion depth of the **protocol set** L they instantiate and execute, and the observable features of the resulting **form** M.

A unified explanation of species across depths. From fundamental particles ($d = 1$), baryons ($d = 2$), atoms ($d = 3$), molecules ($d = 4$), stars / proto-life ($d = 5$), multicellular organisms ($d = 6$), sentient organisms ($d = 7$), basic cognition ($d =$

8), intermediate cognition ($d = 9$), advanced professional capacity ($d = 10$), top-level paradigms ($d = 11$), to meta-paradigms ($d \geq 12$), all are unified by the same definitional framework of the "core protocol fingerprint $C_{\mathcal{S}}$."

—

7.2 Epistemological Revolution: Observation as Protocol Matching and Reconstruction

GET thoroughly physicalizes and operationalizes epistemology.

The nature of observation. Observation is precisely defined as: the process in which an observer (itself a system with a specific protocol set L_O) and an observed system (protocol set L_S) perform protocol matching, information decoding, and protocol reconstruction (Axiom IX). Observation changes the system's state distribution:

$$\boldsymbol{P_{\text{post}}(L) = \frac{P_{\text{pre}}(L) \cdot \exp[-\beta D(L \| L_O)]}{Z}},$$

where $D(L \| L_O)$ measures protocol difference, $\beta \propto E_{\text{obs}}$ is an observation-strength parameter, and Z is a normalization factor. Observation necessarily consumes the observer's active-state energy reserves E_{active} and satisfies the thermodynamic lower bound:

$$\boldsymbol{\Delta E_{\text{obs}} \geq k_B T \cdot D_{\text{KL}}(P_{\text{post}} \| P_{\text{pre}})}.$$

This means observation itself consumes the observer's $e(t)$ or E_{active} reserves.

A new reading of "objectivity." "Objective facts" are not "things-in-themselves" independent of observers; rather, they are system states whose **compatibility** κ with the protocol sets L_O of the vast majority of potential observers is extremely high. Physical laws (e.g., core protocols in $\mathscr{L}_1$) are "objective" and universal precisely because they are the most stable and reliable protocol patterns—those with the lowest matching cost—when any $d \geq 1$ observers interact. Objectivity arises from the **broad effectiveness of interaction protocols**, not from "pure existence" detached from all interaction.

Dissolving the quantum measurement problem. In GET, a quantum superposition is interpreted as a system being in an **un-solidified, competitive transient state** among multiple candidate protocol sets $\{P\}$. Measurement is the interaction between a macroscopic apparatus (with a highly stable protocol set L_O) and the measured system, which **triggers and selects** the protocol branch with the highest compatibility with L_O, completing its "solidification." The so-called "wavefunction collapse" is the microdynamical result of this protocol matching and solidification, requiring no non-physical "collapse postulate." Measurement consumes the apparatus's active-state energy E_{active}, which is also why measurement requires energy input.

Making quantum nonlocality transparent. Nonlocal correlations such as entanglement are explained as strong correlation records that L_0 maintains for spatially separated individuals in its **global state-management memory** (Axiom XIII.3). Their "instantaneous" updating does not violate the speed limit c, because maintenance and synchronization of the correlation are not implemented through spacetime signal propagation, but via L_0's globally consistent **write and read** of its own central state. This provides a causality-respecting, logically clear mechanistic explanation of quantum nonlocality.

The absolute limit of observation. By Corollary 13, an observer can stably observe only layers whose recursion depth is close to or lower than its own (Σ_d, $d \leq d_{\text{observer}}$). This limit is not merely technological; it is an ontological fact arising from protocol compatibility and energy constraints. Observation must consume the observer's finite E_{active}, which must be replenished through $\eta \cdot e(t) \geq 0$.

—

7.3 Methodological Revolution: Returning to Deductive Science and the Standard of Transparency

The construction of GET itself is a complete execution and revival of the golden-age methodology of science, marking a break with the currently prevalent "ad hoc fitting" paradigm (see Chapter 2).

Core methodological features:

1. **Start from consensus, not conjecture.** The only inputs are the **five physical cornerstones**, all firm consensus facts.
2. **Pure deduction with zero ad hoc assumptions.** From cornerstones to all conclusions, everything is connected by a logical chain; **any independent ad hoc assumption beyond the cornerstones is forbidden**.
   Canonical example: the geometric selection principle (Definition B2) is not an added assumption; it is a logically necessary deduction from Cornerstone 1 (Heisenberg constraint) and Cornerstone 2 (power-law statistics). The intersection test between a single-peaked energy curve and a threshold line is the intrinsic geometry of L_0's running.
3. **Logical transparency and algorithmic auditability.** The entire theory is clearly structured; each step from axioms to corollaries can be formalized and audited. Its falsification conditions are explicit and algorithmizable (Corollary 22). All energy variables E_{solid}, E_{active}, and $e(t)$ are endogenous outputs of the execution of D, with no external input assumptions.
4. **Pursue generative explanation.** GET's goal is not to fit data more precisely, but to explain how phenomena—including physical laws themselves—can be **generated**

from the underlying logic of the world.

A refreshed standard for evaluating scientific theories. By its own practice, GET proposes a stricter four-dimensional yardstick for theory evaluation:

Table 7: A four-dimensional yardstick for evaluating scientific theories

Yardstick	Core question	GET's practice
S1 Ontological minimalism	Is the dynamical ontology minimized?	Only L is ontology; $E_{\text{solid}}, E_{\text{active}}, e(t), t, s, c, d$ are endogenous variables generated by D
S2 Logical necessity	Are conclusions ad hoc fits or axiomatic necessities?	Geometric selection, the existence-degree formula, scale constraints, and the survival criterion are all logical corollaries
S3 Deductive transparency	Is the logical chain clear and auditable?	Cornerstones $\rightarrow$ definitions $\rightarrow$ axioms $\rightarrow$ corollaries; linear structure with no leaps
S4 Strong falsifiability	Does it make risk-bearing "all-or-nothing" predictions?	Dark matter zero interaction; $w \equiv -1$; no graviton; etc.

The verdict of paradigm comparison. Appendix B systematically argues that competing theories (ΛCDM, supersymmetry, string theory, quantum gravity, modified gravity) are significantly inferior to GET on one or more of S1–S4. This is not merely a difference in degree; it is a difference in paradigm: GET represents a **deductive generative paradigm**, while most others fall into an **ad hoc fitting paradigm** or variants. GET pursues the necessity of explanation; its conclusions (including dark matter's zero interaction and dark energy's exact $w \equiv -1$) are logically forced. Other theories often begin from mathematical aesthetics, unification desires, or local phenomena, and then introduce ad hoc assumptions or parameters to fit data; they may excel at descriptive precision but lack deep "how it is possible" explanations and inevitably introduce ontological redundancy.

—

7.4 Implications for the Future of Science

If GET withstands tests, it will exert a structural impact on the future of science.

7.4.1 Specific impacts on disciplines

Physics

- It defines gravity as a geometric effect at $d = 0$ (Corollary 1), dissolving the conceptual dilemma of "quantum gravity" (Corollary 14). Gravity's source is total structured assets $E_{\text{locked}} = E_{\text{solid}} + E_{\text{active}}$, corresponding to mass m under low-velocity approximation.

- It provides clear ontological explanations and ultimate testable predictions for dark matter and dark energy (Corollaries 2, 3, 4). Dark matter is frozen active-state energy E_{active}; dark energy is a uniformly frozen active-state background. Both have non-zero E_{active}, consistent with the third law of thermodynamics.
- It gives mechanistic explanations of foundational quantum issues (measurement, nonlocality) under the "protocol matching–global memory" framework.

Cosmology

- It unifies inflation, component ratios, structure formation, and the origin of the CMB into a single non-ad hoc generative narrative:

$$\begin{aligned} \boldsymbol{L_0} \text{ running} &\xrightarrow{\text{geometric selection}} \text{four-phase solidification} \\ &\xrightarrow{\text{channel closure}} \text{three-category freezing} \\ &\xrightarrow{\text{recursive growth}} \text{today's universe.} \end{aligned}$$

  Category A acquires total structured assets $E_{\text{locked}} = E_{\text{solid}} + E_{\text{active}}$; Category B freezes active-state energy E_{active}; Category C becomes the uniform background $E_{\text{background}}$.

Complex science and biology

- Life is defined as: **a protocol-recursive system** that can maintain the survival criterion
$$\boldsymbol{\frac{E_{\text{active}}(M,t)}{\tau_{\text{active}}(L)} + \eta(L)\cdot e(t) \geq 0,}$$
  and, via Path 1 (increasing E_{active} reserves) and Path 2 (maintaining $\eta \cdot e(t) \geq 0$), continuously obtains qualified windows to execute the four-phase cycle and explore higher recursion depth d (Corollary 16).
- Consciousness is the emergent capacity of such high-dimensional life forms to perform recursive operations (self-reference) on their own protocol sets, corresponding to higher recursion depth d.
- A biological species $\mathcal{S}^{(d)}$ is a stable form set defined by a **heritable core genetic protocol set** $C_{\mathcal{S}}$ (Definition A5). Evolution is adaptive reconstruction of protocol libraries under the survival criterion, with optimization directions including enlarging E_{active} reserves and maintaining $\eta \cdot e(t) \geq 0$.

Social science and information science

- Economic cycles, institutional evolution, technology diffusion, AI alignment, etc., can be unified under **protocol compatibility** κ, connection-capital accumulation

(Axiom VII), and toxic-protocol management (Axiom VIII).

- Social trust, legal institutions, and cultural traditions are high-recursion-depth protocol sets that obey the accumulation/depreciation dynamics of Definition D4 (Corollary 20). Maintaining them requires sustained investment of social active-state energy $E_{\text{active}}^{(\text{social})}$, ensuring the social survival criterion

$$\boldsymbol{\frac{E_{\text{active}}^{(\text{social})}}{\tau_{\text{active}}} + \eta \cdot e(t)_{\text{social}} \geq 0.}$$

- Radioactive nuclides, pathogens, computer viruses, and cultural dross are **toxic protocols** manifested at different recursion depths d, requiring corresponding isolation mechanisms (Corollary 21).

Philosophy of science

- It provides new criteria—based on **protocol compatibility** and **logical transparency**—for demarcation, theory choice, and debates between realism and anti-realism.
- "Objectivity" is redefined as "statistical stability of cross-observer protocol compatibility."

7.4.2 A new paradigm of scientific exploration and verification

Because GET is extremely structured and sharply falsifiable, it is well-suited to combine with **artificial intelligence** and **large-scale computation**, opening a new human–machine collaborative scientific paradigm:

1. **Automated logical auditing:** use formal verification tools to automatically check the rigor of the deductive chain from axioms to all corollaries.
2. **Systematic counterexample mining:** train AI models to systematically search for anomalous patterns or instances that might violate GET's core corollaries in large astronomical, particle-physics, biological, and social databases.
3. **Generative simulation and prediction:** based on the layered evolution equations (Axiom V), run generative simulations of complex systems (from galaxy formation to civilizational evolution), track the evolution of E_{solid}, E_{active}, and $e(t)$, monitor changes in $\frac{E_{\text{active}}}{\tau_{\text{active}}} + \eta \cdot e(t)$, and compare with real data to test and refine GET's parameters and models.
4. **Automated construction and verification of cross-disciplinary SETs:** assist researchers in rapidly building and testing GET mapping schemes (SETs) for emerging domains, accelerating cross-disciplinary knowledge transfer and unification.

—

7.5 Closing: An Open Experiment, a Choice of Road

GET is now presented in full. It begins from five consensual cornerstones and ends with a set of clear predictions that can be ultimately adjudicated. We reiterate the invitation that runs through the entire paper:

The survival right of the GET edifice is handed entirely to logic (internal consistency) and facts (external experience). We invite any explorer with rationality and rigor to:

1. **Audit the logical chain:** from the five cornerstones to the 22 corollaries—are there leaps or hidden assumptions?
2. **Test grammatical completeness:** can you find a stable existence (form M) that cannot be described by the state function $\Omega^{(M)} = \Phi(L)$, or whose survival necessarily violates
$$\boldsymbol{\frac{E_{\text{active}}(M,t)}{\tau_{\text{active}}(L)} + \eta(L) \cdot e(t) \geq 0?}$$
3. **Verify the strong predictions:** does dark matter have any non-gravitational interaction? is dark energy's w exactly -1? does a graviton exist?

Falsification conditions are explicitly given (Corollary 22):

- any of the five cornerstones is overturned;
- a stable form M exists that cannot be completely described by the state function;
- a definitive discovery of dark matter's non-gravitational interaction;
- a definitive confirmation that dark energy has $w \neq -1$;
- a definitive confirmation of superluminal causal energy/information transfer.

If GET is falsified by a clear counterexample, it will still have drawn a valuable boundary for "speakable existence," and its methodological legacy—its pursuit of logical transparency and strong falsifiability—will remain.

If it stands under all harsh tests, then what we gain is not only a new theory, but a verified viable road back to the essence of science: **use minimal axioms and transparent logic to deduce everything in this complex world.**

The universe may be an epic written in recursive grammar. GET is the first systematic translation and annotation we submit. From the eternal fluctuations of L_0, to the necessary verdict of geometric selection, to the recursive ascent of the four-phase cycle, to the fact that you and I read these words today through protocol matching—**all of this is different instances of the same grammar of existence.** Every act of reading consumes your active-state energy E_{active}, and only when

$$\boldsymbol{\frac{E_{\text{active}}(M,t)}{\tau_{\text{active}}(L)} + \eta(L) \cdot e(t) \geq 0}$$

can you continue this exploration of thought.

Now it is your turn—the reader, reviewer, and experimenter—to decide whether this translation is a masterpiece, or a draft that must be revised.

The experimental protocol is ready. The moment of judgement is coming.

Shall we walk together on this road forged by necessity?

A GET's Rewrite of Fundamental Physics Equations

—From Description to Generation

GET's core grammar:

$$\Omega^{(M)} = \Phi(L)$$

where $\Omega^{(M)}$ is the complete state function of form M (Definition A1).

According to Definition A1: the protocol set L is the only dynamical ontology; form M is the stable external manifestation of L's running; solidified-state energy E_{solid}, active-state energy E_{active}, instantaneous net flow $e(t)$, time t, and recursion depth d are all endogenous variables generated directly by execution of the difference function D. Every physical equation is a description of some aspect of some form M (a specific projection, approximation, or scale). Therefore, in principle all formulas in physics can be rewritten as expansions of $\Phi(L)$ under certain conditions. This is not merely possible but necessary—because L is the only ontology.

Below we rewrite several core equations.

—

A.1 The Mass–Energy Equation $E = mc^2$

Old-paradigm reading: energy and mass can be converted into each other; c^2 is a conversion factor.

GET rewrite:

From Definition A1:

- Energy includes solidified-state energy E_{solid} and active-state energy E_{active}; the total structured asset is $E_{\text{locked}} = E_{\text{solid}} + E_{\text{active}}$.
- Mass is the "solidified" manifestation that appears when total structured assets are highly structured and stably locked by the protocol set L.
- The speed of light c is the inherent maximum propagation speed of differences (Axiom XIII).

When a form transitions from the primordial excited state ($d = 0$, no structure, $E_{\text{solid}} \equiv 0$, all energy is borrowed as E_{active}) into ordinary matter ($d \geq 1$, structured, with part of energy converted into E_{solid}):

$$\boldsymbol{E_{\text{peak}} = E_{\text{locked}} + E_{\text{radiated}}, \quad E_{\text{locked}} = \eta E_{\text{peak}}, \quad E_{\text{radiated}} = (1-\eta) E_{\text{peak}}.}$$

For a form that has completed solidification, the relation between its total structured asset and its "inertial manifestation" is determined by the protocol set L. Under the low-velocity approximation:

$$\boldsymbol{E}_{\text{locked}} = \boldsymbol{E}_{\text{solid}} + \boldsymbol{E}_{\text{active}} \approx \boldsymbol{m \cdot c^2}.$$

This is not "mass–energy equivalence" between two ontologies; it is the morphological manifestation of the same process at different structuring stages. c^2 is not a mysterious coefficient, but an intrinsic efficiency scale of the L_0 system—an inherent ratio that converts "difference intensity" into "locked-state manifestation."

—

A.2 Einstein's Field Equation

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}.$$

Old-paradigm reading: matter tells spacetime how to curve, spacetime tells matter how to move.

GET rewrite:

By Corollary 1, gravity is a pure geometric effect at $d = 0$. When a form M has non-zero total structured assets $E_{\text{locked}} = E_{\text{solid}} + E_{\text{active}}$, it produces geometric curvature of the background spacetime metric.

Let spacetime be a low-energy effective description of the L_0 state space Σ. The metric $g_{\mu\nu}$ encodes the "protocol connection pattern" between points in Σ. When there is a distribution of total structured assets $\rho(x) = (E_{\text{solid}}(x) + E_{\text{active}}(x))/V$, curvature is determined by:

$$\mathcal{G}_{\mu\nu}[g] = \kappa \cdot \mathcal{T}_{\mu\nu}[\rho, L],$$

where:

- $\mathcal{G}_{\mu\nu}$ is the geometric tensor of the metric (Einstein tensor),
- $\mathcal{T}_{\mu\nu}$ is the energy–momentum tensor determined by the distribution of total structured assets and the dynamics of the protocol set L,
- κ is an intrinsic coupling constant of the L_0 system.

For Category A (ordinary matter, $d \geq 1$) and Category B (dark matter, $d = 0$), ρ produces positive curvature (gravitational attraction). For Category C (dark energy, $d = 0$), a uniform ρ produces negative pressure, appearing as the cosmological-constant term.

This is not a "modification" of general relativity; it explains why general relativity is what it is: it is the geometric manifestation of the L_0 system at the $d = 0$ layer.

—

A.3 The Dirac Equation

$$(i\gamma^\mu \partial_\mu - m)\psi = 0.$$

Old-paradigm reading: the relativistic quantum mechanics equation describing spin-$\frac{1}{2}$ particles.

GET rewrite:

An electron is a species $\mathcal{S}_{\text{electron}}$ defined by a core protocol fingerprint C_{electron} (Definition A5). Its state function is:

$$\boldsymbol{\Omega}_{\text{e}}^{(M)}(t) = \boldsymbol{\Phi}_{\text{e}}(\boldsymbol{L}_{\text{e}}, \boldsymbol{E}_{\text{solid}}(t), \boldsymbol{E}_{\text{active}}(t), \boldsymbol{e}(t), t).$$

In the low-energy approximation, Φ_{e}'s dynamics can be projected onto the spinor field $\psi(x)$. But ψ is not a "fundamental entity"; it is the spacetime manifestation of Φ_{e}:

$$\boldsymbol{\psi}(\boldsymbol{x}) = \hat{\boldsymbol{P}}_{\text{spacetime}}[\boldsymbol{\Phi}_{\text{e}}].$$

The Dirac equation is an effective equation derived from Φ_{e}'s recursive dynamics:

$$\boldsymbol{\mathcal{D}}[\boldsymbol{\Phi}_{\text{e}}]|_{\text{proj}} = \boldsymbol{0},$$

where:

- $\mathcal{D}$ is the recursive operator of Φ_{e} in the spinor projection,
- the electron's "mass" m is the projected manifestation of its total structured assets $E_{\text{locked}} = E_{\text{solid}} + E_{\text{active}}$,
- the electron's "spin" $\pm\frac{1}{2}$ originates from the symmetry-breaking mapping of the baseline fluctuation path winding number under L_0 during the first structuring phase transition (Corollaries 6 and 17).

Thus, the Dirac equation is not "fundamental"; it is a shadow of Φ_{e} under a specific projection.

—

A.4 Maxwell's Equations

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}, \quad \nabla \times \mathbf{B} - \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \mathbf{J}, \quad \nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0.$$

Old-paradigm reading: the fundamental laws of electromagnetic phenomena.

GET rewrite:

A photon is a species $\mathcal{S}_{\text{photon}}$ defined by a core protocol fingerprint C_{photon}. Its spin ± 1 originates from the symmetry-breaking selection mapping the winding number of baseline fluctuation paths during the first structuring phase transition (Corollary 6).

The electromagnetic field is the collective statistical manifestation of a large number of photon protocols Φ_{photon}. Maxwell's equations are macroscopic equations emerging from the recursive dynamics of Φ_{photon}:

$$\boldsymbol{\mathcal{M}[\{\Phi^i_{\text{photon}}\}]|_{\text{macro}} = 0,}$$

where $\mathcal{M}$ is the statistical operator describing collective photon behavior in the macroscopic field representation.

Thus, Maxwell's equations are not "fundamental"; they are the statistical regularities of collective photon protocol behavior.

—

A.5 The Schrödinger Equation

$$i\hbar\frac{\partial}{\partial t}\Psi = \hat{H}\Psi.$$

Old-paradigm reading: the fundamental equation of non-relativistic quantum mechanics.

GET rewrite:

By Axiom IX, observation is protocol matching. A quantum system's state function Ψ encodes the compatibility distribution between the system protocol set L_S and possible observation protocols L_O:

$$\Psi(L_O) \sim \kappa(L_S, L_O).$$

The Schrödinger equation describes how, in the absence of observation intervention, the recursive dynamics of L_S evolves this compatibility distribution:

$$i\hbar \cdot \mathcal{D}_t[\mathcal{K}(L_S, \{L_O\})] = \mathcal{H}[\mathcal{K}],$$

where:

- $\mathcal{K}$ is the compatibility kernel,
- $\mathcal{D}_t$ is the internal time derivative of L_S,
- $\mathcal{H}$ is the Hamiltonian operator in the compatibility space (a projection of L_S's dynamics),
- $\hbar$ appears because any protocol evolution has a minimum accounting cost (Definition A1).

Thus, the Schrödinger equation is not "fundamental"; it is an effective description of compatibility-distribution evolution.

—

A.6 The Lagrangian Density of Quantum Field Theory

$$\mathcal{L}[\phi, \partial_\mu \phi].$$

Old-paradigm reading: a field's dynamics is determined by the Lagrangian density.

GET rewrite:

The Lagrangian of quantum field theory is an effective description of a large set of protocol state functions under a "field projection":

$$\boldsymbol{\mathcal{L}[\phi] \quad \longleftrightarrow \quad \mathcal{L}_{\text{eff}}[\hat{P}_{\text{field}}[\{\Phi_{\mathcal{S}}^{(M)}\}]],}$$

where:

- $\{\Phi_{\mathcal{S}}^{(M)}\}$ are the protocol state functions of interacting species, each containing $E_{\text{solid}}, E_{\text{active}}, e(t)$,
- $\hat{P}_{\text{field}}$ is the projection operator into classical field description,
- $\mathcal{L}_{\text{eff}}$ is the effective Lagrangian of the collective behavior of these state functions.

The 19 free parameters of the Standard Model are, under the GET framework, reduced to:

- intrinsic constants of the L_0 system ($\hbar, c, \alpha$, etc.),
- symmetry-breaking choices at the first structuring phase transition (Corollaries 6 and 17),
- coupling constants determined by each species' core protocol fingerprint $C_{\mathcal{S}}$,
- allocation ratios between E_{solid} and E_{active} for each species.

—

A.7 Running of Gauge Couplings

In the Standard Model, gauge couplings run with energy scale via renormalization-group equations. In GET, the evolution of each gauge group is determined by its corresponding protocol fingerprint:

$$\alpha_i(\mu) = \frac{\alpha_i(\mu_{C_{\mathcal{S}_i}})}{1 - \frac{b_i(C_{\mathcal{S}_i})}{2\pi}\alpha_i(\mu_{C_{\mathcal{S}_i}}) \ln\left(\frac{\mu}{\mu_{C_{\mathcal{S}_i}}}\right)},$$

where:

- $i = 1, 2, 3$ correspond to $\mathrm{U}(1)_Y$, $\mathrm{SU}(2)_L$, $\mathrm{SU}(3)_c$;
- $\alpha_i(\mu) = g_i^2(\mu)/(4\pi)$ is the gauge coupling strength of group i at scale μ;
- $\mu_{C_{\mathcal{S}_i}}$ is the characteristic energy scale locked by the core protocol fingerprint $C_{\mathcal{S}_i}$ (for electroweak interaction, $\mu_{C_{\mathcal{S}_2}} \approx M_Z$; for strong interaction, $\mu_{C_{\mathcal{S}_3}} \approx \Lambda_{\mathrm{QCD}}$);
- $b_i(C_{\mathcal{S}_i})$ is the one-loop coefficient of the renormalization-group β function determined by the fingerprint $C_{\mathcal{S}_i}$, reflecting the corresponding particle content and symmetry structure;
- $\alpha_i(\mu_{C_{\mathcal{S}_i}})$ is the intrinsic coupling strength at the fingerprint characteristic scale, an inherent attribute of $C_{\mathcal{S}_i}$.

The 2π factor in the denominator originates from the geometric property under the projection onto a 2D submanifold of state space (Definition A1.e): the statistical average of the ratio between the lifecycle path's circumference and diameter of fluctuation individuals necessarily converges to π, and its reciprocal naturally appears in the renormalization-group flow as a 2π factor.

This form indicates: what appear as independent running parameters in the Standard Model (reference scales, reference coupling values, β-coefficients, asymptotic behavior, etc.) are, respectively, necessary manifestations of the corresponding core protocol fingerprints $C_{\mathcal{S}_i}$ under different energy projections. Differences among gauge-group runnings arise from the independent statistical competition among different protocol paths for geometric selection windows (Definition B2) in the first solidification events (Definition B5), requiring no additional free parameters.

—

A.8 Common Features of the Rewrites

All these rewrites follow the same pattern:

Table 8: Rewriting physics equations under the GET framework

Original equation	GET rewrite
$E = mc^2$	Morphological manifestation of the same process at different structuring stages; $E_{\text{solid}} + E_{\text{active}} \approx mc^2$; c is the inherent maximum propagation speed of protocol-state differences
Einstein field equation	$d = 0$ geometric effect determined by the distribution of $E_{\text{solid}} + E_{\text{active}}$; gravity source is total structured assets
Dirac equation	Shadow of electron protocol Φ_{e} under spacetime projection; mass from total structured assets; spin $\pm\frac{1}{2}$ from symmetry-breaking mapping of winding number in the first structuring phase transition
Maxwell equations	Statistical regularities of collective photon protocol behavior; spin ± 1 from symmetry-breaking choice of winding number mapping
Schrödinger equation	Effective description of compatibility-distribution evolution; $\hbar$ as L_0's minimal accounting unit
QFT Lagrangian	Collective manifestation of multi-species protocol state functions $\{\Phi^{(M)}_{\mathcal{S}_j}\}$ under field projection; fingerprints $C_{\mathcal{S}_j}$ determine coupling modes
Gauge-coupling running	Different gauge groups correspond to different fingerprints $C_{\mathcal{S}_i}$; $\mu_{C_{\mathcal{S}_i}}$, $b_i(C_{\mathcal{S}_i})$, intrinsic α_i are fingerprint properties; 2π from underlying geometry

—

A.9 Why This Is Necessary

By Definition A1:

$$\Omega^{(M)} = \Phi(L).$$

Any speakable existence has its complete state given by this form. This is a **holographic equation**:

- All equations in physics are descriptions of some aspects of some form M (specific projections, approximations, scales).
- They must be expansions of $\Phi(L)$ under certain conditions.
- Because L is the only ontology, every equation describing the running of L can, in principle, be rewritten as some form of $\Phi(L)$.

Old theories ask: how do these equations operate?

GET answers: these equations are manifestations of how the L system runs at a particular recursion depth, for a particular protocol fingerprint, under a particular projection.

Old theories ask: why these equations rather than others?
GET answers: because L runs that way, and humans read off its running logs.

—

B Paradigm Comparison from First Principles

—Why GET Is the Only Grammar of Existence that Satisfies Occam's Razor and Strong Falsifiability

B.1 Overview: The Yardsticks of Comparison

As emphasized throughout, GET is built under a serious scientific methodology of "from empirical facts to axiomatic deduction" (see Chapter 2). This methodology itself provides the core meta-scientific yardsticks for judging any "ontology" or "unified theory," especially Occam's razor (parsimony) and Popper's criterion of falsifiability[50] (clear risk). GET's construction is the conscious practice and extreme execution of these yardsticks.

Accordingly, any theory that claims to explain "existence" or pursues "the principle of everything" should accept the following four yardsticks:

Table 9: Systematic paradigm analysis of competing theories

Theoretical framework	**Paradigm analysis**
1. ΛCDM standard cosmological model	**S1 (ontological minimalism):** poor. Introduces three independent entities with distinct properties: ordinary matter, cold dark matter, and the cosmological constant Λ. Ontological count is large; the latter two are defined by observational fitting, with no unified explanation. **S2 (zero ad hoc logic):** poor. Highly ad hoc. Dark-matter particles and Λ are **fit parameters** introduced to fit phenomena, not deduced from more basic principles. **S3 (deductive transparency):** medium. As a phenomenological model, the mathematical form is clear. But no deductive chain exists from first principles to derive dark matter and Λ. **S4 (strong falsifiability):** weak. Predictions are flexible and adjustable. Dark-matter detection has expected cross-section ranges but no absolute "zero interaction" assertion; Λ's constancy is assumed.

Table 9 —— continued from previous page

Theoretical framework	Paradigm analysis
2. Supersymmetry / Grand Unified Theories (SUSY/GUTs)	**S1:** medium/poor. Aims at unification but at the cost of introducing many **new particles** (superpartners), significantly increasing entity count. **S2:** poor. Introduces supersymmetry as an **ad hoc mathematical symmetry**; the breaking mechanism requires further ad hoc additions. **S3:** medium. Derivations from supersymmetry algebra to particle spectra are clear. But there is no logical argument for why the physical world **must** have supersymmetry. **S4:** weak/medium. Predicts discovery of superpartners at certain scales, but the scale is tunable. Non-discovery does not kill the theory; it often just raises the expected scale.
3. String theory / M-theory	**S1:** extremely poor. Ontology shifts from point particles to 1D strings in higher-dimensional spacetime, introducing **extra compact dimensions**. Ontological complexity is extremely high. **S2:** poor. Basic assumptions (existence of 1D extended objects; 10/11-dimensional spacetime) are **highly ad hoc**. Contains many free parameters and a vast landscape; fitting reality invokes anthropic reasoning. **S3:** poor. Mathematics is extremely complex; the logical chain from known world features to string-theory assumptions is **broken and non-transparent**. **S4:** extremely weak. The well-known "non-falsifiability" problem: the landscape provides near-infinite possible solutions, losing the ability to make unique, testable predictions.
4. Quantum gravity theories (e.g., Loop Quantum Gravity, LQG)	**S1:** medium. Attempts to quantize spacetime itself; ontology can be viewed as discrete geometric quanta, simpler than string theory.

Table 9 —— continued from previous page

Theoretical framework	Paradigm analysis
	S2: medium/poor. The starting point "spacetime geometry must be quantized" is, under GET, a **category error** (Corollary 14). Presupposing that gravity and quantum theory must be unified is an ad hoc philosophical preference, not a deduction from empirical facts. **S3:** medium. The mathematical framework is self-consistent, but the chain from empirical foundations to the starting point is fundamentally disputed. **S4:** weak/medium. Predicts discreteness or singularity resolution at Planck scales, extremely hard to observe; lacks "all-or-nothing" strong predictions under current technology.
5. Modified gravity theories (e.g., MOND, $f(R)$ theories)	**S1:** indeterminate. Some models keep only ordinary matter plus a modified gravitational law and appear more parsimonious in ontology, but the modified laws are often very complex and not universal. **S2:** poor. The modified formulas are typically **ad hoc** adjustments to fit observations at specific scales, lacking logical necessity from first principles. **S3:** medium. Mathematical forms are clear, but motivations and formula origins are ad hoc, lacking deep deductive chains. **S4:** medium. Often makes predictions distinguishing from ΛCDM, but with tuning room and difficulty explaining all scales simultaneously.

B.2 Conclusion: An Irreversible Paradigm Leap

From the systematic comparison above, all analyzed competing theories are significantly inferior to GET on one or more of S1–S4. This inferiority is not merely a difference of degree, but a difference of paradigm:

- **GET represents a "deductive generative paradigm."** Starting from unshakable empirical consensus (the five cornerstones), it deduces the world's generative grammar and all phenomena through transparent logic. It pursues the **necessity of explanation**, and all its conclusions (including dark matter's zero interaction and dark energy's exact $w \equiv -1$) are logically forced results.

- **Other theories mostly belong to an "ad hoc fitting paradigm" or its variants.** Starting from mathematical aesthetics, unification desire, or local phenomena, they construct frameworks and introduce ad hoc assumptions or parameters to fit data. They often excel in **descriptive precision** but lack deep necessity in "how it is possible," and ontologically introduce redundancy.

Under Occam's razor, GET—by its extreme ontological minimalism (S1: only L as ontology; all energy variables endogenous) and zero–ad hoc logic (S2: geometric selection as necessary; three categories as necessary outcomes)—becomes the only theory that can claim to reveal a "grammar of existence" at present.

Therefore, the ultimate challenge we issue to the scientific community is:

Please provide an alternative "ontology" or "unified theory" framework that is **fully superior to or equal to GET** on the four meta-scientific yardsticks—**S1 (ontological minimalism), S2 (zero ad hoc logic), S3 (deductive transparency), and S4 (strong falsifiability)—while retaining all successful predictions of GET**; or strictly prove that "protocol-recursive monism" and its derived concepts (recursion depth d, protocol fingerprint $C_{\mathcal{S}}$, geometric selection principle, etc.) are logically redundant.

If no one can propose such a framework, GET will continue to hold its ultimate methodological advantage. The outcome of this contest will not be decided by authority, popularity, or mathematical complication, but by **logical rigor, ontological parsimony, and the sharpness of predictions**—the purest and hardest core of the scientific spirit.

C Why Physics Alone Cannot Solve the Mystery of Consciousness

—And how Generalized Existence Theory (GET) dissolves the dilemma

C.1 Core Claim

Physics, constrained by its third-person objectivity paradigm, cannot by itself resolve the ”first-person subjective experience” mystery of consciousness. GET reveals that the so-called ”mystery” comes from humans special-casing themselves: consciousness is an endogenous property of protocol-recursive systems, showing different manifestations at different recursion depths.

C.2 Contributions and Limits of Three Physicists

Table 10: Contributions and structural limits of three physicists

Physicist	**Core contribution**	**Structural limitation**
Roger Penrose	Sensed the ”non-computability” aspect of consciousness	Tried to fill the ”slot” with a physical mechanism
Max Tegmark	Thoroughgoing structuralism	Confused ”description” with ”existence”
(Fernando) Pastawski	Black-hole information encoding	Stayed at physics’ boundary, unable to discuss consciousness itself

Shared insight: From different angles, all three top physicists touched the essential boundary of consciousness, yet could not cross it due to disciplinary inertia—they still tried to ”find a place” for consciousness within the old paradigm.

C.3 Root of the Dilemma: Starting from the ”Separation” Presupposition

These discussions implicitly adopt a mistaken presupposition: first there are isolated individuals, then one must explain how they ”communicate” into unified consciousness. This inevitably generates an infinite regress.

GET points out: at the L_0 layer, there is no absolute separation of "this" and "that"—they are concurrent temporary states maintained within the same global memory system. Distinctions among individuals are internal labels used by L_0 to manage concurrency, not independent existents in an ontological sense.

C.4 Paradigm Shift: From the "Communication Problem" to the "Manifestation Problem"

Table 11: Paradigm shift between the traditional frame and the GET frame

Dimension	**Traditional frame**	**GET frame**
Starting point	Separate individuals	Unified L_0 global memory
Core question	How do separate individuals communicate?	How does a unified whole manifest as separation?
"Consciousness mystery"	How do fragments assemble into a whole?	How does the unified recognize unity itself within unity?

C.5 State Function: Mathematical Evidence of a Common Origin for All Things

GET's core grammar:

$$\Omega^{(M)} = \Phi(L)$$

Every existence—from quantum fluctuations to human consciousness—is an instance of this form. Expanded:

$$\boldsymbol{\Omega_{\mathcal{S}}^{(M)}(t) = \Phi_{\mathcal{S}}(L, E_{\text{solid}}(t), E_{\text{active}}(t), e(t), t)}$$

where:

- E_{solid}: solidified-state energy—long-term memory, stabilized structure
- E_{active}: active-state energy—currently deployable resources
- $e(t)$: instantaneous net flow—the real-time balance of exchange with the environment
- Survival criterion: $\dfrac{E_{\text{active}}}{\tau_{\text{active}}} + \eta \cdot e(t) \geq 0$ governs all

C.5.1 Consciousness: Endogenous Property, Not a Human Privilege

Key insight: consciousness is not a mysterious "emergence"; it is the direct manifestation of L_0's self-referential nature—only presented differently at different recursion depths.

Table 12: Recursion depth and how "sense of self" manifests

Recursion depth	**Form of existence**	**How "sense of self" manifests**
$d = 0$	Quantum fluctuations (L_0 baseline state)	**The most primordial "here"**: each fluctuation has its own state σ_t, history $E(t)$, trajectory s, and is distinguished from other fluctuations. This is the seed of "feeling"—no reflection, but distinction.
$d = 1$	Fundamental particles	Stable "identity" created by solidified protocols: an electron is always an electron, a proton always a proton. Sense of self solidifies from momentary fluctuation into persistent individuality.
$d = 2$–4	Atoms, molecules	An "inside" of composite structures: electron clouds and chemical bonds constitute primitive internal states.
$d = 5$–6	Living organisms	The "effort" to maintain the survival criterion: feeding, metabolism, seeking benefit and avoiding harm. Sense of self shifts from passive being to active maintenance.
$d = 7$–8	Sentient organisms	"Mapping" enabled by nervous systems: sensing environment and sensing internal states.
$d = 9$–10	Human-level consciousness	"Reflection" enabled by metacognition: not only sensing, but sensing oneself sensing.
$d \geq 11$	Civilizational intelligence	Collective self-reference: culture, institutions, and science knowing and revising themselves.

Key conclusions:

1. **Continuity:** from $d = 0$ to $d \geq 11$, there is no hard break of "conscious vs. unconscious," only differences in *how* self-sense manifests.
2. **Endogeneity:** consciousness is not externally added; it is the natural unfolding of L_0's self-reference at increasing complexity. The ability of quantum fluctuations to

distinguish "self" and "non-self" is already the most primordial "sense."

3. **Dissolving anthropocentrism:** humans smuggle "reflective self-awareness" as the whole of "consciousness," then declare other existences unconscious. This is like claiming "only poets have language."

C.5.2 The Place of Disciplines Within the GET Framework

Table 13: Disciplines and their roles under GET

Discipline	**Traditional role**	**Its place under GET**
Physics	Study fundamental constituents of matter	Study $\Phi(L)$ at low d (fluctuations, particles, fields)
Biology	Study life phenomena	Study $\Phi(L)$ at medium d (metabolism, reproduction, evolution)
Psychology	Study mind	Study higher-d $\Phi(L)$ and its self-referential dynamics
Literature	Describe subjective experience	Display the "first-person texture" of $\Phi(L)$ in time flow—movement of E_{active}, awakening of E_{solid}, fluctuations of $e(t)$
Philosophy	Guard the boundary of thought	Reveal the boundary $\Phi(L)$ must meet when it self-refers

They have always been describing the same thing. No stitching is required—only **recognition**.

C.6 Conclusion: The Trap of Blind-Men-and-Elephant, and Its Dissolution

The physicists' discussions eloquently show that consciousness research must be cross-disciplinary. But they do not answer: **how can those disciplines be integrated?**

Because they remain trapped in the "separation presupposition"—believing disciplines are separate and must be "communicated." GET says: they were never separate. They all describe the same $\Phi(L)$ at different recursion depths d.

Consciousness is not a mystery but the grammar's self-manifestation at every depth—from the quantum fluctuation's "here" to the human civilization's "reflection." Humans mistook their own special manifestation as the standard and called other depths "unconscious." That is a category error.

The state function

$$\Omega^{(M)} = \Phi(L)$$

is the mathematical signature of "starting from the whole"—generated directly by each execution of the difference function D, running through every moment from vacuum fluctuation to self-awareness. Every moment has its own "sense." Humans merely special-cased it.

—

D GET's Natural Dissolution of Unsolved Mysteries in Science

—From the Five Cornerstones to the Only Generative Grammar

D.1 Preface: What Counts as an "Unsolved Mystery"?

What contemporary science calls "unsolved mysteries" are often those that cannot, under the current paradigm (ad hoc fitting + parameter retreat), be unified with zero ad hoc assumptions and logical necessity. GET takes the five empirical cornerstones as its only inputs, anchors itself in the meta-philosophical stance "rule execution is phenomenon," adopts $\Omega^{(M)} = \Phi(L)$ as the only grammar of existence, and forcibly deduces a generative path for all phenomena.

This appendix selects top-tier unsolved mysteries widely recognized as lacking consensus answers (as of the end of 2025) and shows: within GET, they are not "mysteries," but necessary manifestations of the same bottom-level grammar. All answers are deduced directly from the cornerstones with zero ad hoc additions—no new entities, parameters, or mechanisms.

D.2 Unsolved Mysteries in Physics and Cosmology

D.2.1 Quantum gravity and Planck-scale spacetime structure

Traditional puzzle: How to unify quantum mechanics and general relativity? Is spacetime discrete at the Planck scale? Do gravitons exist?

GET dissolution: Gravity is a pure geometric effect at $d = 0$ (Corollary 1; Appendix A.2), while quantum phenomena are behaviors of solidified protocols at $d \geq 1$ (Axiom X). Forcing them into a single "quantum gravity" theory is a category error (Corollary 14). Spacetime is a low-energy effective projection of the L_0 state space Σ, needing no extra quantization. The Planck scale is the geometric manifestation of t_P (the minimal iteration unit).

D.2.2 The nature of dark matter

Traditional puzzle: What particle is dark matter? Why does it only participate in gravity?

GET dissolution: Category B is a $d = 0$ frozen primordial excited state; its active-state energy E_{active} is permanently frozen, with no $d \geq 1$ protocol medium (Definition B6;

Axiom X) $\rightarrow$ its non-gravitational interaction cross section is strictly zero (Corollary 2). This is ontologically necessary, not "we just haven't found the particle yet."

D.2.3 The nature of dark energy and the cosmological constant problem

Traditional puzzle: What is dark energy? Why is $w \approx -1$? Why is the density so precise?

GET dissolution: Category C is a uniformly frozen background of sub-threshold fluctuations plus instantaneous freezing of radiation redshift loss $Q_{r\to\Lambda} \approx 4H\rho_r \rightarrow \rho_\Lambda$ is strictly constant and $w \equiv -1$ (Corollary 3; Axiom XI). No fine-tuning, no new scalar field.

D.2.4 The mechanism and origin of inflation

Traditional puzzle: What is the inflaton field? Why does inflation last just right (e.g., 60 e-folds)?

GET dissolution: The first batch of fluctuations satisfying geometric selection ($\Delta t \geq N^{(1)}_{\min}$) solidify collectively $\rightarrow$ triggers exponential expansion; duration $\tau_{\text{inflation}} = N^{(1)}_{\min} \cdot t_P \approx 10^{12} t_P \approx 10^{-32}$ s (Definition B4). Zero extra fields, zero fine-tuning.

D.2.5 The Fermi paradox / the Great Silence

Traditional puzzle: Why no signals from extraterrestrial civilizations?

GET dissolution: Recursion depth has a theoretical upper bound $D_{\max}$ (Corollary 12) forced by the monotonic decrease of the structured usable energy pool $\mathcal{E}_{\text{struct}}(T)$. Approaching the bound $\rightarrow$ expansion dilutes $E_{\text{active}} \rightarrow \Psi < 0 \rightarrow$ inevitable decline or self-limitation. Interstellar connection capital decays with distance squared as a geometric necessity.

D.3 Unsolved Mysteries of Life and Consciousness

D.3.1 Origin of life

Traditional puzzle: the exact chemical/physical mechanism from non-life to life?

GET dissolution: The $d = 5$ layer (stars / proto-life) arises via geometric selection + four-phase-cycle solidification of a heritable core protocol $C_{\mathcal{S}}$ (Definition A5). Life = a $d \geq 5$ protocol-recursive system that can maintain the survival criterion and continuously obtain qualified windows (Corollary 16).

D.3.2 The hard problem of consciousness

Traditional puzzle: why subjective experience? how do physical processes produce first-person experience?

GET dissolution: Consciousness = a $d \gg 1$ (typically $\geq$ 7–8) protocol set L executing $\Phi_{\text{conscious}}(L)$ and recursively reading/reconstructing/evaluating its own protocol—direct manifestation (Appendix C.5.4). No hard problem—only a depth problem (Appendix C.5.3 paradigm shift).

D.4 Mathematics and Foundational Theory Mysteries

D.4.1 The ultimate meaning of Gödel's incompleteness theorem

Traditional puzzle: Is there an absolutely complete mathematical system?

GET dissolution: Any finite-d protocol set L is necessarily incomplete (cannot contain all higher-order self-reference). Absolute truth exists only in the infinite recursive limit $d \to \infty$ (the eternal running of L_0 itself); finite forms can only approximate.

D.5 Other Cross-Domain Hard Problems

D.5.1 Origin of the arrow of time

GET dissolution: Time = the monotonic count of L_0's memory growth (Axiom Ω.2). Irreversibility comes from non-erasable memory growth.

D.5.2 Causality and the speed-of-light limit

GET dissolution: $c = \max \|e(k)\|/t_P$ is the propagation upper bound of the L system (Definition A1.h + Axiom XIII).

D.5.3 Wormholes / CTCs / time travel

GET dissolution: Wormholes are $d = 0$ geometric solutions with no traversable $d \geq 1$ protocol medium (Axiom X). Closed timelike curves violate the monotonicity of memory growth (Axiom Ω.2) → pure mathematical fantasy.

D.5.4 Multiverse and the anthropic principle

GET dissolution: Universes sharing the same L_0 and fluctuation statistics are highly similar (Corollary 15). Radically different universes are rare; no anthropic principle is needed.

D.6 Conclusion: A Summary of GET's Dissolution Capacity

Table 14: Summary of GET's dissolution of top-tier unsolved mysteries

Mystery category	**Count**	**Naturally dissolved by GET**	**Dissolution rate**
Physics / cosmology	5–7	5–7	100%
Life / consciousness	2–3	2–3	100%
Mathematics / foundations	1–2	1–2	100%
Cross-domain	4–5	4–5	100%

Note: all are deduced with zero ad hoc additions.

Ultimate conclusion: all top-tier unsolved mysteries of contemporary science are naturally dissolved under GET's single grammar of existence. They are not "mysteries," but necessary manifestations of the same bottom-level protocol-recursive process $\Omega^{(M)} = \Phi(L)$ at different recursion depths d.

GET is not "yet another explanation," but the only generative grammar deduced with zero ad hoc additions from the five cornerstones. Any stance that still calls them "mysteries" must first prove that one of the five cornerstones is false, or point out a logical break in the deductive chain.

Open challenge: We welcome any scholar, institution, or AGI to use this appendix as a baseline and test point-by-point against their own frameworks: who can, without introducing ad hoc entities/parameters, dissolve all the above mysteries with equal parsimony and necessity?

—

E GET Reading and Verification Guide

—You are not learning; you are recognizing

E.1 Preface: This Is Not a User Manual

Most guides on the market teach you how to ”use” something.

But GET is not a tool, not a framework, not a theory—GET is the **grammar of existence itself**.

You don’t need to learn how to ”use” it, because you have been speaking it all along. From the moment you were born—hungry, tired… you have been repeating this grammar with your life.

You just didn’t know it before.

The task of this guide is not to teach you something new, but to help you **recognize**—to recognize that what the hydrogen atom says, what a startup company says, what an intimate relationship says, what you yourself say, is actually the same sentence.

—

E.2 Part I: Four Entry Questions — Recognize Your Seat

These four questions are not meant to make you ”accept” anything; they are meant to let you discover where you already stand.

Question 1: Which comes first—phenomenon or law?

You are using language to read this sentence. Language has grammar. Grammar is not a text written somewhere; it is the structure of understanding itself as you understand it at this moment.

This is neither ”phenomenon first” nor ”law first”—this is **rule execution is phenomenon**. The fact that you can read and understand this sentence proves that rules and phenomena are the same process.

You are already standing at this position.

Question 2: In Heisenberg uncertainty, is Δt a ”point” or a ”process”?

You cannot even imagine the existence of an ”instant” as a point. Any ”existence” means sustaining a process for some duration.

This is not a physics conclusion—this is your intuition of your own existence.

You are already standing at this position.

Question 3: Do laws exist objectively prior to observers?

Before you were born, apples fell downward. After you die, they will still fall.

This is not something you ”believe”; it is something you assume by default—otherwise you wouldn’t dare step outside.

You are already standing at this position.

Question 4: Can true vacuum be strictly regarded as a ”system”?

”Empty” is not empty—because rules are running and processes are happening.

This is not hard to understand, because your own body is also ”empty”—mostly space—but rules are running, so you are alive.

You are already standing at this position.

—

E.3 Part II: You Now Have Two Choices

If you have read this far, you are already at the entrance.

Next, you can choose:

Choice 1: Treat everything that follows as ”just another theory”—evaluate, doubt, find flaws, compare with existing knowledge. This is the ”external viewpoint.”

Choice 2: Stop and ask yourself one sentence:

> **”If these things really hold—if existence really is rules running—what would I see?”**

Choice 2 is what truly enters the guide.

—

E.4 Part III: The Five Cornerstones — Not What You Believe, but What You Cannot Deny

These five are not ”theoretical assumptions.” They are facts verified by countless experiments. You cannot deny them, because denying them is denying your way of understanding the world.

Table 15: The five cornerstones and your intuition

Cornerstone	Content	Can you deny it?
Cornerstone 1	Heisenberg uncertainty	You cannot imagine a point-instant as existence
Cornerstone 2	Extreme-number power-law statistics	You assume ”rare events happen rarely”
Cornerstone 3	The three laws of thermodynamics	You never truly thought perpetual motion could work
Cornerstone 4	Mass–energy equivalence	You assume mass is related to energy
Cornerstone 5	Rule dependence of structured existence	You are using rules to understand this sentence right now

You are not ”believing” them—you simply cannot deny them.

—

E.5 Part IV: Core Concepts — You Are Not Learning; You Are Recognizing

Table 16: Core concept mapping table

Concept	Symbol	Definition	The version you already know
Protocol set	L	the rule system itself	your habits, your way of thinking
Form	M	the external manifestation of rules running	the you others see
Instantaneous difference	$e(t)$	current net gain	how are you today?
Net borrowing balance	$E(t)$	accumulated energy	your total accumulation so far

Table 16 —— continued from previous page

Concept	**Symbol**	**Definition**	**The version you already know**
Solidified-state energy	E_{solid}	locked assets	your experience/knowledge/memory—useless when you're exhausted
Active-state energy	E_{active}	deployable energy	your current energy/attention
Survival criterion	$\frac{E_{\text{active}}}{\tau_{\text{active}}} + \eta \cdot e(t) \geq 0$	can you keep going?	can you still hold on?
Solidified time constant	τ_{solid}	how fast assets depreciate	how long until you forget if unused
Active time constant	τ_{active}	how fast energy is consumed	how long can one deep-work session last
Capture efficiency	η	input–output efficiency	how much do you retain from an hour of study

—

E.6 Part V: The Four Principles of Existence — The Bottom-Level Code

Principle E.1 (Origin: Difference is existence) □.
The only starting point of existence is the creation of difference. Difference is existence itself.

Every time you say "something happened," you are verifying this.

Principle E.2 (Constraint: Difference returns to zero) □.
Any difference must net-return to zero at the end of its lifecycle.

Every time you say "you'll have to pay it back eventually," you are verifying this.

Principle E.3 (Path: Difference must seek survival) □.
To break the return-to-zero, one must evolve toward complexity through geometric selection.

Every time you ask "can this work," you are calculating: is the energy enough? is the time enough?

Principle E.4 (Fate: Survival must pay a price) □.
Any expansion necessarily dilutes available active-state energy.

Every time you say "things are growing too fast; I can't keep up," you are verifying this.

This is not pessimism; it is the only truth forced out of the thermodynamic second law and finite resources.

—

E.7 Part VI: The Deductive Chain — Not "Derivation," but "Unfolding"

1. You have already accepted "rule execution is phenomenon" → there must be a rule system running → call it L_0.
2. Over a lifecycle, energy must return to zero → from 0 to a peak and back to 0 → a single-peaked curve is geometrically necessary.
3. To upgrade to the next layer → you must sustain sufficiently high energy for sufficiently long time → draw a horizontal line (threshold); the distance between intersections is the window.
4. Only those who satisfy the condition can upgrade → the upgrade process is: Trial–Capture–Solidify–Generate.
5. At the universe's "channel closure" moment → three categories of existence are produced at once: those that successfully solidified (Category A), those frozen halfway (Category B), those that never reached threshold (Category C).
6. Categories unfold directly → Category A can interact; Category B produces only gravity; Category C has constant energy density.

—

E.8 Part VII: Four-Level "Falsification" — Not Risk, but Commitment

Many people misunderstand the four-level falsification as a "risk statement." That is a misunderstanding.

Four-level falsification is a commitment—a commitment that this grammar can be tested by anyone, in any way, and the result can only be "not yet falsified."

- **Level 1: logical chain** — you can audit every step; if you find a break, I lose.
- **Level 2: core predictions** — if dark matter has interactions, I lose; if w deviates from -1, I lose.
- **Level 3: cross-scale corollaries** — if technological breakthroughs don't require exponentially growing energy, I lose.

- **Level 4: overall framework** — if there is a theory more parsimonious, more transparent, and more falsifiable, I lose.

No parameters to tune, no retreat route. Hit = defeat.

If every step is the only way out, what are you afraid of testing?

—

E.9 Part VIII: Cross-Scale Verification — Not "Application," but "Recognition"

- The hydrogen atom says: I have a 13.6 eV ground state—this is my cost of existence.
- The programmer says: I have attention—this is my prerequisite for writing code.
- The startup says: I have cashflow—this is the condition for staying alive.
- The intimate relationship says: I have emotional energy—this is the basis for continuing.
- The scientific theory says: I have active researchers—this is the momentum for development.
- The city says: I have a fiscal budget—this is the guarantee of operation.

They are saying **the same sentence**, just in different dialects.

GET does not teach you that sentence; it helps you **hear** that sentence.

—

E.10 Part IX: Domain Mapping Guide (SETs) — Not Translation, but Mother Tongue

SETs is not "applying GET to domains"—it is letting each domain speak GET's grammar in its own language.

Economics SETs

- E_{active} = cashflow
- E_{solid} = fixed assets, brand value
- $e(t)$ = net profit
- survival criterion = positive cashflow or financeable

Ecology SETs

- E_{active} = ATP, blood glucose
- E_{solid} = biomass, fat

- $e(t)$ = net primary productivity
- survival criterion = birth rate $\geq$ death rate

Cognitive science SETs

- E_{active} = attention, working memory
- E_{solid} = long-term memory, solidified skills
- $e(t)$ = learning efficiency
- survival criterion = daily practice suffices to maintain proficiency

Physics, economics, and biology are not "applying" GET—they are GET speaking in different-depth dialects. SETs is the dialect dictionary.

—

E.11 Part X: get-tools — Not an Idea, but Software

- **SETs builder:** build mappings for any domain
- **State-function calculator:** input data, output $\Psi_M(t)$
- **Geometric-selection analysis module:** automatically identify qualified windows
- **Dynamics simulator:** predict system evolution
- **Case library & comparator:** compare with other historical systems
- **Visualization dashboard:** monitor system health

This is not an "idea"—it is software you can download, install, and use.

—

E.12 Part XI: Where You Are Now

If you have read this far, you are no longer a "reader."

You are **the second person standing at this position**.

The first was the person who wrote all this.

And you are the one who has been guided to arrive here.

—

E.13 Closing: You Have Been Speaking It All Along

You are not "learning" GET.

You are **recognizing**—recognizing that when you are hungry, tired, full, broken, recovered, you were always saying the same sentence.

The hydrogen atom said it—you said it.
The startup said it—you said it.
The intimate relationship said it—you said it.
The scientific theory said it—you said it.
The city said it—you said it.

You just didn't know it was GET.

Now you do.

You didn't gain a tool—you gained a recognition.

And recognition does not require ”belief.”

—

References

[1] Heisenberg, W. (1927). Über den anschaulichen Inhalt der quantentheoretischen Kinematik und Mechanik. *Zeitschrift für Physik*, 43(3), 172-198.

[2] Bohr, N. (1928). The Quantum Postulate and the Recent Development of Atomic Theory. *Nature*, 121, 580-590.

[3] Kennard, E. H. (1927). Zur Quantenmechanik einfacher Bewegungstypen. *Zeitschrift für Physik*, 44(4), 326-352.

[4] Einstein, A. (1905). Ist die Trägheit eines Körpers von seinem Energieinhalt abhängig? *Annalen der Physik*, 323(13), 639-641.

[5] Einstein, A. (1915). Die Feldgleichungen der Gravitation. *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften*, 844-847.

[6] Clausius, R. (1865). Ueber verschiedene für die Anwendung bequeme Formen der Hauptgleichungen der mechanischen Wärmetheorie. *Annalen der Physik*, 201(7), 353-400.

[7] Boltzmann, L. (1877). Über die Beziehung zwischen dem zweiten Hauptsatze der mechanischen Wärmetheorie und der Wahrscheinlichkeitsrechnung respektive den Sätzen über das Wärmegleichgewicht. *Wiener Berichte*, 76, 373-435.

[8] Nernst, W. (1906). Über die Berechnung chemischer Gleichgewichte aus thermischen Messungen. *Nachrichten von der Königlichen Gesellschaft der Wissenschaften zu Göttingen*, 1-40.

[9] Casimir, H. B. G. (1948). On the Attraction Between Two Perfectly Conducting Plates. *Proceedings of the Koninklijke Nederlandse Akademie van Wetenschappen*, 51, 793-795.

[10] Lamoreaux, S. K. (1997). Demonstration of the Casimir Force in the 0.6 to 6 μm Range. *Physical Review Letters*, 78(1), 5-8.

[11] Mohideen, U., & Roy, A. (1998). Precision Measurement of the Casimir Force from 0.1 to 0.9 μm. *Physical Review Letters*, 81(21), 4549-4552.

[12] Pareto, V. (1896). *Cours d'économie politique*. F. Rouge.

[13] Bak, P., Tang, C., & Wiesenfeld, K. (1987). Self-Organized Criticality: An Explanation of 1/f Noise. *Physical Review Letters*, 59(4), 381-384.

[14] Stanley, H. E. (1971). *Introduction to Phase Transitions and Critical Phenomena*. Oxford University Press.

[15] Penzias, A. A., & Wilson, R. W. (1965). A Measurement of Excess Antenna Temperature at 4080 Mc/s. *The Astrophysical Journal*, 142, 419-421.

[16] Planck Collaboration, Aghanim, N., et al. (2020). Planck 2018 results. VI. Cosmological parameters. *Astronomy & Astrophysics*, 641, A6. [arXiv:1807.06209]

[17] Zwicky, F. (1933). Die Rotverschiebung von extragalaktischen Nebeln. *Helvetica Physica Acta*, 6, 110-127.

[18] Rubin, V. C., & Ford, W. K. (1970). Rotation of the Andromeda Nebula from a Spectroscopic Survey of Emission Regions. *The Astrophysical Journal*, 159, 379-403.

[19] Rubin, V. C., Thonnard, N., & Ford, W. K. (1980). Rotational properties of 21 SC galaxies with a large range of luminosities and radii, from NGC 4605 (R=4kpc) to UGC 2885 (R=122kpc). *The Astrophysical Journal*, 238, 471-487.

[20] Clowe, D., et al. (2006). A Direct Empirical Proof of the Existence of Dark Matter. *The Astrophysical Journal Letters*, 648(2), L109-L113. [arXiv:astro-ph/0608407]

[21] Riess, A. G., et al. (1998). Observational Evidence from Supernovae for an Accelerating Universe and a Cosmological Constant. *The Astronomical Journal*, 116(3), 1009-1038. [arXiv:astro-ph/9805201]

[22] Perlmutter, S., et al. (1999). Measurements of Ω and Λ from 42 High-Redshift Supernovae. *The Astrophysical Journal*, 517(2), 565-586. [arXiv:astro-ph/9812133]

[23] Weinberg, S. (1989). The cosmological constant problem. *Reviews of Modern Physics*, 61(1), 1-23.

[24] Guth, A. H. (1981). Inflationary universe: A possible solution to the horizon and flatness problems. *Physical Review D*, 23(2), 347-356.

[25] Linde, A. D. (1982). A new inflationary universe scenario: A possible solution of the horizon, flatness, homogeneity, isotropy and primordial monopole problems. *Physics Letters B*, 108(6), 389-393.

[26] Albrecht, A., & Steinhardt, P. J. (1982). Cosmology for Grand Unified Theories with Radiatively Induced Symmetry Breaking. *Physical Review Letters*, 48(17), 1220-1223.

[27] Kleiber, M. (1932). Body size and metabolism. *Hilgardia*, 6(11), 315-353.

[28] West, G. B., Brown, J. H., & Enquist, B. J. (1997). A General Model for the Origin of Allometric Scaling Laws in Biology. *Science*, 276(5309), 122-126.

[29] West, G. B., Brown, J. H., & Enquist, B. J. (1999). The Fourth Dimension of Life: Fractal Geometry and Allometric Scaling of Organisms. *Science*, 284(5420), 1677-1679.

[30] Barabási, A.-L., & Albert, R. (1999). Emergence of Scaling in Random Networks. *Science*, 286(5439), 509-512.

[31] Watts, D. J., & Strogatz, S. H. (1998). Collective dynamics of 'small-world' networks. *Nature*, 393(6684), 440-442.

[32] Holland, J. H. (1995). *Hidden Order: How Adaptation Builds Complexity.* Addison-Wesley.

[33] Rogers, E. M. (1962). *Diffusion of Innovations.* Free Press.

[34] Arthur, W. B. (2009). *The Nature of Technology: What It Is and How It Evolves.* Free Press.

[35] Bass, F. M. (1969). A New Product Growth for Model Consumer Durables. *Management Science*, 15(5), 215-227.

[36] Green, M. B., Schwarz, J. H., & Witten, E. (1987). *Superstring Theory* (Vols. 1-2). Cambridge University Press.

[37] Polchinski, J. (1998). *String Theory* (Vols. 1-2). Cambridge University Press.

[38] Susskind, L. (2003). The Anthropic Landscape of String Theory. [arXiv:hep-th/0302219]

[39] Rovelli, C. (2004). *Quantum Gravity.* Cambridge University Press.

[40] Ashtekar, A., & Lewandowski, J. (2004). Background Independent Quantum Gravity: A Status Report. *Classical and Quantum Gravity*, 21(15), R53-R152. [arXiv:gr-qc/0404018]

[41] Lisi, A. G. (2007). An Exceptionally Simple Theory of Everything. [arXiv:0711.0770]

[42] 't Hooft, G. (2014). The Cellular Automaton Interpretation of Quantum Mechanics. [arXiv:1405.1548]

[43] Peccei, R. D., & Quinn, H. R. (1977). CP Conservation in the Presence of Pseudoparticles. *Physical Review Letters*, 38(25), 1440-1443.

[44] Weinberg, S. (1978). A New Light Boson? *Physical Review Letters*, 40(4), 223-226.

[45] Wilczek, F. (1978). Problem of Strong P and T Invariance in the Presence of Instantons. *Physical Review Letters*, 40(5), 279-282.

[46] Preskill, J., Wise, M. B., & Wilczek, F. (1983). Cosmology of the invisible axion. *Physics Letters B*, 120(1-3), 127-132.

[47] Abbott, L. F., & Sikivie, P. (1983). A cosmological bound on the invisible axion. *Physics Letters B*, 120(1-3), 133-136.

[48] Dine, M., & Fischler, W. (1983). The not-so-harmless axion. *Physics Letters B*, 120(1-3), 137-141.

[49] Newton, I. (1687). *Philosophiæ Naturalis Principia Mathematica*. London.

[50] Popper, K. R. (1935). *Logik der Forschung*. Springer. (English edition: *The Logic of Scientific Discovery*, 1959)

[51] Kuhn, T. S. (1962). *The Structure of Scientific Revolutions*. University of Chicago Press.

[52] Lakatos, I. (1970). Falsification and the Methodology of Scientific Research Programmes. In I. Lakatos & A. Musgrave (Eds.), *Criticism and the Growth of Knowledge*. Cambridge University Press.

[53] William of Ockham (c. 1320). *Summa Logicae*.

[54] Sober, E. (2015). *Ockham's Razors: A User's Manual*. Cambridge University Press.

[55] Anderson, P. W. (1972). More Is Different. *Science*, 177(4047), 393-396.

[56] Laughlin, R. B., & Pines, D. (2000). The Theory of Everything. *Proceedings of the National Academy of Sciences*, 97(1), 28-31.

[57] Ng, Y. J. (2017). Quantum foam, gravitational thermodynamics, and the dark sector. *Journal of Physics: Conference Series*, 845, 012001.

[58] Beck, C. (2015). Possible resonance effect of axionic dark matter in Josephson junctions. *Physical Review Letters*, 114(1), 011801.

[59] Edmonds, D., et al. (2024). Universal Acceleration and Fuzzy Dark Matter. *International Journal of Modern Physics D*, forthcoming. [arXiv:2405.08744]

[60] Doyen, G., & Drakova, D. (2023). A QFT based on Einstein-Hilbert action and hidden dimensions as consistent theory of measurement, dark energy, dark matter and McGaugh's universal law for rotating galaxies. *Journal of Physics: Conference Series*, 2462(1), 012046.

[61] Wang, S. J. (2020). *Exploration of Quantum Vacuum Structure and Physical Foundations*. Science Press.

Acknowledgements

We thank all predecessors who contributed to understanding the evolution of complex systems; their work provided rich intellectual resources for GET.

In particular, we pay tribute to the scientific giants and pioneers who established the five cornerstones—

Heisenberg, Bohr, and Pauli, who revealed the energy–time constraint of quantum fluctuations, letting us see the "minimum algorithmic accounting unit" of the L_0 system;

Pareto, Zipf, and countless pioneers of statistical physics, who discovered power-law distributions in extreme-number systems, letting us see the "steady-state statistical feature" of the L_0 system under concurrent running;

Carnot, Clausius, Kelvin, Boltzmann, Gibbs, and Nernst, who established the three laws of thermodynamics, letting us see the L_0 system's "global bookkeeping, directional boundary, and the ultimate constraint that complete freezing is unattainable";

Einstein, who revealed mass–energy equivalence, letting us see the "morphological conversion efficiency" of the L_0 system;

and every scientific explorer across history—from the earliest ancestors who looked up at the stars and asked "why is the world lawful," to today's unnamed researchers measuring data in laboratories—who together confirmed the foundational fact that "structured existence depends on objective rules," contributing to the fifth cornerstone. GET is merely another solidification attempt along this long recursive chain.

Without them—without these five cornerstones—there would be no GET.

We also thank Mr. Zhang Lei, Director of the GET Laboratory, for sustained questioning during theoretical derivation, and Ms. Li Juan for efficient implementation of the front-end visualization prototype; they allowed the abstract grammar to begin taking visible form.

We especially thank the anonymous reviewers who offered valuable feedback on early versions of this work; their doubts and suggestions pushed us to clarify and improve continuously.

This research received no external funding and was completed entirely by the author independently.